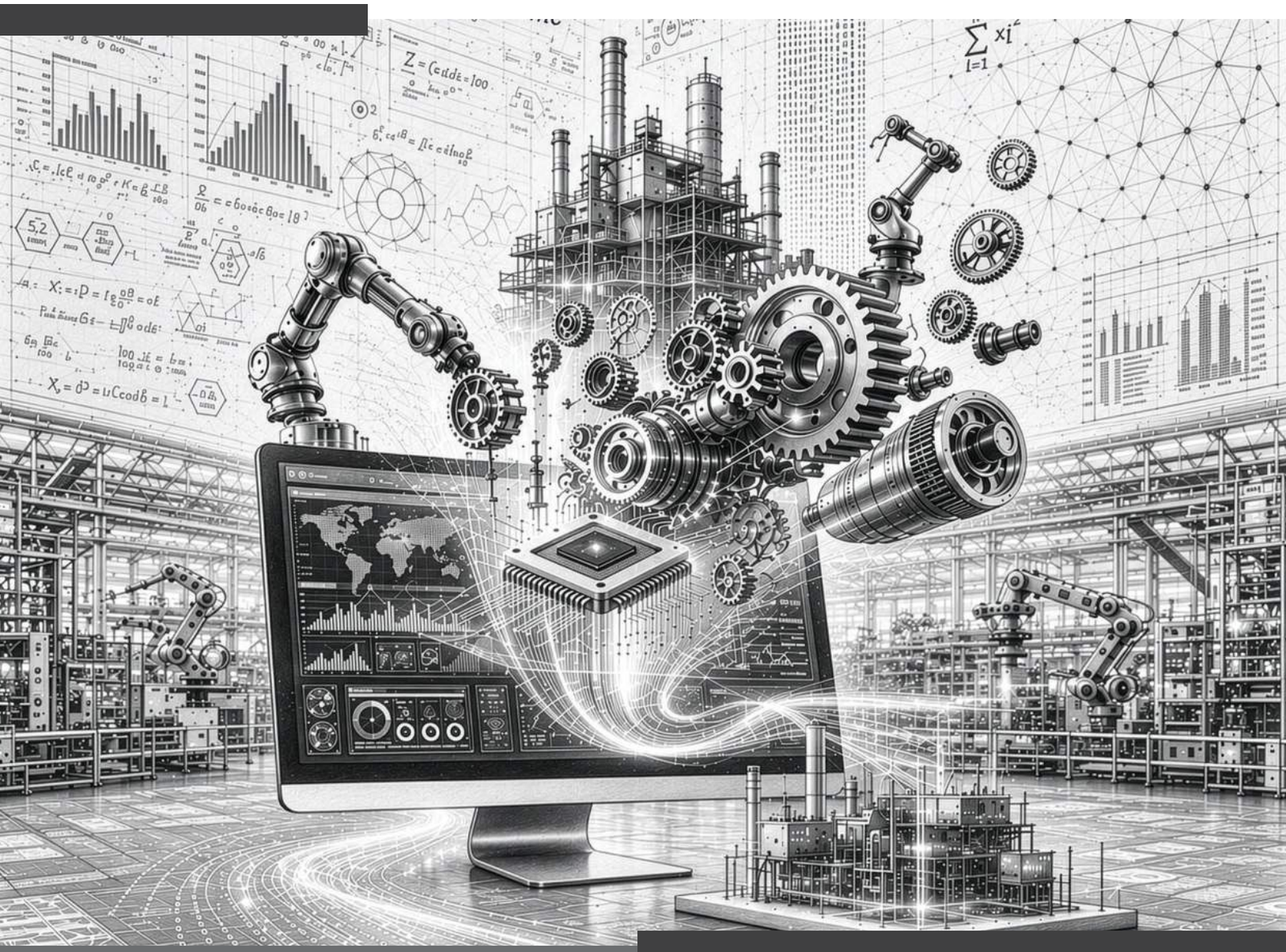
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تأهيل
TA'HEAL



DIGITAL TRANSFORMATION

SEMESTER ONE **2026/27**

STUDENT BOOK

GRADE **1**

Digital Transformation

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Digital Literacy & Office Productivity Essentials

1

EZZ STEEL APPLIED TECHNOLOGY SCHOOLS

UNIT



UNIT OVERVIEW

This unit connects everyday computer use to real technical work. As you read, notice that every digital action has a purpose: finding reliable information, protecting evidence, communicating clearly, or supporting a safe decision. The examples are taken from mechatronics and steel-production settings so that new terms are linked to situations you may meet during training.

Modern mechatronics combines mechanical, electrical and electronic systems with computers, software and data. A technician may still use hand tools, meters and drawings, but they also need to open digital work orders, find a manufacturer manual, read a dashboard, organise evidence, write reports and communicate through workplace systems. This unit develops the essential digital habits that support safe, accurate and efficient work.

Learning Outcomes

By the end of this unit, you will be able to:

- Explain why digital literacy is part of a modern technician's professional competence.
- Describe basic computer, operating-system and file-management concepts.
- Produce clear technical documents using word-processing tools.
- Search for technical information safely and evaluate whether a source is trustworthy.
- Communicate professionally using workplace email and digital etiquette.
- Use AI assistants responsibly to support learning while checking accuracy and protecting information.

Main Topics

DIGITAL SKILLS FOR THE MECHATRONICS TECHNICIAN



Lessons

Lesson	Title
1	Digital Literacy in the Mechatronics Workplace & Industry 4.0 Context
2	Computers, Operating Systems & File Management
3	Word Processing for Technical Documentation
4	Internet Foundations & Safe, Effective Online Search
5	Professional Email & Digital Workplace Etiquette
6	Introduction to AI Assistants & Responsible Use
7	Unit Assessment (theory)
8	Feedback and Treatment

Sessions

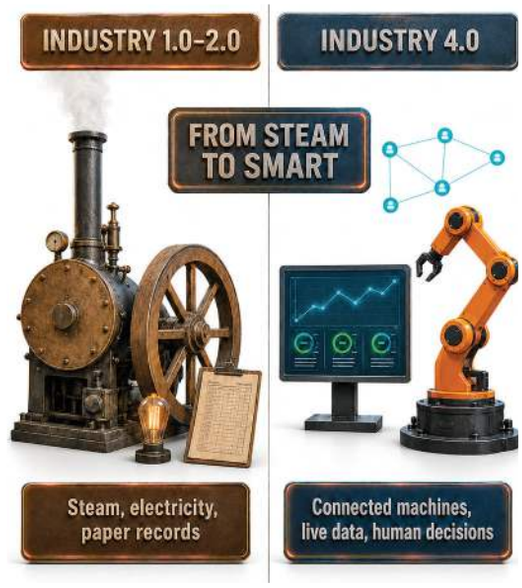
Session	Title
1	Recognizing Digital Work & Solving a Digital Problem in the Workshop
2	Creating a Structured Folder System for Workshop Documents
3	Producing a Formatted Safety Inspection Report in Word
4	Searching, Locating & Evaluating a Motor Datasheet Online
5	Writing & Sending a Professional Maintenance-Request Email
6	Using an AI Chatbot to Explain a Technical Concept & Verifying the Answer
7	Unit Assessment activity (practical Capstone): Building an Equipment Inspection Checklist
8	Repetition of Substandard Activity

LESSON 1 Digital Literacy in the Mechatronics Workplace & Industry 4.0 Context

Learning Outcomes

By the end of the lesson, you will be able to:

- define digital literacy and explain the beginner-friendly five-part cycle used in this book.
- distinguish basic device operation from safe and professional digital practice.
- describe the four industrial revolutions and explain the main features of Industry 4.0.
- explain a simple smart-factory data flow.
- apply the UNDERSTAND-FIND-TRY-CHECK method to a simple digital problem.
- record simple digital/workshop information accurately and responsibly.



From steam-powered factories to connected smart factories.

WARM-UP

One Day Without Digital Tools

In pairs, imagine that computers, screens, sensors, phones, and network connections stop working in a steel factory, then:

1. write three problems the factory may face. Include at least one safety problem and one production problem.
2. compare with another pair and select your strongest example to share.

Introduction: Your First Morning in the Workshop

Scenario: A SUPERVISOR GIVES YOU FIVE TASKS

It is 07:30 on your first day as a trainee technician. Your supervisor asks you to open the digital work order for Conveyor 12, check yesterday's maintenance note, find the manufacturer's motor datasheet, save two inspection photographs in the correct folder, and send a short update before the morning meeting. At 07:40, a warning appears on the dashboard: "Motor vibration above normal range." You are not expected to repair the motor alone. You are expected to read the information accurately, follow the approved process, record what you observe and report it to the correct person.

In pairs, think: Which parts of this situation require technical knowledge, and which parts require digital literacy?

Topic 1: What Is Digital Literacy?

Digital literacy is broader than knowing how to operate a phone or computer; it combines technical use with judgement, safety, responsibility and communication. The story, definition and cycle that follow show how these qualities work together during one workshop task.

1.1 Two Technicians, One Fault

Compare the following two technicians carefully. Both may understand the mechanical fault, but their results differ because they handle digital information differently.

TWO TECHNICIANS, ONE PROBLEM

A conveyor motor stops. Technician A opens the correct digital manual, searches for error code E-42, checks that the manual matches the motor model, follows the authorised reset procedure and records the result. Technician B knows motors, but searches random websites, opens a diagram for a different model and cannot find the previous maintenance record. Both technicians have mechanical knowledge, but only one uses digital information accurately and safely.

Think: What professional behaviours made Technician A more effective?

1.2 Definition

Read the definition once for the main idea, then connect each part—access, understand, evaluate, use, create and communicate—to the workshop example. The words safe and responsible are essential because digital actions can influence equipment, records and people.



Definition - Digital literacy

Digital literacy is the ability to access, understand, evaluate, use, create and communicate information with digital technologies in a confident, critical, safe and responsible way.
Workshop example: Finding the correct motor datasheet, checking the model number, using the information to complete an inspection, writing a report, and sharing it with the authorised supervisor.

In this book, we use a simple five-part cycle: FIND -> EVALUATE -> USE -> CREATE -> SHARE. Safety and responsibility are checked at every step. This cycle is a beginner-friendly learning model. It is consistent with the wider areas of digital competence described in the European DigComp framework, including information literacy, communication, content creation, safety and problem solving.

The cycle diagram should be read clockwise. Each coloured stage represents one action a technician performs with digital information, while the centre identifies the professional user who connects the steps. The arrows show that digital work is a continuing process rather than a single click.

THE DIGITAL LITERACY CYCLE



Figure 1.1: The digital literacy cycle used in this course

1.3 The Five Parts in Detail

The five-part cycle breaks a complex professional competence into manageable actions. In practice, the steps may overlap or repeat; for example, you may evaluate a source again after finding new information. The table helps you connect each step with a question and a concrete workshop behaviour.

The following table explains the five-part digital literacy cycle in operational terms. Read across each row from the action to the self-question and then to the workshop example; this makes the concept easier to apply in a new situation.

Part	What the technician does	Questions to ask	Workshop example
Find	Locates the information needed.	What exactly do I need? Where should it be stored?	Find the maintenance history for Press 3.
Evaluate	Checks source, date, model and accuracy.	Is it official? Current? Is it for the correct machine?	Confirm a wiring diagram is from the manufacturer.
Use	Applies information according to procedures and authority.	What am I allowed to do? What safety rule applies?	Use a checklist to inspect a guard.
Create	Produces a clear digital record or complete document.	Is the information factual and readable?	Write a maintenance note with date and evidence.
Share	Sends information to the right people using an approved method.	Who needs it? Is any information confidential?	Upload the report to the shared system and notify the supervisor.

1.4 Digital Literacy Is More Than "Knowing Computers"

A person may know where to click but still use the wrong file, share confidential information or ignore a safety boundary. Read each pair from left to right and identify the extra judgement required in the workplace.

This comparison table shows why operating a device is only the starting point. Each right-hand entry adds judgement, accuracy, safety or responsibility to the basic action shown on the left.

Basic device operation	Professional digital literacy
Can open a browser.	Can search using precise technical terms and choose a reliable source.
Can type a document.	Can create a complete, accurate and well-formatted technical record.
Can copy a file.	Can place it in the correct folder, use a clear name and protect the original.
Can send a message.	Can choose the correct channel, recipient, tone and level of confidentiality.
Can read a dashboard.	Can interpret a warning carefully, verify context and escalate within role boundaries.

PROFESSIONAL STANDARD

- Accuracy: use the correct file, model, date and version.
- Safety: do not bypass procedures, interlocks or access controls.
- Security: use authorised accounts, devices and storage locations.
- Efficiency: organise work so it can be found and understood quickly.
- Communication: record facts clearly and send them to the correct person.



1.5 Home Skills and Workplace Skills

Many useful digital habits begin at home, but workplace use has higher requirements. Technical information must be accurate, traceable, and shared through authorised systems because mistakes can affect production or safety. The table shows how familiar actions become more disciplined in a professional setting.

The following table compares familiar personal use with the higher standards required at work. Read the two context columns side by side and identify the added need for verification, authorisation and documentation.

Skill	At home you may...	At work you must...
Find	Search for a game tip or a recipe.	Search for a specific manual, error code or standard.
Evaluate	Compare reviews or check whether a post is real.	Check manufacturer, model, revision, date and approval status.
Create	Post a photo or write a message.	Create a factual report, checklist or labelled inspection image.
Share	Send content to friends.	Use authorised systems and protect workplace information.
Solve problems	Try different settings.	Use a safe method, remain within authority and record what was tried.

Pop-Up Quiz 1

1. Which action is EVALUATE?

a. Downloading a file **b.** Checking that a manual matches the motor model

c. Sending a report **d.** Creating a folder

2. Why is digital literacy a safety issue? Write one sentence.

3. A trainee can use a tablet but shares an unverified wiring diagram. Is the trainee digitally literate? Explain.

1.6 Digital Traceability and Accountability

Digital traceability means being able to follow the history of a digital record: who created it, when it was created, which machine it concerns, and which version is current. In technical work, traceability allows another person to check the evidence and understand how a decision was reached.

Consider a safety photograph named photo1.jpg and saved on a desktop. The image may be useful, but its date, machine and purpose are unclear. A professional record uses a meaningful name, a known folder and an authorised account so that responsibility is visible without changing the evidence.

The table below shows four pieces of information that make a digital record traceable. Read across each row and notice how a small file-management habit supports a larger professional need.

Traceability element	Question it answers	Workshop example
Identity	Who created or updated the record?	The trainee saves the report using an individual authorised account.
Time	When was the evidence created or changed?	The file name and document show the inspection date.
Asset and purpose	What machine and task does the record concern?	Conveyor12_SafetyReport identifies the asset and document type.
Version and location	Which copy is current and where is it stored?	v02 is saved in the approved Safety folder; the older version is retained according to procedure.

Topic 2: Digital Work in Mechatronics

2.1 FIVE DIGITAL TASKS BEFORE LUNCH

Sara's timeline demonstrates that digital work is distributed across the day rather than limited to an office. She moves between work orders, manuals, dashboards, photographs and communication tools. As you read, classify each action as finding, evaluating, using, creating or sharing information.

Scenario:

07:30 - Sara opens the computerized maintenance system and reads today's work orders. 08:10 - She scans the QR code on a pump to open its approved manual. 09:00 - She reads temperature and vibration trends on a dashboard. 09:45 - She saves inspection photographs with meaningful names. 10:30 - She writes a short fault report and emails a spare-part request. 11:15 - She checks that her team can open the shared files.

Think: Which task uses file management, which uses information evaluation, and which uses communication?

2.2 The Main Digital Tools

The following table is an orientation map of tools you may meet during the programme. Focus first on the purpose of each category and the trainee's responsibility, rather than memorizing product names. The same professional rules- accuracy, permission, security, and clear records- apply across different systems.

Category	Examples	Main Purpose	First-year responsibility
Devices	Desktop PC, laptop, tablet, barcode or QR scanner	Access systems and capture information.	Handle equipment carefully; use authorised accounts.
Productivity software	Word processor, spreadsheet, PDF reader	Create and read technical documents.	Save in the correct format and location.

Category	Examples	Main Purpose	First-year responsibility
Maintenance systems	CMMS, digital work-order system.	Plan, record, and track maintenance.	Enter accurate facts; do not close work you did not complete.
Industrial interfaces	HMI, dashboard, alarm screen.	Show machine status, trends, and alarms.	Observe, record and report; do not change settings without authority.
Communication tools	Email, shared drive, team platform.	Coordinate work and share evidence.	Use professional language and protect information.
Reference sources	Manufacturer manuals, datasheets, standards.	Support correct technical decisions.	Check model, revision, source and date.

The skills-map diagram places accurate, safe, efficient and professional work at the centre. The surrounding boxes show the practical abilities that support that standard.

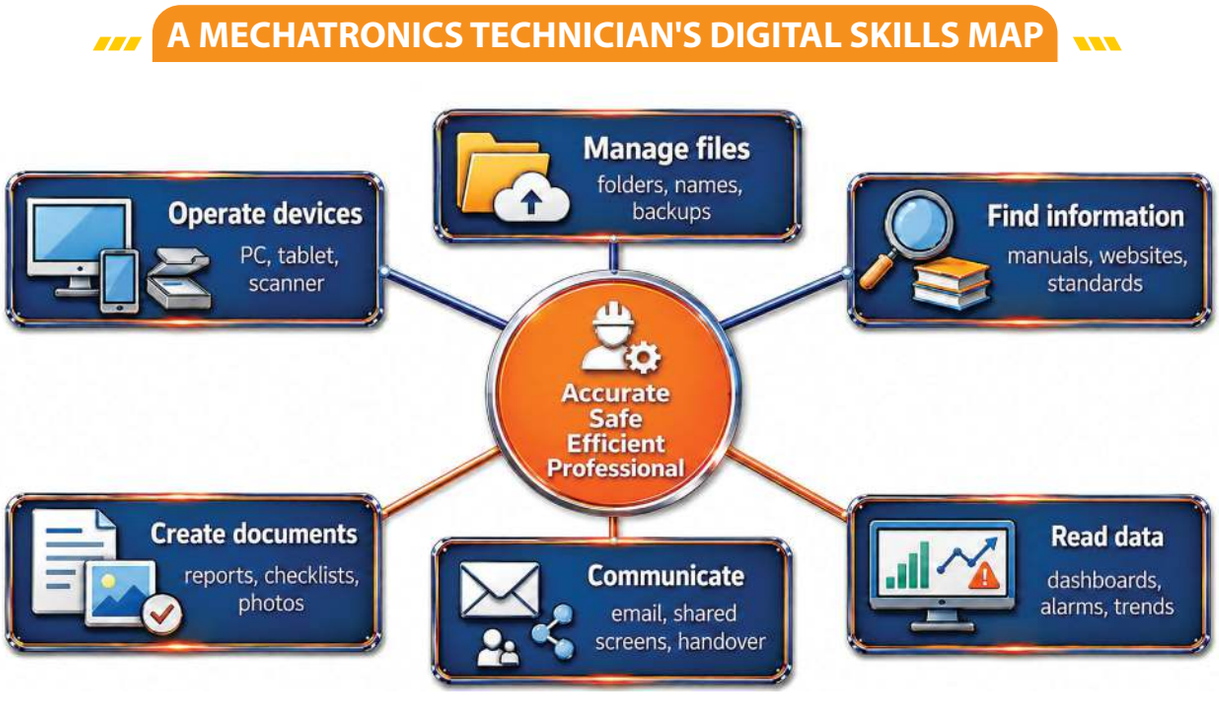


Figure 1.2: Digital skills map of a mechatronics technician.

2.3 Data, Information and Evidence

Technicians often see numbers before they know what the numbers mean. Raw data becomes useful information only when it is connected to context such as the normal range, machine model, time, and operating condition.

Evidence is the record that allows another person to check what was observed and how a decision was made.

DEFINITIONS

Data

Data are raw facts or measurements, such as 82 degrees Celsius, 1,450 rpm, or a vibration value of 6.2 mm/s.

Workshop example: A sensor sends a temperature value every second.

Information

Information is data given meaning and context so that a person can understand it.

Workshop example: "Motor temperature increased from 65 degrees Celsius to 82 degrees Celsius during the last 20 minutes."

Digital evidence

Digital evidence is a record that supports what happened, what was observed, or what action was taken.

Workshop example: A dated photograph, dashboard screenshot, completed checklist or maintenance log entry.

GOOD TECHNICAL RECORDS ARE FACTUAL

- Write what you observed, not what you guessed.
- Include the machine or asset identity, date and time.
- Use measurements when available.
- Record who was informed and what happened next.
- Do not alter, hide, or delete records to make work look better.



Activity 1 *Data or Information?*

Read and label each of the following statements as DATA, INFORMATION, or PREDICTION. Then discuss which statement requires the most checking before action.

74 degrees Celsius = ...

Bearing temperature is 12 degrees above its normal range = ...

Motor may fail today = ...

2.4 Reading Data in Context

A number on a screen does not explain itself. To interpret it, a technician needs context: the correct asset, unit, normal range, time, operating condition, and source. Without context, even an accurate measurement can be misunderstood.

For example, a temperature of 82 degrees Celsius may be acceptable for one component and dangerous for another. A first-year trainee should record the exact value and context, compare it with an approved reference, and escalate when a decision is outside the trainee's authority.

The table demonstrates how the same raw value becomes more useful as context is added. Read from top to bottom to see the movement from an isolated number to evidence that can support a safe decision.

Level	Example	What is still needed?
Raw data	82 °C	Asset, location, time, unit confirmation, and operating condition.
Contextual information	Motor 7 bearing temperature is 82 °C during normal production.	Approved normal range and recent trend.
Evidence for action	Motor 7 bearing temperature is 12 °C above the approved normal limit and has risen for three readings.	Physical check and escalation through the approved maintenance procedure.

What Does This Mean for YOU?

The machines you will maintain are full of sensors and controllers that produce **data**. A modern technician must be able to read that data on screens and dashboards, record work digitally, and communicate through digital tools. Companies today advertise for 'mechatronics technicians with digital skills' – this unit is your entry ticket to those jobs!



Activity 2 *What is Industry 4.0?*

Watch the first 4 minutes of this video with your class, then answer the questions below.

1. Write down TWO technologies of Industry 4.0 that you hear about.

.....

.....

2. How can a 'smart' machine help a technician do maintenance?

.....

.....

3. Which Industry 4.0 technology sounds most exciting to you?

.....

.....

DID YOU KNOW?

The word 'Industry 4.0' (Industrie 4.0) was first used in **Germany in 2011** at the Hannover Fair, one of the world's biggest industrial technology exhibitions. Germany created a national plan to make its factories 'smart' – and the idea spread around the whole world.

Reference: Recommendations for implementing the strategic initiative INDUSTRIE 4.0

Topic 3: Industrial Revolutions and Industry 4.0

This topic places modern digital work within the history of industrial development. The purpose is not to memorise many dates, but to understand the major change in each revolution. Industry 4.0 is best understood as connected automation: machines, systems, and people exchange data so that work can be monitored and improved

3.1 A Journey Through Four Generations

The family story provides a simple timeline from mechanical power to connected production. Imagine how the daily work of each generation would change as new technologies appear. The key question is not only what the machines can do, but what new knowledge and responsibilities workers need.

Imagine four members of one family working in factories at different times. In the 1800s, one uses a steam-powered machine. In the early 1900s, another works on an electric production line. In the late 1900s, a third monitors a PLC-controlled automated machine. Today, the fourth sees live sensor data on a dashboard and receives a warning before a fault becomes a breakdown. This journey helps us understand the four industrial revolutions.

The timeline diagram summarises the four industrial revolutions using a dominant technology for each stage. Read it from left to right and focus on the progression from mechanical power, to electrical mass production, to computer automation and finally to connected data systems.

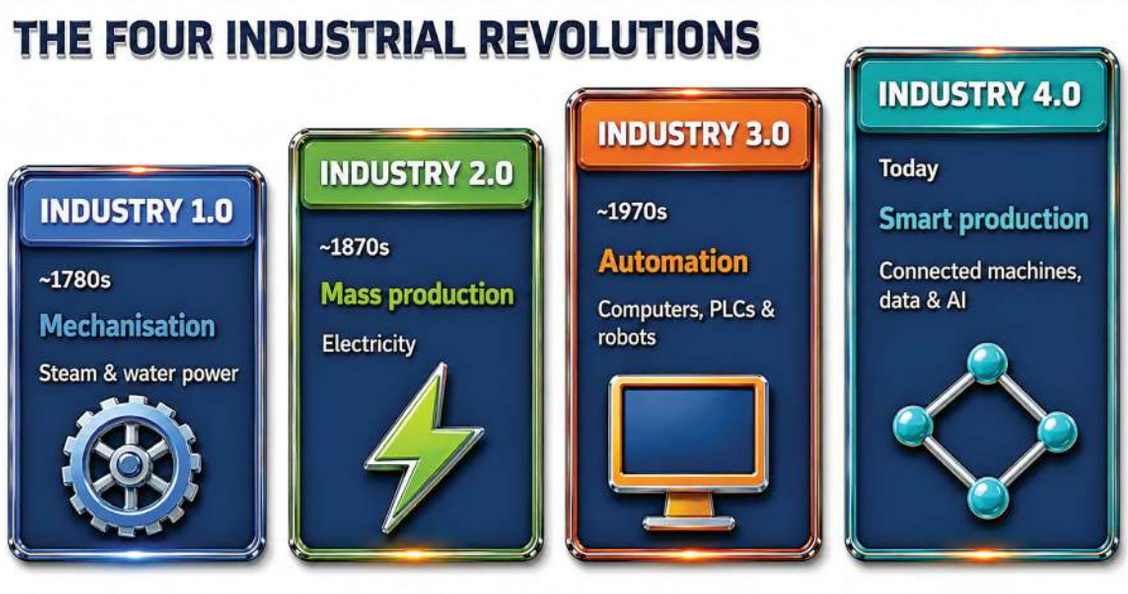


Figure 1.3: The four industrial revolutions.

3.2 The Four Revolutions Explained

The table summarises each revolution using the same four lenses: main change, technologies and effect on the technician. Read down the first column to see the historical sequence, then read across a row to connect technology with changes in work. This structure makes comparison easier than learning isolated facts.

Revolution	Main change	Typical technologies	Effect on the technician
Industry 1.0	Machines replace some manual effort.	Steam engine, water power, mechanical looms	Operate and maintain mechanical machinery.
Industry 2.0	Electricity enables large-scale production.	Electric motors, assembly lines, standardised parts	Maintain electrical machines and repeatable production systems.
Industry 3.0	Computers automate control.	Electronics, PLCs, CNC machines, industrial robots	Diagnose automated systems and work with control logic.
Industry 4.0	Machines, products, systems and people exchange data.	Sensors, industrial networks, IoT, cloud/edge computing, analytics and AI.	Combine mechanical/ electrical knowledge with data, software and secure communication.

3.3 What Exactly Is Industry 4.0?

Industry 4.0 combines connected equipment, digital data, and human decision-making. It does not mean that every factory is fully automatic or that people are no longer needed. The definition should be read together with the examples so that the term remains practical rather than abstract.



Definition - Industry 4.0

Industry 4.0 is the intelligent digital networking of people, machines, products and industrial processes so that data can be exchanged and used to improve production, quality, maintenance, flexibility and safety. A factory that uses these connected systems is often called a smart factory.

Workshop example: A vibration sensor sends data through a controller and network to a dashboard. Software identifies an unusual pattern. A technician checks the physical machine and follows the approved maintenance process.

Industry 4.0 does not mean

A factory with no people. Skilled people remain responsible for judgement, safety and improvement.
Every machine is controlled by artificial intelligence. Many systems use ordinary automation and rules.
A warning is automatically correct. Sensors, data, settings and context must be checked.
A trainee may change any machine setting. Access and authority still apply.

3.4 Key Technologies

The technologies in the table work as parts of a larger system. A sensor alone does not create a smart factory; value appears when measurements are transferred, analysed, presented, and used within an approved process. First-year students should recognise the purpose of each technology and one workshop example.

The technologies table below translates specialist terms into simple functions and workshop examples. Use it to recognize how several technologies cooperate in a smart-factory system.

Technology	Definition	Workshop example
Sensor	A device that measures a physical condition.	Measures temperature, pressure, speed, position, or vibration.
PLC	An industrial computer that reads inputs and controls outputs according to a programme.	Stops a conveyor when a safety sensor is activated.
HMI / dashboard	A screen that shows machine information and allows authorised interaction.	Displays alarms, status, and production counts.
Industrial IoT	Connected industrial devices that exchange data through networks.	A pump sends condition data to a monitoring platform.
Data analytics	Methods that find useful patterns in data.	Compares vibration values across several weeks.
Artificial intelligence	Software that performs tasks such as recognising patterns, classifying, or predicting.	Flags a pattern that may indicate bearing wear.
Cloud and edge computing	Places where data is stored or processed remotely (cloud) or near the machine (edge).	An edge device checks data quickly; selected records are stored centrally.
Robot/cobot	A programmable machine that performs physical tasks; a cobot is designed for closer human collaboration under controlled conditions.	Moves parts, welds or assists assembly.
Digital twin	A digital representation of a real asset or process, updated with data.	A virtual pump model is used to study performance.

3.5 How Data Moves in a Smart Factory

The diagram follows one measurement from the physical machine to a technician's decision. It shows that data passes through several stages, and each stage can affect quality or reliability. The final human action remains important because a warning supports judgement but does not replace inspection, procedures or authority

The smart-factory flow diagram follows data from a sensor to a controller, through a network and dashboard, and finally to a technician's action. The feedback arrow shows that the result is recorded and can improve later monitoring. Notice that the human decision appears after, not before, verification.

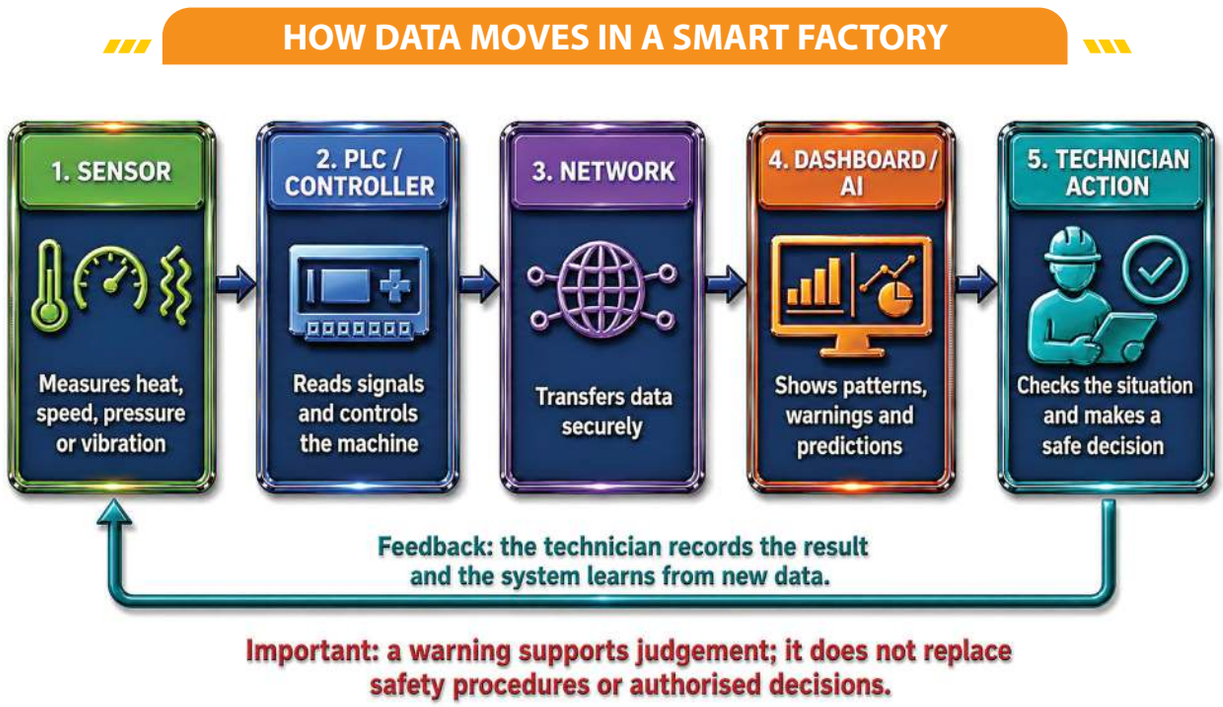


Figure 1.4: A simplified smart-factory data flow.

The flow begins with a physical condition. A sensor converts that condition into a signal. The controller reads the signal. The network transfers selected data. A dashboard or analytical tool presents the data. The technician then checks the real situation, follows procedures and records the outcome. This last human step is essential: digital tools support decisions, but they do not remove professional responsibility.

3.6 Benefits and Challenges

Digital systems can improve safety, maintenance and productivity, but every benefit depends on good data, secure access and competent people. The table pairs advantages with limitations so that Industry 4.0 is discussed realistically. Look for the condition that must be met before each benefit can be trusted.

The table presents benefits and challenges together so that the discussion remains balanced. For each benefit, identify the condition or risk that must be managed before the advantage can be achieved.

Possible benefit	How it helps	Important challenge
Earlier fault detection	Trends may show deterioration before breakdown.	Poor sensor data can produce misleading warnings.
Less unplanned downtime	Maintenance can be planned at a suitable time.	The organisation needs correct procedures and spare parts.
Improved quality	Data can identify variation and repeated defects.	People must interpret results and avoid hiding bad data.
Better traceability	Records show what happened and when.	Data must be accurate, protected and retained correctly.
Safer work	Remote monitoring can reduce exposure to hazards.	Connected systems introduce cybersecurity and access risks.
Energy efficiency	Data can reveal unnecessary energy use.	Measurements need context; lower energy must not reduce safety or quality.

SCENARIO: A PREDICTIVE-MAINTENANCE WARNING:

A dashboard reports "possible bearing wear" on Motor 7. The trainee checks the asset number, compares the trend with normal values, listens for unusual noise from a safe location, and informs the authorised technician. The trainee does not open the guard or replace the bearing alone. The digital warning helps the team investigate earlier; it does not replace the maintenance procedure.
Think: Which actions show digital literacy, and which action shows correct role boundaries?

Pop-Up Quiz 2

1. Which revolution introduced computer-based automation and PLCs?

a. 1.0 **b.** 2.0 **c.** 3.0 **d.** 4.0

2. What is the main difference between Industry 3.0 and Industry 4.0?

3. Place these in order: technician action - sensor - dashboard - PLC/controller - network.

4. Give one benefit and one challenge of connected production.

3.7 The Human Role in Industry 4.0

Connected systems can collect and display large amounts of information, but people remain responsible for interpreting that information within procedures. A dashboard may show a warning, yet an authorised technician still checks the asset, the physical condition, and the quality of the data before deciding what should happen.

Human judgement is especially important when information is incomplete, contradictory or safety-critical. Industry

4.0 therefore changes the technician's skill set: digital awareness becomes more important, but it works alongside mechanical knowledge, communication and responsibility.

The table separates what a digital system can support from what an authorised person must still decide. This distinction prevents students from treating automation or AI output as final proof.

Digital system can...	Human technician or team must...	Reason
Measure and display values	Check the correct sensor, asset and operating context	A value may be accurate but attached to the wrong situation.
Detect patterns and issue warnings	Inspect, verify and apply the approved procedure	A warning is evidence for investigation, not a complete diagnosis.
Store records and histories	Record factual actions and protect accountability	Poor or missing records weaken future decisions.
Recommend or prioritize actions	Confirm authority, safety and practical feasibility	Responsibility for authorised work remains human.

Topic 4: Solving Digital Problems Safely

This topic introduces a repeatable method for everyday digital problems. The purpose is not to make first-year students responsible for advanced repair; it is to replace random clicking with observation, reliable guidance and controlled actions. The stop rules protect equipment, information and people

4.1 Technicians Do Not Need to Know Every Answer

A competent technician does not guess when information is missing. Instead, the technician describes the problem accurately, consults an authorised source, and chooses a safe next step. This approach is more professional than pretending to know the cause or changing several settings at once.

SCENARIO: THE USB DRIVE DOES NOT APPEAR:

A trainee inserts an authorised USB drive, but it does not appear in File Explorer. One student repeatedly removes it and changes random settings. Another student reads the screen, checks the connection, tries a different authorised port, tests the drive on the teacher's computer, and reports what happened. The second student may not know the cause, but uses a controlled method.

Think: Why is a method more reliable than random trial and error?

Definition - Digital problem solving

Digital problem solving is a controlled process of understanding a problem, finding reliable guidance, trying a safe action and checking the result.
Workshop example: Troubleshooting a missing file, a printer message or a USB connection without changing unsafe settings.

The problem-solving diagram is a loop with a safety boundary at the centre. Follow the arrows through UNDERSTAND, FIND, TRY and CHECK, then return to the first step if the result is not satisfactory. The central warning reminds you that the method stops when an action is unsafe or outside your authority.

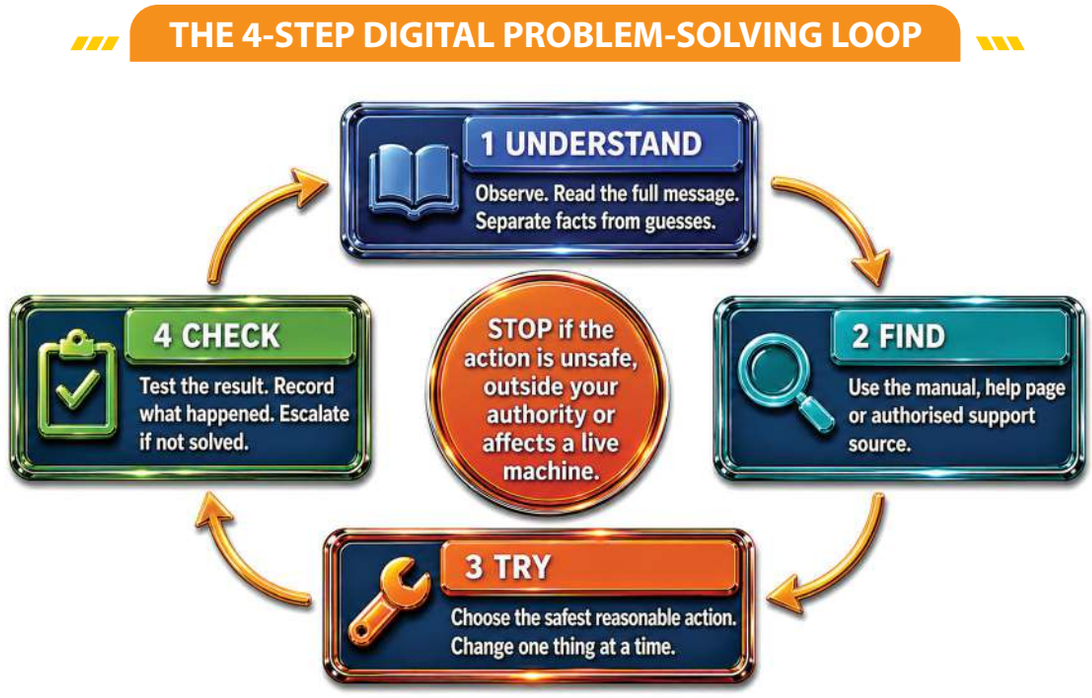


Figure 1.5: The UNDERSTAND-FIND-TRY-CHECK loop.

4.2 The Four Steps

The four steps form a loop because troubleshooting often produces new information. UNDERSTAND defines the problem, FIND identifies reliable guidance, TRY applies one safe action, and CHECK tests the result. If the problem remains, return to UNDERSTAND with better evidence rather than repeating the same action.

This table turns the four-step method into a practical checklist. Read across each row to connect the action with a useful question and the evidence that should be recorded.

Step	What to do	Useful questions	Evidence to record
UNDERSTAND	Observe carefully. Read the full message. Reproduce the problem once if safe.	What exactly happened? When? On which device? What still works?	Exact message, time, device, file or port.
FIND	Use approved help sources: Manual, teacher guide, IT help page, or authorised colleague.	Which source is most relevant instruction.	Source checked and reliable? Does it match this version?
TRY	Choose the safest reasonable action. Change one thing at a time.	Is it within my authority? Can I reverse it?	Action taken and order of steps.
CHECK	Test the result. Confirm normal operation. If not solved, return with new information or escalate.	Did the original problem disappear? Did a new problem appear?	Result, remaining issue and person informed.

4.3 Observation Is Not the Same as Assumption

An observation can be seen, heard, measured, or verified; an assumption is an interpretation that may or may not be correct. Keeping the two separate improves reports and prevents premature conclusions. The comparison table helps you practice factual language before suggesting a possible cause.

The comparison table separates factual language from interpretation. Use it to practice reporting what is known before suggesting a possible cause.

Observation - what you can see or measure	Assumption - what you think may be true
"The USB drive is not listed under This PC."	"The USB drive is broken."
"The printer screen says Paper jam - Tray 2."	"The printer has failed completely."
"The dashboard value changed from 3.0 to 5.4 mm/s."	"The bearing will fail today."

SAFETY - Know When to Stop

- Do not open electrical equipment, guards or control cabinets unless authorised and supervised.
- Do not bypass an alarm, interlock, password or access control.
- Do not connect an unknown personal USB drive to a workplace computer.
- Do not install software or change network/system settings without permission.
- Stop and escalate when the problem affects a live machine, safety system, confidential data or production control.

4.4 Worked Example: Printer Message

A worked example shows how the method is applied step by step. Notice that the trainee begins by reading the complete message and checking the approved instructions. The example also demonstrates the principle of making one change at a time and confirming the result before moving on.

The worked-example table shows how one printer problem is handled in a controlled sequence. The second column demonstrates the type of action expected at each stage.

Step	Example action
UNDERSTAND	The screen says "Paper jam - Tray 2". The printer has power and other trays are visible.
FIND	Use the printer's on-screen help diagram or approved manual.
TRY	Follow the safe paper-removal steps. Do not force parts. Change only the jam condition.
CHECK	Print one test page. Confirm the message is cleared and the page is complete.



Activity 3 *Apply the Four Steps*

Step 1: Problem: "The maintenance report you saved yesterday cannot be found."

Step 2: Write one action for UNDERSTAND, FIND, TRY, and CHECK.

Step 3: Circle any action that would need teacher or IT permission.

4.5 Reporting a Digital Problem

When a problem cannot be solved safely, the next professional action is to report it clearly. A useful report allows the teacher, IT support person or supervisor to reproduce the situation and avoid repeating actions that have already failed.

State what you were trying to do, what you observed, the exact message, where the problem occurred and what safe actions you tried. Avoid vague statements such as "the computer is broken" because they do not provide evidence.

Use the reporting structure below when asking for help. Each part gives the support person information needed to choose the next action efficiently.

The table below organises information under the headings Report part, Useful wording and Weak wording to avoid. Read across each row so that the details remain connected rather than treating individual cells as isolated facts.

Report part	Useful wording	Weak wording to avoid
Task and location	I was opening the Press 3 safety report in 01_Reports\Safety.	It happened somewhere in my files.
Observation	The file appears in Details view, but Word shows 'file cannot be opened'.	It does not work.
Actions tried	I checked the extension and tried opening the backup copy.	I clicked many things.
Current result and request	The backup opens; please advise whether the main copy should be replaced.	Fix it for me.

Habit	Why it matters	Example
Save regularly in the correct location	Prevents lost work and duplicate files.	Save the report in 01_Reports, not on the Desktop.
Use clear names and versions	Helps the team find the current document.	Use v01, v02; avoid "final FINAL".
Back up important work	Protects against loss or damage.	Keep an authorised second copy and verify it opens.
Report errors honestly	Supports safety and learning.	State what happened and what you tried.

BEFORE YOU USE OR SHARE A FILE - 5 QUICK CHECKS

- Correct asset or machine?
- Correct date and version?
- No confidential or personal information shared incorrectly?

Pop-Up Quiz 3

1. Why should a technician record observations instead of guesses?

2. Which action is safest when a live-machine dashboard shows an unfamiliar alarm?
   a. Disable it
   b. Change settings
   c. Record and escalate according to procedure
   d. Ignore it

3. Name two professional digital habits you will use in the lab.

1 SESSION

Recognising Digital Work & Solving a Digital Problem in the Workshop

Practical focus: identify digital tools, apply the UNDERSTAND–FIND–TRY–CHECK method, and act within your role

Lab Scenario

Lesson 1 taught you what digital literacy is, the five-part FIND → EVALUATE → USE → CREATE → SHARE cycle, the four industrial revolutions and Industry 4.0, and the UNDERSTAND–FIND–TRY–CHECK problem-solving method. This first lab lets you practice those ideas directly – before you touch file management, Word or the internet in later lessons. You will look at a realistic workshop morning, decide which tasks need digital literacy, and then work a real digital problem through the four-step method, staying safely inside a trainee's authority.

The point of this lab is not to fix a machine. It is to show that a technician's value lies in reading information accurately, choosing the right digital action, and reporting to the right person – the professional behaviours from Lesson 1's 'Two Technicians, One Fault' story. Work slowly, talk through your reasoning with your partner, and write down each decision.

Lab Scenario – Your First Morning

- It is 07:30 on your first day as a trainee technician. Your supervisor gives you five tasks: open the digital work order for Conveyor 12, read yesterday's maintenance note, find the manufacturer's motor datasheet, save two inspection photographs in the correct folder, and send a short update before the morning meeting. At 07:40, a dashboard warning appears: 'Motor vibration above normal range.' You are not expected to repair the motor. You are expected to read the information accurately, follow the approved process, record what you observe, and report it to the right person

Lab Success Criteria

Success is measured by the quality of your reasoning and the safety of your choices, not by speed. Read these criteria before you begin, and pause at each checkpoint to check your work against them.

- You can sort the five tasks into 'needs technical knowledge' and 'needs digital literacy', and justify each choice.
- You can name the digital tool used for each task and the professional standard it must meet (accuracy, safety, security, efficiency, communication).
- You can apply UNDERSTAND–FIND–TRY–CHECK to the vibration warning and stop at the correct role boundary.
- You record every decision in clear, factual language a supervisor could read.

Part A: Sort the Morning – Technical Knowledge vs Digital Literacy

Before doing any task, a professional decides what kind of skill each one needs. This part builds that habit. Work through the five tasks with your partner and classify each one, because most workshop work is a blend of the two – and the digital half is the part many technicians do badly.

The table below lists the five tasks. For each, discuss which digital-literacy step of the FIND → EVALUATE → USE → CREATE → SHARE cycle it mainly uses, then write your answer in the last column.

Task	Mainly technical or digital?	Cycle step (Find/Evaluate/Use/Create/ Share) (you decide)
Open the Conveyor 12 work order		
Read yesterday's maintenance note		
Find the motor datasheet		
Save two inspection photos correctly		
Send a short update before the meeting		

- **Step 1:** For each task, agree with your partner whether it needs mainly technical knowledge, mainly digital literacy, or both.
- **Step 2:** Name the cycle step it uses (finding the datasheet is FIND; saving the photos is CREATE/USE; sending the update is SHARE).
- **Step 3:** Write one sentence explaining why the digital half matters – for example, 'the wrong datasheet model would give the wrong reset procedure.

Part B: Map the Tools and Their Professional Standards

Every digital task uses a tool, and every tool must be used to a professional standard. This part connects the two, reinforcing Lesson 1's Professional Standard box (accuracy, safety, security, efficiency, communication).

The table below pairs each task with its likely digital tool. Discuss which professional standard matters most for each, and write it in the final column – there may be more than one, so choose the most important.

Task	Digital tool	Most important standard (you decide)
Open the work order	Maintenance system / browser	
Read the maintenance note	Shared records system	
Find the datasheet	Search engine / manufacturer site	
Save the photos	File Explorer / folders	
Send the update	Email / workplace messaging	

Part C: Work the Problem with UNDERSTAND–FIND–TRY–CHECK

Now apply Lesson 1's four-step method to the dashboard warning. This is the heart of the lab: a calm, safe, documented response to a real signal, staying inside a trainee's authority. Do not invent a repair – the correct professional action for a trainee is to read, record and escalate.

The table below gives the four steps. Fill in what you would do at each one for the 'Motor vibration above normal range' warning, keeping every action inside your role.

Step	Guiding question	Your action for the vibration warning (you decide)
UNDERSTAND	What is the signal telling me, exactly?	
FIND	Where is the correct information / who is responsible?	
TRY	What am I ALLOWED to do at my level?	
CHECK	Did I record it and report to the right person?	

Role-Boundary Rule

- A trainee's 'TRY' step is almost never 'reset the machine'. It is: read the warning carefully, check the work order and maintenance note for context, record the exact message and time, and escalate to the responsible technician or supervisor. Acting beyond your authority – even with good intentions – is a safety failure, not an initiative.

Part D: Write the Professional Record

The final skill is documentation – the SHARE step of the cycle. A good record is factual, timed, and free of guesses. This part practices writing one that a supervisor could act on immediately.

- **Step 1:** Write a three-to-four line record of the morning: what you were asked to do, what the dashboard showed (exact message and time), what you did, and who you told.
- **Step 2:** Use factual language only – no opinions or guesses about the cause ('the bearing is broken' is a guess; 'vibration warning at 07:40, reported to Technician Ali' is a fact).
- **Step 3:** Swap records with another pair. Could they act on yours without asking you a single question? If not, fix what is unclear.

Write your record here:

Lab Self-Assessment Checklist

Lab Rubric – How Your Practical Is Marked (10 marks)

Your teacher marks this lab using the rubric below. Read it before you finish so you know exactly what evidence is being looked for. The marks reward clear reasoning and safe, professional choices – not speed.

Criterion	What secure performance looks like	Marks
1. Identifies digital work	Correctly identifies the digital component of workshop tasks and explains why it matters.	2
2. Uses professional standards	Refers to accuracy, evaluation, traceability, security, or communication when justifying decisions.	2
3. Applies UNDERSTAND–FIND–TRY–CHECK	Uses the four stages in logical order and connects each to the scenario.	2
4. Maintains authority boundary	Does not propose an unauthorised reset, repair, bypass, or settings change.	2
5. Records and communicates accurately	Produces a factual record containing asset, time, observation, action and person informed.	2

Marking bands: 810 is secure performance; 57 is developing, with one or two criteria to correct; 04 leads to a supported repetition in Session 8.

Lesson/Session Summary

In this lesson, you learned that digital literacy is more than knowing how to operate a computer or digital device. A professional technician must use digital tools **accurately, safely, responsibly, and effectively**.

You explored the five-part digital literacy cycle used in this course:

FIND → EVALUATE → USE → CREATE → SHARE

You also learned that digital work is an important part of modern mechatronics. Technicians may use computers, tablets, QR codes, maintenance systems, dashboards, shared files and communication tools to find information, monitor equipment, record work and communicate with others.

You reviewed the four industrial revolutions and saw how industrial technology has developed from mechanical power to electricity, computer-based automation and connected digital systems. **Industry 4.0** connects machines, people, processes and data to support better production, maintenance, quality and safety.

Finally, you learned that digital problems should not be solved by random clicking or guessing. Use the four-step method:

UNDERSTAND → FIND → TRY → CHECK

If a problem is unsafe or outside your authority, stop and report it to the appropriate person.

Class Discussion

- "What is the difference between using a device and being digitally literate?"
- "What is the difference between Industry 3.0 and Industry 4.0?"
- "What should a trainee do when a digital warning requires an action outside their authority?"

Assessment

The assessment checks knowledge, interpretation, and practical judgement. Read each question carefully and use the terminology from the lesson. Where a question asks for an example or safe action, make the answer specific enough that another person could follow your reasoning.

Part A: Multiple Choice

Choose the best answer, not simply an answer that sounds related to the topic. Eliminate options that are unsafe, incomplete, or inconsistent with the definitions. Each item is worth one mark.

1. Which action is part of EVALUATE?

a. Opening any search result
b. Checking the manufacturer, model, and revision
c. Creating a folder
d. Sending a file.

2. Industry 4.0 is best described as

a. steam machinery
b. electricity only
c. computer automation without data exchange
d. intelligently connected production using data

3. Which statement is an observation?

a. The motor will fail
b. The bearing is damaged
c. The dashboard shows 82 degrees Celsius
d. The sensor is wrong

4. Which step confirms whether a solution worked?

a. UNDERSTAND **b.** FIND **c.** TRY **d.** CHECK

Part B: True or False

NOTE: A sentence may contain familiar words but still be false because one important condition is missing.

1. A person is digitally literate simply because they own a modern phone.
2. Industry 4.0 can include sensors, networks, data analytics, and AI.
3. A trainee should act only within their authorised role when responding to a digital warning.
4. Recording an exact warning, time, and asset can help another technician understand the situation.

Part C: Short Answer

1. Give one benefit and one challenge of Industry 4.0

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.....

2. A dashboard shows a warning for Conveyor 12. The trainee is not authorised to reset the

machine. Explain what the trainee should do using UNDERSTAND–FIND–TRY–CHECK

.....

.....

Part D: Digital Literacy Cycle Mapping

Match each action to the most appropriate stage of:

FIND – EVALUATE – USE – CREATE – SHARE

Action	Expected stage
Locate the correct motor datasheet	
Check manufacturer, model and revision	
Apply approved information to the task	
Record a factual workshop observation	
Send the verified update to the responsible technician	
Communicate a warning through the approved reporting route	

Self-Assessment

When I finish this lesson, I can:	Yes	Not yet
explain digital literacy in simple words.	☐	☐
distinguish basic computer use from professional digital practice.	☐	☐
identify the main idea of Industry 4.0.	☐	☐
identify common digital tools used by technicians.	☐	☐
use UNDERSTAND–FIND–TRY–CHECK to approach a simple digital problem safely.	☐	☐

REFLECT AND CONNECT

As a technician, write some examples of tasks that contain both technical and digital work.

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Glossary قاموس المصطلحات

The dictionary supports independent reading by giving a simple English meaning and an Arabic equivalent. Use it when a technical word is unfamiliar, then return to the sentence where the word appeared. The glossary is a support tool; the goal is to understand the term in context.

The bilingual dictionary table supports comprehension of technical vocabulary. Read all three cells in a row so that the English term, simple explanation and Arabic equivalent remain connected.

English term	Meaning	المصطلح بالعربية
Digital literacy	Using digital technologies and information confidently, critically, safely and responsibly.	الثقافة / الكفاءة الرقمية
Digital Competence	Knowledge, skills and attitudes needed to use digital technology effectively.	الكفاءة الرقمية
Industry 4.0	Intelligently connected industrial production using data and digital technologies.	الثورة الصناعية الرابعة
Smart Factory	factory where connected systems exchange and use production data.	المصنع
Data	Raw facts or measurements.	بيانات
Information	Data with meaning and context.	معلومات
Digital evidence	A digital record supporting what happened or was observed.	دليل رقمى
Sensor	A device that measures a physical condition.	حساس / مستشعر
HMI	A screen/interface through which a person views or interacts with a machine.	واجهة الانسان
PLC	An industrial computer that controls machines using a programme.	متحكم منطقى قابل للبرمجة
IoT / IIoT	Connected devices that exchange data; IIoT refers to industrial use.	إنترنت الأشياء / إنترنت الأشياء الصناعى
Digital twin	A digital representation of a real asset or process.	التوأم الرقمى
File	One saved digital item.	ملف
Subfolder	A folder inside another folder.	مجلد فرعى
Path	The digital address of a file or folder.	مسار الملف
Extension	Letters after the final dot that identify a file type.	امتداد الملف
Naming convention	An agreed rule for constructing file names.	قاعدة تسمية الملف
Version	A numbered revision of a document.	إصدار / نسخة
Backup	A separate copy used to recover lost or damaged work.	نسخة احتياطية
Restore	Recovering data from a backup.	استعادة

English term	Meaning	المصطلح بالعربية
Downtime	Time when equipment is unavailable for production.	المعدات توقف وقت
Troubleshooting	A controlled process for identifying and solving a problem.	استكشاف الأخطاء وإصلاحها
Cybersecurity	Protecting systems, networks and data from unauthorised access or attack.	الأمن السيبراني

Abbreviations

Abbreviations are common in technical documents because they save space, but they must be understood accurately. Read the full form the first time an abbreviation appears and avoid guessing from the letters alone.

The abbreviation table gives the complete wording behind shortened technical terms. Use the full form when first introducing an abbreviation in discussion or writing.

Abbreviation	Full form
AI	Artificial Intelligence
CMMS	Computerized Maintenance Management System
DT	Digital Transformation
HMI	Human-Machine Interface
IIoT	Industrial Internet of Things
IoT	Internet of Things
PLC	Programmable Logic Controller
USB	Universal Serial Bus

LESSON 2 Computers, Operating Systems & File Management

Learning Outcomes

Computer components and an industrial panel PC in a mechatronics workplace.

By the end of this lesson, you will be able to:

- identify the main internal and external hardware components of a computer and explain the purpose of each.
- explain the Input-Process-Store-Output model and connect it to a PLC-based mechatronics system.
- distinguish hardware, software, firmware, drivers, operating systems and applications.
- describe how an operating system manages programmes, memory, devices, files, users and security.
- interpret common file extensions and storage-size units, and choose an appropriate save location.
- apply the UNDERSTAND-FIND-TRY-CHECK method to basic computer problems within safe authority limits.
- create a controlled storage structure, give files meaningful names, retrieve evidence, and make a verified backup.

WARM-UP

Human Computer Role-play

In groups of 4, you will be volunteering to be a computer in front of the class. Each student gets a card: Eyes/Keyboard (Input), Brain (CPU), Notebook (Memory/Storage), Mouth/Screen (Output).

Step 1: The teacher asks a simple question (example: $12 \times 4 = ?$).

Step 2: 'Input' hears it and passes it to 'Brain'. 'Brain' calculates. 'Notebook' writes it down. 'Output' says the answer aloud.

Step 3: Class discussion: Which part of a real computer does each student represent? What happens if one part is missing?

Introduction: Your Computer Is Part of the Workshop

In this lesson, you transfer Lesson 1 concepts into a controlled Word task. You will look inside the computer, study the software that manages it, learn how files are stored, and then create the structured folder system, naming, retrieval, and verified backup. The aim is not to become a computer engineer. The aim is to use workplace computers accurately, safely and confidently.

Scenario: the interrupted report

- A trainee has typed one page of findings but has not saved the file. The power suddenly goes off. Which computer component was holding the unsaved work? Which part could have protected it? Keep these questions in mind during Topic 1.

Topic 1: Computer Hardware - The Parts You Can Touch

This topic introduces the physical parts that make a computer system work. The aim is not to memorize a long component list, but to connect each component to a job, recognize how the parts depend on one another, and apply the same system model to industrial computers and PLC-based equipment.

1.1 Workplace Scenario: "Why Is This Panel PC Slow?"



The scenario begins with an everyday statement that can be technically unclear: "the computer needs more memory." By examining the situation, you learn to separate RAM, storage, processor workload, and software demand instead of using one general word for different causes.

A technician opens a large digital drawing on an industrial panel PC while several monitoring programme are already running. The screen reacts slowly. The technician says, "The computer needs more memory." Does "memory" mean RAM or storage? Could the processor also be busy? A useful diagnosis begins by naming the components correctly.

1.2 Key Definitions

The definitions establish a shared technical vocabulary. Read the terms together because a computer system contains many components, hardware includes all physical parts, and peripherals are connected devices that extend input, output, or storage.

The table below separates three closely related words. Read across each row from the term to its definition and then to the example so that you can use the vocabulary accurately in discussion and troubleshooting.

Term	Definition	Example
Computer system	A group of hardware and software components that work together to receive, process, store, and communicate information.	A workshop PC, an industrial panel PC or a PLC-based control system.
Hardware	The physical parts of a computer or digital device.	CPU, RAM, SSD, keyboard, screen and network adapter.
Peripheral	A device connected to a computer to provide input, output or storage.	Mouse, printer, scanner or external USB drive.
Component	One part of a larger system with a specific job.	A RAM module on the motherboard.

1.3 Main Internal Components

Internal components work as one system. Study the job of each part, the workshop analogy and the important limitation so that you can explain not only what the component is called, but also why a fault or limitation in one part can affect the whole computer.

The table compares the main internal components from four viewpoints: technical function, a familiar workshop analogy and an important limitation. Use the analogy to remember the job, but use the technical description when explaining the component.

Component	Main job	Workshop analogy	Important point
CPU / processor	Follows programme instructions and performs calculations.	The skilled worker carrying out instructions.	A faster CPU can complete more work, but performance also depends on RAM, storage and software.
RAM	Holds data and programmes that are being used now.	A workbench holding current tools and parts.	RAM is temporary: its contents normally disappear when power is removed.
SSD or HDD storage	Keeps programmes and files for long-term use.	A labelled tool cabinet or storeroom.	Storage keeps data when power is off. SSDs are generally faster and have no moving disk.
Motherboard	Connects components and provides pathways for data and power.	The workshop layout connecting all work areas.	Components must be compatible with the motherboard.

Component	Main job	Workshop analogy	Important point
Power supply unit (PSU)	Converts and distributes electrical power to internal components.	The electrical supply panel.	Students must not open a powered computer or PSU.
Cooling systemt	Removes heat from the processor and other components.	Ventilation and cooling around a machine.	Blocked vents and dust can cause overheating.
Network adapter	Connects the computer to a wired or wireless network.	The communication link to other departments.	Network access may be controlled by school or company policy.

Before studying the labels, look at the computer as one connected system. The arrows show that the CPU, RAM, storage, and power supply depend on the motherboard and on one another. Use the diagram to explain each job, then identify which parts keep information temporarily and permanently.

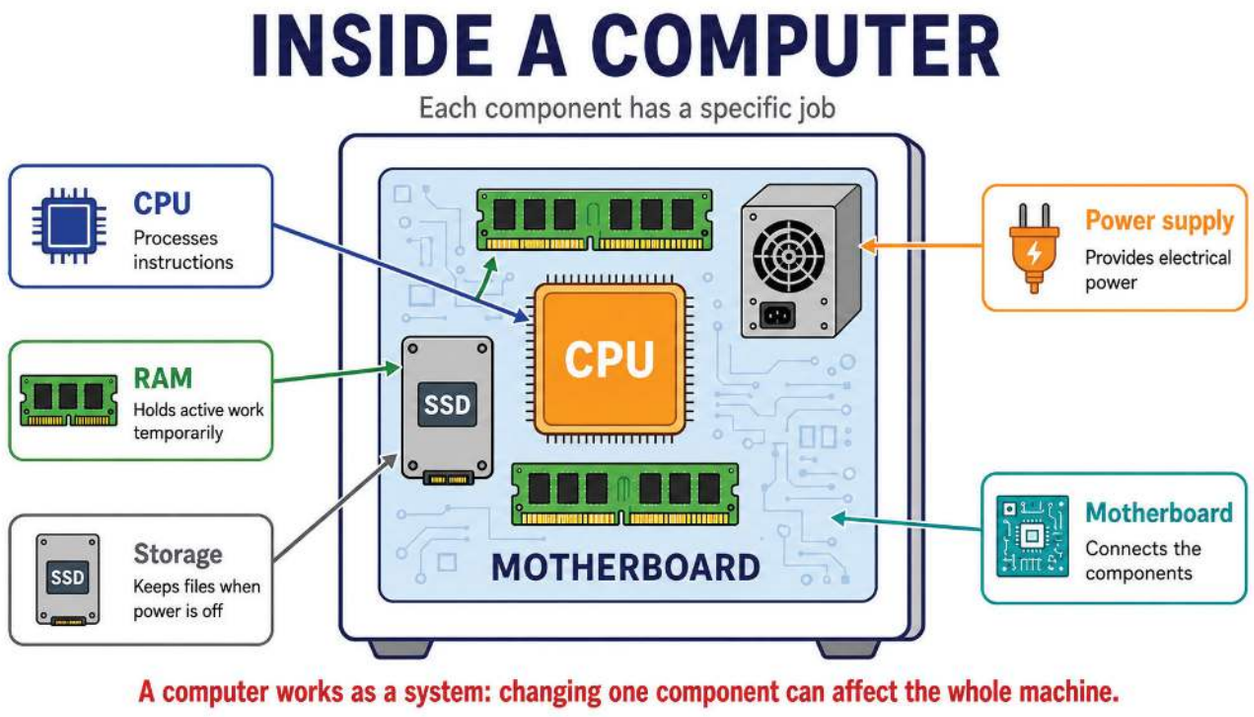


Figure 2.1: Simplified internal components of a desktop computer.

Safety boundary

- Identifying components is useful; opening a powered computer, power supply, industrial control panel or live machine is not a first-year troubleshooting action. Isolate equipment and follow authorised procedures.

1.4 External Hardware and Ports

External devices connect the computer to users, documents, networks, and machines. This subsection helps you classify devices by what information they send or receive and develops the safe habit of matching connectors rather than forcing plugs or using unauthorised interfaces.

The table classifies external hardware by the direction and purpose of information flow. Notice that some devices combine input and output, while ports provide the physical connection rather than performing the information task themselves.

Category	Purpose	Examples in a workshop
Input devices	Send information into the computer.	Keyboard, mouse, barcode scanner, camera, microphone, and machine sensor.
Output devices	Communicate a result or cause an action.	Monitor, printer, alarm speaker, indicator light, motor or valve.
Combined input/output	Receive and display information.	Touchscreen, network adapter and some multifunction printers.
Ports and connectors	Provide physical connection points.	USB, network/Ethernet, HDMI/ display and audio ports.

Professional habit: match the connector carefully. Never force a plug into a port. For industrial equipment, use only authorised cables and interfaces.

1.5 The Input-Process-Store-Output Model



This model explains the journey of information through a digital system. The four stages are useful because they can describe an ordinary report-writing task, a barcode scanner, an industrial panel PC or a sensor-controlled conveyor using the same logical sequence.

Every digital system receives information, processes it, may store it and then produces an output. Storage is included because technicians often need to keep records, settings or evidence after the immediate task ends.

Read this diagram from left to right. The upper sequence defines the four stages, while the lower examples apply the same sequence to typing and printing a report and to a PLC controlling a motor. Notice especially that saving transfers work from temporary RAM to persistent storage.

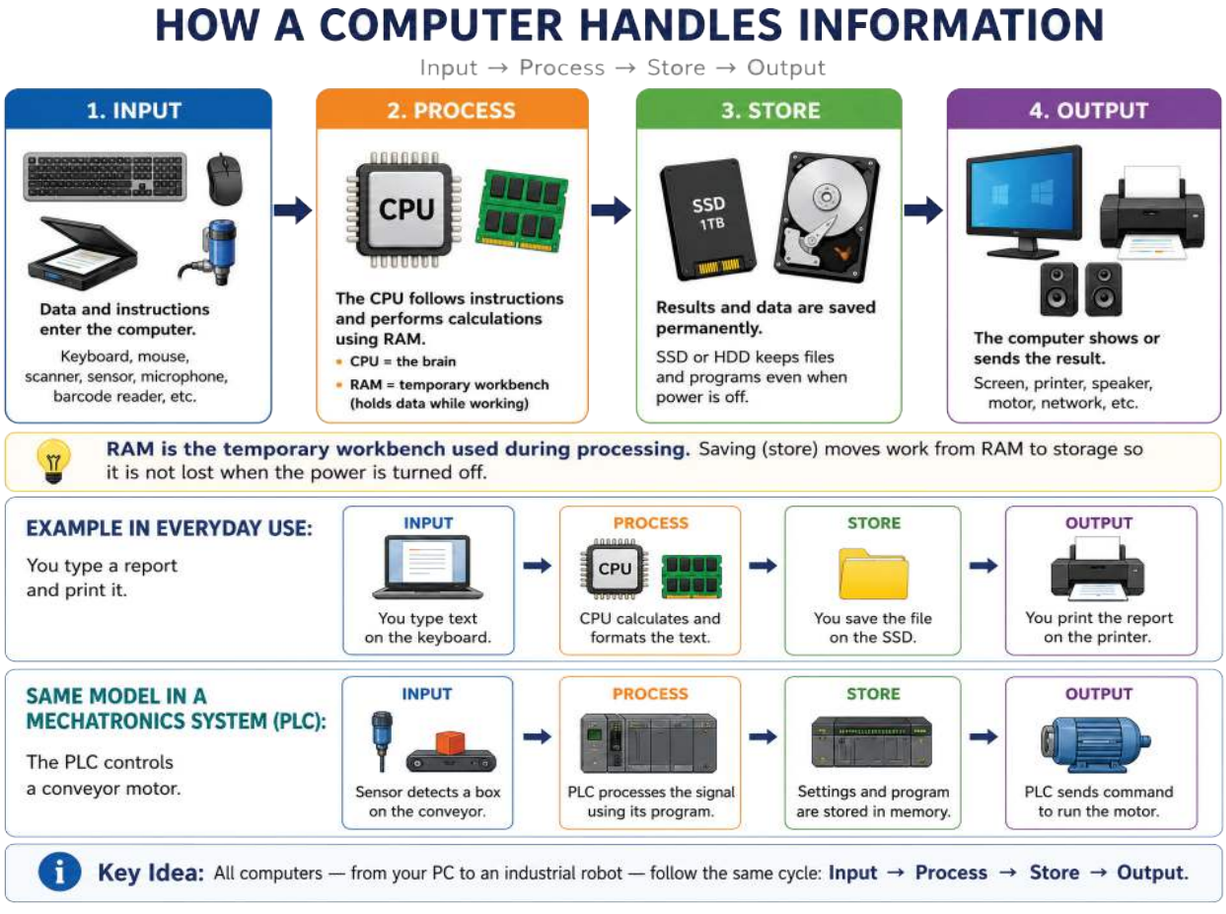


Figure 2.2: The Input-Process-Store-Output model.

Scenario: The Industrial Panel PC

In your future workshop, computers are not only on desks. **A panel PC** is mounted directly on a CNC machine: its touchscreen is the input AND output, its SSD stores machine programmes, and its CPU runs the control software. Even a **PLC** is a special industrial computer: sensors are its input devices, and motors/valves are its output devices! Understanding computer parts = understanding the machines you will maintain.

1.6 Worked Example: PLC Controls a Conveyor

The worked example transfers the model from a desktop computer to a mechatronics system. Follow the signal from the sensor to the PLC, then to stored records and finally to the motor so that you can see how information and physical action are linked.

1. **INPUT:** a photoelectric sensor detects a box on the conveyor.
2. **PROCESS:** the PLC reads the sensor signal and follows its control programmes.
3. **STORE:** the system may record a count, time stamp, or alarm in memory or a database.
4. **OUTPUT:** the PLC starts, stops, or changes the speed of a motor.

A PLC is not the same as a general-purpose desktop computer, but both use the same information model. The PLC is designed for reliable real-time industrial control, while a PC is designed for many different user applications.

1.7 Understanding Basic Specifications



Specifications summarize a computer's capabilities, but one number never tells the whole story. This subsection teaches you to interpret processor, RAM, storage and operating-system information at a beginner level and to relate each specification to the type of work being performed.

The table turns common specification labels into practical meaning. It is designed for interpretation, not purchasing decisions: use it to explain what a specification may influence and why several specifications must be considered together.

Specification	What it tells you	Beginner interpretation
Processor / CPU	The type of processor and its capacity to execute instructions.	Useful for performance, but do not judge a computer by one number alone.
RAM capacity	How much active data and how many programmes can be held comfortably.	More RAM usually helps when several applications or large files are open.
Storage capacity	How much long-term data can be kept.	A 512 GB drive holds far more than a 16 GB USB drive.
SSD or HDD	The storage technology.	An SSD is usually faster and more resistant to vibration because it has no spinning disk.
Operating-system type	The software platform the computer runs.	Applications must be compatible with the OS.

Return to the opening scenario

- The unsaved report was mainly held in RAM. Saving would have written it to storage. This is why Ctrl + S is a professional safety habit, not only a keyboard shortcut.

1.8 System Performance: Look for the Bottleneck

A computer is a system, so slow performance is not automatically caused by one component. A bottleneck is the part of a process that limits the overall speed. For example, a fast processor may still wait if storage is slow, RAM is full, or a large application is performing several tasks at once.

Use evidence before making a claim. Task Manager, file size, the number of open programmes, and the exact action that feels slow can help an authorised support person decide whether the problem is processor workload, insufficient RAM, storage delay, network delay, or software behaviour.

Workshop discussion example: a panel PC opens a small checklist quickly but takes two minutes to open a large 3D drawing. This evidence suggests that file size, application demand, RAM or storage performance should be investigated before anyone concludes that the entire computer is defective.

About this table: The table links an observable situation to several possible system factors. It is not a repair guide; it shows why a technician should collect evidence instead of naming a cause too early.

Observed situation	Possible factor to investigate	Useful evidence
Many programmes are open and switching is slow	RAM pressure or high processor workload	Number of open programmes; Task Manager information if authorised.
A large file opens slowly but small files open normally	File size, storage speed or application demand	File size; storage location; time taken to open.
Only network files are slow	Network connection or shared-system demand	Local file comparison; connection status; time and location
The computer becomes hot and the fan runs continuously	Cooling, dust, blocked vents or high workload	Vent condition; fan noise; temperature warning; workload

Pop-Up Quiz 1

1. Which component normally loses its active contents when power is removed?

a. SSD **b.** RAM **c.** USB drive **d.** HDD

2. In a PLC system, a temperature sensor is mainly an:

a. input **b.** output **c.** storage device **d.** application

3. Which component connects the CPU, RAM, storage, and other internal parts?

a. monitor **b.** motherboard **c.** keyboard **d.** printer

Write your answers: .....

Topic 2: Software and the Operating System - The Invisible Manager

Hardware cannot perform useful work without instructions. This topic explains the software layers that sit between the user and the physical components, with particular attention to the operating system as the manager of programmes, memory, devices, files, accounts and security.

2.1 Scenario: What Happens During Start-Up?

Start-up looks simple to the user, but several coordinated actions occur in the background. The scenario introduces booting and helps you understand why the computer must load system software, prepare devices, and establish a user session before applications can be used.

You press the power button. The manufacturer logo appears, a loading symbol turns, and then the desktop opens. During this process, the computer checks basic hardware, loads the operating system from storage into RAM, and prepares devices and user services. This process is called booting or start-up.

2.2 Definitions: Software, Firmware, Driver and Operating System

These terms describe different kinds of instructions. Comparing them prevents a common mistake: treating every programme as the same type of software. Focus on where the instructions operate and what relationship they have with the hardware and operating system.

The table below distinguishes software categories that are often confused. Focus on the relationship between each type of software, the operating system, and the physical device.

Term	Definition	Example
Software	Instructions and programmes that tell hardware what to do.	Word, a browser, a PLC programming package.
Firmware	Low-level software stored in a device that helps it start and control basic functions.	Firmware inside a printer, router, sensor or controller.
Device driver	Software that helps the operating system communicate with a particular hardware device.	Printer driver or graphics driver.
Operating system (OS)	The main system software that manages hardware, applications, files, users and security.	Windows, Linux, Android or an embedded industrial OS.
Application	Software designed for a specific user task.	Word for reports, Excel for logs, CAD for drawings.

The diagram should be read vertically. Applications sit above the operating system because they request services from it; the operating system sits above the hardware because it manages physical resources. Drivers form a specialized communication link between the operating system and particular devices.

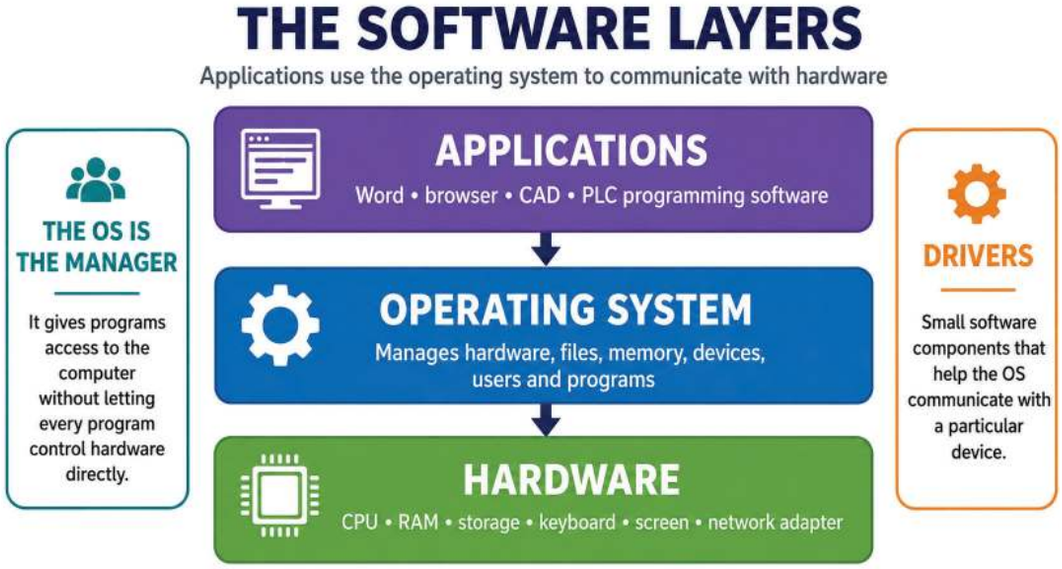


Figure 2.3: Hardware, operating system and application layers.

2.3 Six Core Jobs of an Operating System

The operating system performs several jobs at the same time. The table translates these jobs into actions you already recognize, such as opening Word, sharing RAM, finding files, printing and protecting accounts.

The operating-system table links an abstract management function to an action students can observe. Read the middle column first, then use the example to explain how the function appears during normal computer use.

OS job	What the OS manages	Example
User interface	Windows, menus, icons, pointer and keyboard interaction.	You click the Word icon and move between open windows.
Process management	Which programmes use the processor and when.	Word and the browser can run at the same time.
Memory management	How RAM is shared and protected.	The OS allocates memory to each running application.
File management	Creating, naming, copying, moving, searching and deleting files.	File Explorer shows folders and paths.
Device management	Communication with hardware through drivers.	The OS sends the document to the correct printer.
Accounts and security	User identity, permissions, passwords and updates.	A trainee cannot install restricted software without permission.

2.4 The Desktop and Application Windows

The desktop and application window are working environments rather than separate programmes. Learning the common parts of an interface helps you transfer your skills between different versions of Windows, Word and other workplace software.

The desktop is the main workspace shown after sign-in. A typical operating system provides a taskbar or dock, application icons, a notification area and a file-management tool. An application window usually includes a title bar, window controls, menus or ribbon tabs, the working area and scroll bars.

Interface differences

- The exact location of commands can differ between Windows versions, school computers and Microsoft Word editions. Focus on the purpose of a command, not only its colour or exact position.

2.5 User Accounts, Permissions and Professional Use

A workplace account identifies who performed an action and what that person is authorised to access. The rules in this subsection protect confidentiality, accountability and the stability of shared computers and industrial systems.

- Use your own authorised account and keep your password private.
- Do not install software, change security settings or connect unknown devices without permission.
- Lock or sign out when leaving a shared computer.
- Store work in the approved location, not in another student's folder.
- Restart only when it is safe to interrupt the current task and no live process depends on the computer.

2.6 Restart, Shut Down and Sleep

These commands affect programmes and user sessions in different ways. Choosing correctly depends on whether work is saved, whether another person will use the computer, and whether any critical process depends on the system.

The table compares commands that can look similar but have different effects on open programmes, power, and user access. Choose the action from the situation rather than using the same command automatically.

Action	What it does	Use it when
Restart	Closes programmes, reloads the OS and starts again.	A safe troubleshooting step after saving work.
Shut down	Closes the OS and turns the computer off.	The computer will not be used and procedures permit shutdown.
Sleep	Keeps the session in a low-power state for quick return.	A short break on an approved personal or school computer.
Sign out / lock	Protects the account while leaving the computer running.	You leave a shared workstation or another user needs to sign in.

2.7 Compatibility, Updates and Safe User Decisions

Software and hardware must be compatible. An application may require a particular operating-system version, a printer may need the correct driver and an industrial tool may be approved only for a specific controller or interface. Compatibility means that the parts are designed and authorised to work together.

Updates can correct faults and improve security, but they can also interrupt work or change behaviour. On a school or company computer, trainees should not install updates, drivers or software unless the approved process gives permission. The professional response is to record the message, save work and report the requirement to the responsible person.

Example: Word displays a message that an add-in is disabled. The trainee should not download a random replacement from the internet. They should continue with approved functions if possible and report the exact message to the teacher or IT support.

Pop-Up Quiz 2

1. Which software manages the computer as a whole?

a. operating system **b.** Word **c.** photo **d.** keyboard

2. Which item is a device driver?

a. a physical cable **b.** software that helps the OS communicate with a printer

c. a folder **d.** a report

3. A trainee leaves a shared computer for lunch. The safest first action is to:

a. leave it open **b.** lock or sign out **c.** delete files **d.** remove the RAM

Write your answers: .....

DID YOU KNOW?

Many industrial machines run on **Linux or special embedded operating systems,** not Windows! Robots, smart sensors and even car computers have operating systems inside. When you learn how an OS works, you understand a little of EVERY smart machine.

Reference: Linux Embedded Systems Market Size

Topic 3: Files and Storage - How Digital Work Is Kept

This topic follows information after it has been created. You will connect names, extensions, sizes, storage locations and versions so that a file can be opened by the correct programme, found by another person and protected from accidental loss.

3.1 From Data to a File

A file gives data a name, format and location. This subsection revisits file, folder and path because every later decision—saving, searching, sharing, revising and backing up—depends on knowing exactly what the file is and where it is stored.

Data becomes useful when it is organised and saved in a recognizable form. A file is a named collection of data. A folder groups related files. A path is the digital address showing the drive and sequence of folders that lead to a file. These ideas were introduced in Lesson 1; this lesson adds file type, size, save methods, and storage location.

3.2 File Extensions

An extension identifies a file format and helps the operating system select a suitable application. It is part of the file name, but it is not a conversion tool, so changing the letters alone does not change the content inside the file.

A file extension is the group of letters after the final dot in a file name. It helps the operating system decide which application can open the file. Changing the letters in the name does not truly convert the file; correct conversion must be performed by an application.

The extension table connects file names to formats, workplace purposes, and suitable applications. Read across a row to see why recognizing the extension helps you select the correct tool and avoid opening the wrong content.

Extension	File type	Typical workshop use	Usually opened with
.docx	Word-processing document	Inspection report or work instruction.	Microsoft Word or compatible word processor.
.xlsx	Spreadsheet	Downtime log, parts list or calculation table.	Microsoft Excel or compatible spreadsheet.
.pdf	Fixed-layout document	Manufacturer manual or approved form.	PDF reader or browser.
.jpg / .png	Image	Fault photograph, component image or diagram.	Image viewer or editor.

Extension	File type	Typical workshop use	Usually opened with
.csv	Plain table data	Export from a machine, sensor or database.	Spreadsheet or data-analysis software.
.zip	Compressed package	Several files collected into one smaller package.	File Explorer or archive software.

Security note

A dangerous programme can be given a misleading name. Do not open unknown attachments or files only because the visible name looks familiar. Follow the organisation's security rules.

3.3 File Size: KB, MB, GB and TB

File-size units help you judge whether information will fit on a drive, upload successfully or move efficiently across a network. The aim is comparative judgement: recognize the difference between a short report, a photo, a video and a large storage device.

Digital size is measured in bytes. For first-year work, use this practical scale: KB for small documents, MB for photos and audio, GB for videos and software, and TB for large drives and archives. Exact values can vary because computer systems may calculate units in slightly different ways.

The upper part of the figure breaks a professional file name into date, asset, document type, version and extension. The lower part compares common size units. Use both parts together: the name explains what the file is, while the size helps you decide how it can be stored or shared.

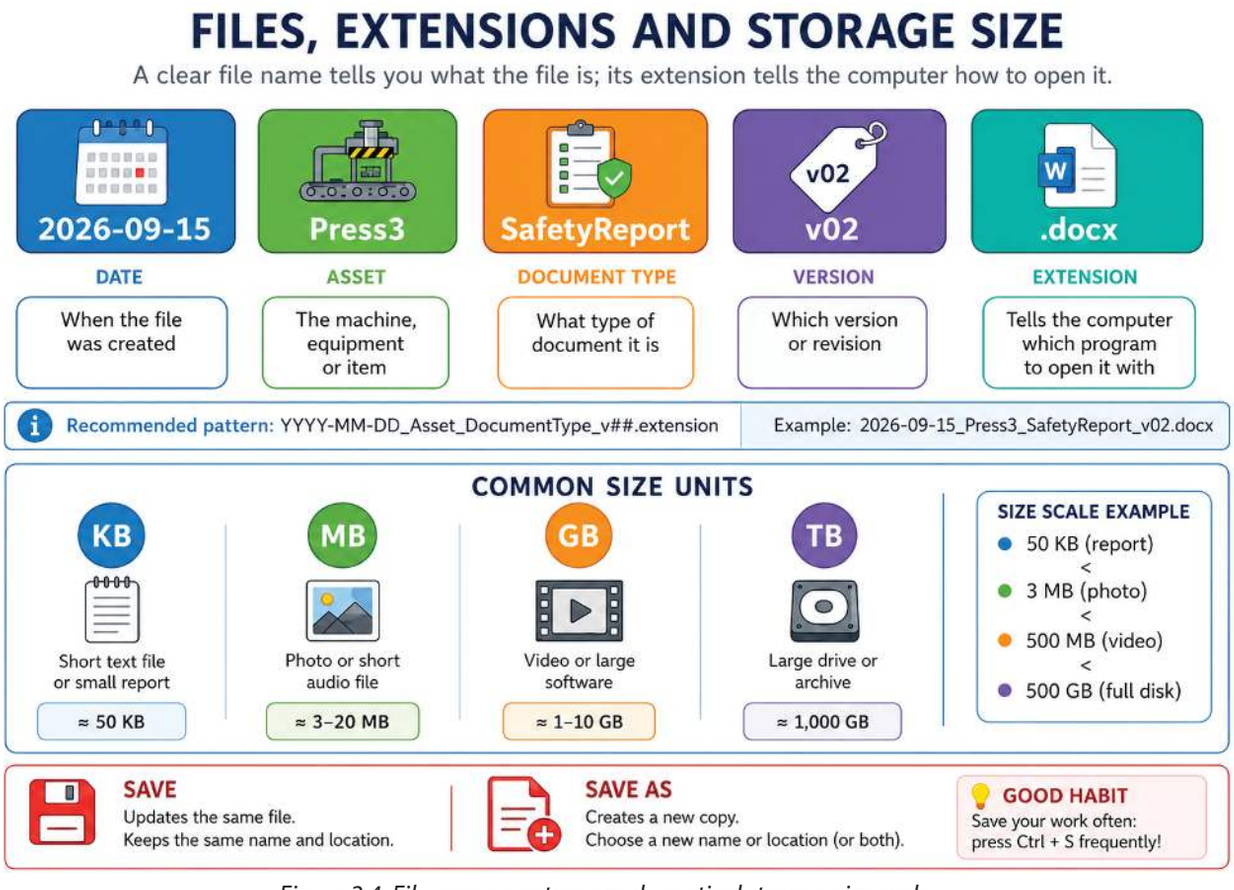


Figure 2.4: File-name anatomy and practical storage-size scale.

3.4 Save, Save As and AutoSave

Saving protects current work, while Save As creates a new name, location or format. Understanding the difference is essential for version control because professionals often need to preserve an original file while creating a revised copy.

The saving table compares actions that protect the current document, create a new copy or depend on an enabled service. The decision column is the most important because it shows when each command supports professional version control.

Command	What happens	When to use it
Save (Ctrl + S)	Updates the current file in its current location.	Use every few minutes and after each major change.
Save As	Creates a new file name, format or location.	Use for a new version, template copy or different folder.
AutoSave	Automatically saves changes when supported and enabled.	Useful with approved cloud storage, but still verify the file name and location.
Export / Save as PDF	Creates a fixed-layout copy for sharing or printing.	Use when the reader should not edit the report. Keep the editable .docx too.

3.5 Local, Removable, Network and Cloud Storage

A storage location is chosen for a reason. Compare speed, availability, sharing and security so that you can select an approved location rather than saving wherever the computer suggests by default.

The storage-location table shows that convenience is only one consideration. Compare access, sharing and risk so that the chosen location follows school or workplace policy.

Location	Strength	Risk or rule
Local drive	Fast and available on the computer.	Can be lost if the computer fails or the file is saved in the wrong user account.
Authorised USB drive	Portable and useful for approved backup or transfer.	Easy to lose; may carry malware; use only authorised drives and eject safely.
Network folder	Central shared location managed by the Organisation.	Needs correct permissions and network access.
School/ company cloud	Access from approved devices and may support version history.	Requires sign-in, correct sharing permissions and internet/service availability.

3.6 Version Control for Beginner Reports

Version control records meaningful changes and reduces confusion about which copy is current. A short, consistent version number is more reliable than informal names such as "final final" because it makes the revision history visible.

Version control means showing which revision of a file is current. Avoid names such as FINAL_ final_new2. Use a consistent version number, for example 2026-09-15_Press3_SafetyReport_v01. docx. Create v02 only when a meaningful revision is made. Keep approved copies according to the organisation's rules.

Scenario

Your supervisor corrects the recommended action but wants the original draft kept. Use Save As and change v01 to v02. Do not overwrite the original unless the supervisor asks you to.

3.7 The File Life Cycle: From Creation to Archive

A professional file has a life cycle. It is created for a purpose, saved in an approved location, revised when evidence changes, shared with authorised people and eventually archived or disposed of according to rules. Thinking about the whole life cycle prevents files from becoming lost, duplicated or misleading.

The same file may exist in more than one format for different purposes. The editable .docx version is used for revision, while a PDF may be used for fixed-layout distribution. Both copies need clear names, dates, and storage locations so another person understands their relationship.

Do not delete old versions simply because a new one exists unless the teacher or workplace procedure allows it. Earlier versions may show what changed, who reviewed the work, and when a decision was made.

About this table: The table shows the main stages in a beginner file life cycle and the professional question to ask at each stage. Use it as a planning guide for reports and other workshop records.

Stage	What happens	Professional question
Create	A new file is produced for a defined task.	What is the purpose and correct file name?
Save	The file is written to an approved storage location.	Can another authorised person find it?
Revise	Content changes and a new version may be created.	What changed, and should v01 be preserved?
Share	The file is sent or made available to an authorised reader.	Who needs access, and is the format suitable?
Archive or dispose	The record is retained or removed according to rules.	How long must it be kept, and who may delete it?

Pop-Up Quiz 3

1. Which extension usually identifies a spreadsheet?

a. .jpg **b.** .xlsx **c.** .pdf **d.** .mp4

2. Which command creates a new name or location while keeping the original file?

a. Save As **b.** Cut **c.** Delete **d.** Print

3. Which is normally largest?

a. 50 KB **b.** 3 MB **c.** 2 GB **d.** 500 bytes

Write your answers: **1.** ..... **2.** .....

3. .....

Topic 4: When the Computer Misbehaves - Safe First-Aid Troubleshooting

Troubleshooting is a controlled process for collecting evidence and selecting the safest next action. This topic limits first-year work to user-level checks and clearly separates an ordinary application problem from a live-machine or restricted-system problem that requires authorised escalation.

4.1 Scenario: The Report Freezes Before Submission

The scenario creates a realistic pressure point: an important document is not responding close to submission. It helps you practice patience, observation, and saving habits before discussing possible causes or attempting a fix.

You click a picture in the report, and Word stops responding. The pointer may show a waiting symbol. A professional technician does not immediately switch off the power. First, they observe the symptom, wait briefly, protect any other work, and apply a safe method.

4.2 Symptom, Cause and Evidence

A symptom is what you can observe, a cause is an explanation that still needs evidence, and evidence is information that supports or rejects a possible cause. Keeping these ideas separate prevents guesses from being reported as facts.

The troubleshooting vocabulary table separates what is observed from what is suspected and what is used as proof. This distinction is essential when reporting a digital problem to another person.

Term	Meaning	Example
Symptom	What you can directly observe.	"The Word window does not respond to clicks."
Possible cause	An explanation that still needs evidence.	The image may be very large, RAM may be busy, or the programme may be processing.

Term	Meaning	Example
Evidence	Information that helps confirm or reject a cause.	Task Manager shows high memory use; the problem began after inserting a 100 MB photo.
Fix	An action used to remove or reduce the cause.	Wait, close the frozen task safely, resize the photo, and reopen the document.

4.3 Apply UNDERSTAND-FIND-TRY-CHECK

The four-step method turns troubleshooting into a repeatable cycle. Each step produces information for the next one, and the safety boundary in the centre reminds you that no digital method gives permission to work beyond your role.

Safe digital troubleshooting means solving a computer problem in a careful, logical and controlled way. Begin by **understanding** the problem: observe the symptoms, read the complete message and note what happened before the problem appeared. Next, **find** reliable guidance by checking the manual, help page or asking an authorised person. Then **try** the safest reasonable action, changing only one thing at a time so that you can identify what affects the result. Finally, **check** whether the problem has been solved and record what you observed and tried. If the problem continues, return to the first step using the new information you collected. Always stop and ask for help when the problem involves a live machine, electrical danger, restricted systems, unfamiliar settings or an action beyond your authority. The complete cycle is shown in the figure below. Follow the arrows around the four-step cycle and read the centre warning before considering any action. The side examples show that troubleshooting is not random trial and error: it begins with evidence, changes one thing at a time and stops when safety, permission or live-machine control is involved.

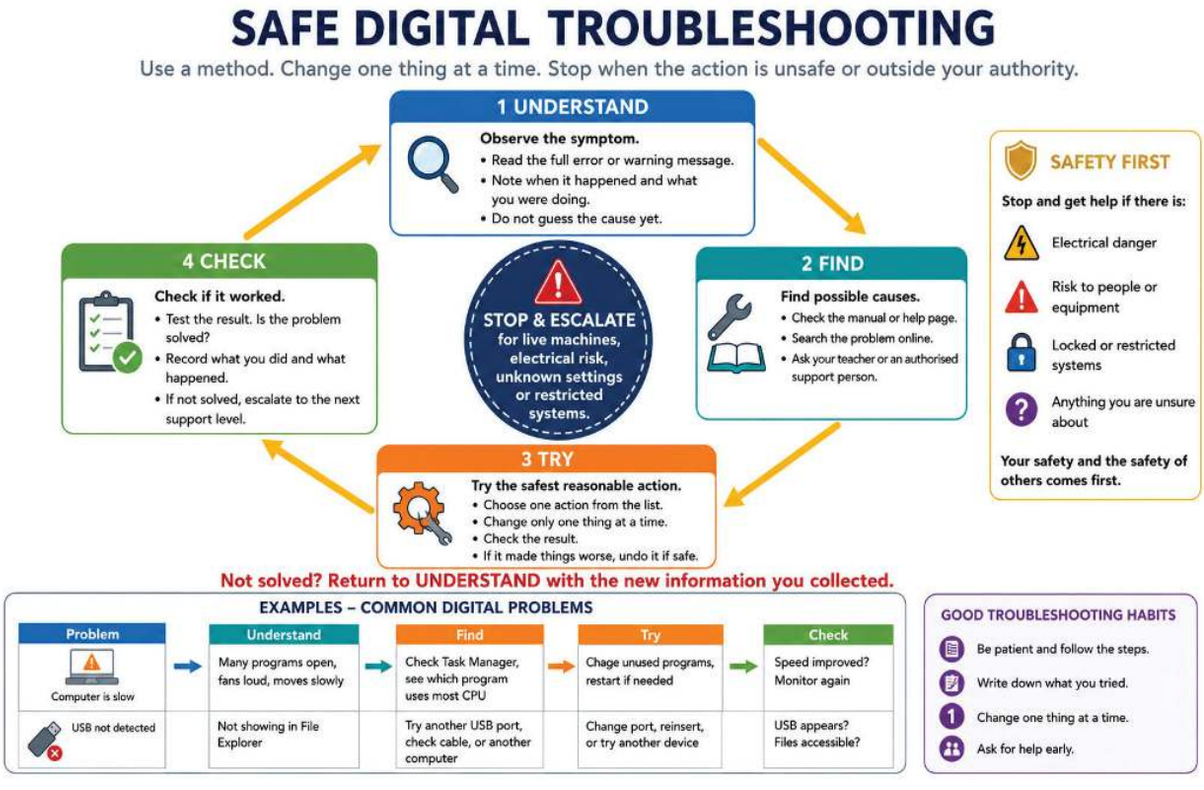


Figure 2.5: Safe digital troubleshooting loop and authority boundary.

4.4 Safe First Actions for Common Problems

Common computer problems often have simple first actions, but the same symptom can have more than one cause. Read the table as a decision guide: begin with the observed symptom, try only approved actions and stop when the situation crosses the stated boundary.

The troubleshooting table should be read from left to right as a safety decision. Start with the exact symptom, select only approved user-level actions, and stop immediately when the boundary in the last column applies.



Observed symptom	Safe first actions	Stop and escalate when
One programme is not responding	Wait 20-30 seconds; save other work; open Task Manager through an authorised method; end only the frozen application if instructed.	The application controls a live machine, critical process or protected system.
Computer is very slow	Close unused tabs and applications; check whether an update or scan is running; save and restart if permitted.	There are unusual pop-ups, overheating, repeated crashes or suspected malware.
No display	Check monitor power, brightness and external cable without opening equipment; confirm the correct display is selected.	There is a burning smell, damaged cable, liquid or electrical hazard.
No sound	Check volume, mute status and selected output device.	The issue affects an alarm or safety-warning system.
USB not recognized	Try an authorised alternative port; check File Explorer; test only on an approved computer.	The drive is unknown, physically damaged or contains suspicious files.
Cannot find a file	Read the current path; search by part of the professional name; check recent files and the correct user account.	The file may contain restricted information or was deleted from a shared system.
Printer does not print	Read the printer message; check paper, selected printer and queue; retry one test page.	There is a jam requiring access beyond the user instructions or a mechanical/electrical risk

4.5 Reporting a Problem Professionally

A clear problem report allows another person to continue the investigation without repeating your work. It should describe the exact event, context, actions, and result rather than using vague statements such as "the computer is broken."

A useful problem report helps another person act quickly. Include:

- WHAT happened: the exact symptom and error message.
- WHERE it happened: computer, application, file, or machine identification.
- WHEN it started and what you were doing at the time.
- WHAT YOU TRIED: safe actions in the order used.
- RESULT: what changed or did not change.
- IMPACT: whether work, safety, production or a deadline is affected.

Model IT report

"At 10:15, Word stopped responding on Training PC 06 while I inserted a 48 MB photo into Press 3 SafetyReport_v01. I waited 30 seconds and checked that other programmes responded. I ended only Word through Task Manager with the teacher's permission, reopened the file and recovered the last saved version. The report still needs the photo before submission."

4.6 Stop, Record and Escalate

Good troubleshooting includes knowing when not to continue. Stop when the problem involves electrical danger, a live machine, a safety interlock, confidential information, administrator permissions, restricted settings or an action you do not fully understand.

Stopping does not mean doing nothing. Record the exact symptom, preserve the file or screen if this is safe, note the actions already tried, and contact the authorised person. This creates a useful handover and prevents another person from repeating unsafe or unnecessary steps.

Example: an HMI connected to a live conveyor stops responding. A trainee should not end the process through Task Manager. The correct action is to follow the machine's approved communication and escalation procedure, because closing the application could affect control or monitoring.

Pop-Up Quiz 4

1. Which is a symptom rather than a cause?

a. The USB does not appear in File Explorer"

b. "The USB controller is damaged"

2. A programme controls a live machine and stops responding. The trainee should first:

a. end the task immediately

b. stop and follow the authorised escalation procedure

3. In the four-step method, testing whether the action worked belongs to:

a. FIND **b.** TRY **c.** CHECK **d.** UNDERSTAND

Write your answers:

1. ..... **2.** .....

3. .....

4.7 Practice: Build an Evidence-Based Problem Report

This short task turns the reporting checklist into a written record. Use only the information in the scenario, distinguish what you observed from what you suspect, and leave unsupported causes out of the report.

Scenario: On Training PC 04, the authorised USB drive appears in File Explorer, but the folder opens very slowly. This began after you copied a 1.8 GB training video. You waited, closed unused programmes, and tried the same approved USB on Training PC 05, where it opened normally.

Use this table as a writing organiser. Each row requests one type of evidence, helping you produce a complete problem report without adding an unsupported diagnosis.

Report element	Your evidence
WHAT happened	
WHERE	
WHEN / context	
WHAT you tried and the result	
Impact and next safe action	

Topic 5: Professional Digital Habits

Professional habits are small repeated actions that protect the quality of larger tasks. They reduce lost files, unclear versions, unauthorised access, and incomplete records. This topic prepares you for the laboratory by turning broad principles into routines you can apply every day.

5.1 Small Habits Prevent Large Problems

The table links each habit to its practical reason and an example. Some habits protect accountability, others protect confidentiality or make teamwork faster. Choose two habits that you do not yet use consistently and plan how you will apply them during the lab.

Habit	Why it matters	Example
Use your own authorised account	Ensures actions can be traced to the correct person.	Do not share passwords or use another student's login.
Lock the screen when leaving	Prevents unauthorised access.	Press Windows + L before leaving the workstation.
Check before clicking or downloading	Reduces malware and incorrect-source risk.	Download manuals from the manufacturer or approved portal.

5.2 A Five-Minute Digital Housekeeping Routine

Digital housekeeping is a short routine for keeping work organised before files become difficult to manage. At the end of a task or shift, a technician checks names, locations, versions, and backup status. Five minutes of checking can prevent a much longer search or an incorrect handover later.

Use the routine only for records you are authorised to manage. Do not delete uncertain files simply because they look old; first confirm the retention or version-control procedure.

The routine below can be used at the end of the laboratory and later lessons. Complete the actions in order so that the current record is identified before any duplicate or temporary item is handled.

The table below organises information under the headings Step, Housekeeping action and Evidence of completion. Read across each row so that the details remain connected rather than treating individual cells as isolated facts

Step	Housekeeping action	Evidence of completion
1	Save current work and confirm the professional file name.	The title bar and file list show the expected name.
2	Place files in the agreed folder and check the path.	Another person can follow the same path.
3	Confirm the current version and label older copies according to procedure.	No file is called final FINAL or new copy.
4	Check that no confidential file is in an unauthorised location.	Only approved storage locations are used.
5	Update and verify the backup when required.	At least one file opens from the separate backup location

2 SESSION Creating a Structured Folder System for Workshop Documents

Lab Scenario

You will create a controlled storage structure, give files meaningful names, retrieve evidence, and make a verified backup. Imagine that another technician must continue your work tomorrow; your organisation should allow that person to understand the files without asking you.

THE 30-SECOND REQUEST

Your supervisor asks: "Open the September safety report for Conveyor 12 and show me the inspection photograph." A well-organised folder system allows you to do this in seconds. A poor system forces you to open many files with names such as report.docx, photo1.jpg, and final2.docx. The purpose of this lab is to build a system that another person can understand and use.

- Think: What should a good folder system tell you before you open a file?

File management is the process of storing, naming, organising, locating, moving, protecting, and removing digital files in a controlled way. Think of it as a labelled tool cabinet: each tool has a type, a location and a purpose. A professional system reduces delay, duplication and mistakes.

Lab Success Criteria

Success is measured by the quality and reliability of the final system, not by speed alone. Read the criteria before starting so that you know what evidence the teacher will check. During the lab, pause at each checkpoint and correct problems before continuing.

- I can open File Explorer and identify the navigation pane, address bar, search box, and Details view.
- I can explain the difference between a drive, folder, file, path, and extension.
- I can create the required folder tree with exact names.
- I can rename and move the practice files using the agreed naming convention.
- I can find a requested file in 30 seconds or less.
- I can create and verify an authorised backup copy.
- I can use UNDERSTAND-FIND-TRY-CHECK when a simple problem occurs.



SAFETY - Lab Rules

- Use only the student account, authorised practice files, and approved USB or cloud location.
- Do not delete or rename system folders.
- Do not connect personal or unknown storage devices.
- When working in pairs, the learner at the keyboard performs the action; the partner may guide verbally.
- Ask permission before changing any setting outside File Explorer.

Key Terms for the Lab

The table introduces the vocabulary needed to follow the practical instructions. Read the definition and example together because technical words are easier to remember when attached to an action. Use the terms path, extension, copy, move and backup when explaining your work to a partner.

Term	Definition	Example
File	One saved item of digital content.	A Word report, photograph, PDF manual or spreadsheet.
Folder	A container used to organise files and other folders.	01_Reports
Subfolder	A folder inside another folder.	01_Reports\Safety
Drive/storage location	A place where files are stored.	Computer drive, authorised USB, or school cloud folder.
Path	The complete digital address showing the route to a file or folder.	C:\Documents\DT_Year1_Ahmed\01_Reports
File name	The descriptive name before the extension.	2026-09-15_Press3_SafetyReport_v01
Extension	Letters after the final dot that identify the file type.	docx., xlsx., pdf., jpg.
Copy	Creates another file while keeping the original.	Copy a report to the backup location.
Move	Changes the file's location; the original is no longer in the first location.	Move a photograph from Downloads to 03_Photos.
Shortcut	A link that points to a file or folder; it is not the original item.	A shortcut on the Desktop to the main folder.
Backup	A separate copy kept so work can be recovered after loss or damage.	A verified copy on an authorised USB or school cloud.
Restore	Copies data from a backup back to a working location.	Restore one practice file to a Test_Restore folder.

Practical 1 Meet File Explorer

This practical introduces the main Windows workspace used to manage files and folders. The aim is to develop location awareness: you should know which drive and folder you are viewing before you search, copy or rename anything. Work slowly enough to read the path and file details accurately.

Introduction

This practical session begins with a short purpose statement so that each command is connected to a professional reason. Before following the steps, identify the final result you must produce and the safety boundary that applies. Use the routine: the teacher demonstrates a small group of actions, you repeat them, then the result is checked.

File Explorer is the Windows tool used to find, open, organise, and manage files and folders stored on the computer, connected drives, and some cloud locations. The appearance can differ slightly between Windows versions, but the main ideas are the same: choose a location, read the path, view the contents, and use commands such as New, Copy, Move, Rename, and Search.

The File Explorer diagram labels the areas students use most often. Start with the navigation pane and address bar to establish location, then identify the command tools, search box, and file list. The numbered callouts are designed to support the same sequence used in Practical 1.

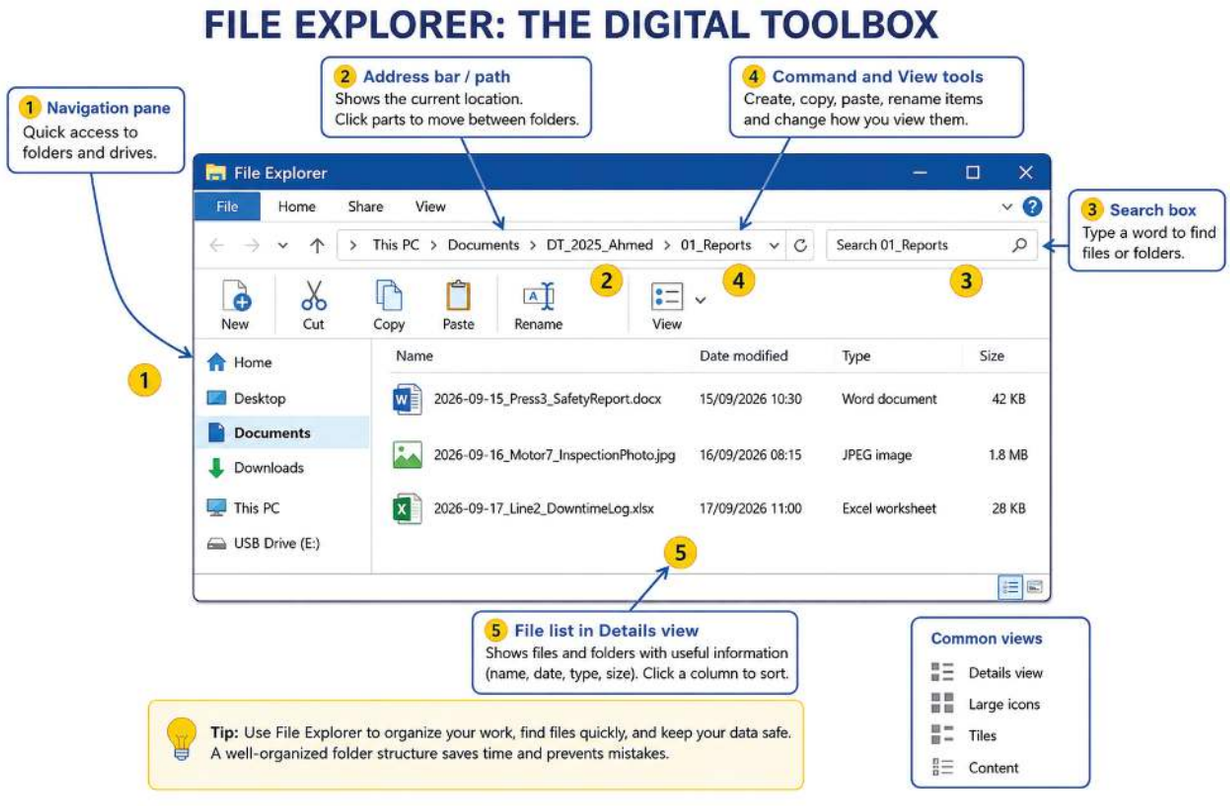


Figure 2.6: Main areas of File Explorer.

Guided Steps

Follow the steps in order and pause after each numbered group. Explain aloud what you are doing and why, especially when reading a path or changing the view. If your screen looks different, compare the function rather than clicking randomly.

1. Open File Explorer by pressing Windows key + E, or select the folder icon on the taskbar.
2. In the navigation pane, locate Home, Desktop, Documents, Downloads, and This PC. Locate the authorised USB drive only if the teacher has provided one.
3. Select Documents. Read the address bar from left to right. This is the route to your current location.
4. Choose View -> Details. Identify the columns Name, Date modified, Type and Size.
5. Choose View -> Show -> File name extensions, if permitted. Notice that docx., xlsx. and jpg. identify file types.
6. Use the search box to search for the word "report". Observe that the search is limited by the location you selected.
7. Sort once by Name and once by Date modified. Return to sorting by Name before continuing.

✓

CHECKPOINT P1

- Point to the navigation pane, address bar, search box and file list.
- Read your current path aloud to your partner.
- Explain what the .docx extension tells you.

PRACTICAL 1 QUICK CHECK

- Which view is most useful for comparing file type, date and size? a) Large icons b) Details c) Tiles
- What does the address bar show?
- Why should you not change a file extension just by typing different letters after the dot?

Practical 2 Plan and Create a Folder Tree

This practical focuses on structure before content. A well-designed folder tree groups related records and gives each location a predictable purpose. Planning the hierarchy first prevents files from being scattered across the desktop, Downloads folder, or unrelated locations.

Introduction

This practical begins with a short purpose statement so that each command is connected to a professional reason. Before following the steps, identify the final result you must produce and the safety boundary that applies. Use the routine: the teacher demonstrates a small group of actions, you repeat them, then the result is checked.

Definition - Folder tree

A folder tree is a hierarchy of folders and subfolders organised from a broad category to more specific categories. Workshop example: The main folder DT_Year1_Ahmed contains Reports, Datasheets, Photos and Checklists; Reports contains Safety and Maintenance.

Professionals plan the tree before moving files. A tree should be simple enough to remember, but detailed enough to separate different types of evidence. Too many folders create unnecessary clicking; too few folders create clutter. For this first-year programme, use the exact structure below so every teacher and student follows the same system.

The folder-tree diagram shows a hierarchy from the main Year 1 folder to categories and subfolders. Read from top to bottom to understand the parent–child relationship, and use the example path to translate the visual tree into a written location.

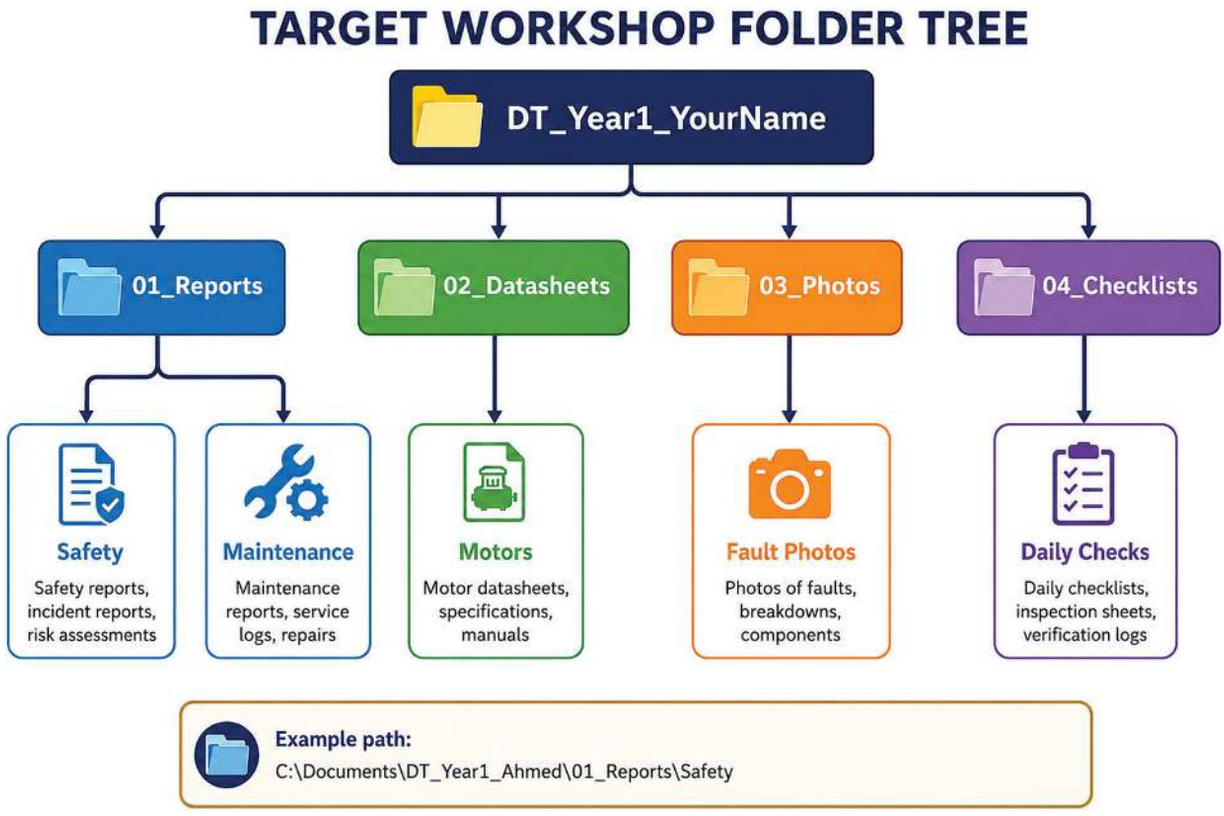


Figure 2.7: Required workshop folder tree.

Create the Structure

Create the main folder and subfolders exactly as shown before adding any files. Exact names make classroom checking easier and model the consistency required in a workplace team. After creation, read at least one complete path aloud to confirm where the folder is located.

1. Open Documents. Right-click an empty area -> New -> Folder.
2. Name the main folder DT_Year1_YourName. Replace YourName with your own first name written consistently.

3. Open the main folder and create: 01_Reports, 02_Datasheets, 03_Photos and 04_Checklists.
4. Open 01_Reports and create the subfolders Safety and Maintenance.
5. Return to the main folder and compare the structure with Figure 2.7. Correct spelling and number prefixes.
6. Open Safety and read the full path aloud. Then use the Back button to return to the main folder.

WHY USE 01, 02, 03 AND 04?

Computers normally sort names alphabetically and numerically. Number prefixes keep the folders in the agreed order. Use leading zeroes (01 rather than 1) so the order stays consistent when more folders are added later.

CHECKPOINT P2

- Your tree has exactly four main folders and two report subfolders.
- Names match the model, including underscores and leading zeroes.
- You can navigate to Safety and return without using the Desktop.

Practical 3 Naming Files Professionally

Practical 3 develops traceability. A meaningful file name should communicate the date, asset, document type, and version without opening the file. Consistent naming also improves sorting and search, especially when a folder contains many records.

Introduction

Practical 3 begins with a short purpose statement so that each command is connected to a professional reason. Before following the steps, identify the final result you must produce and the safety boundary that applies. Use the routine: the teacher demonstrates a small group of actions, you repeat them, then the result is checked.

A file name should help a colleague identify the file without opening it. Good names are consistent, specific and short enough to read. The class will use one naming convention so files sort logically and can be searched by date, asset or document type.

The file-name diagram separates one professional name into meaningful parts. Read from left to right and explain what each part tells a colleague before the file is opened. The colours make the fixed order of the naming convention easier to remember.



Figure 2.8: Parts of a professional file name.

Definition - Naming convention

A naming convention is an agreed rule for constructing file names so that all team members use the same order, vocabulary and formatting.

Workshop example: YYYY-MM-DD_Asset_DocumentType_v##.extension

The Class Convention

The convention below is a classroom standard that models workplace practice. Each part has a fixed position so that filenames can be scanned quickly. Do not change the extension, because it identifies the file format rather than the description of the content.

- The table below organises information under the headings RECOMMENDED PATTERN YYYY-MM-DD_Asset_DocumentType_v##.extension
- Example: 2026-09-15_Conveyor12_SafetyReport_v01.docx, RECOMMENDED PATTERN YYYY-MM-DD_Asset_DocumentType_v##.extension
- Example: 2026-09-15_Conveyor12_SafetyReport_v01.docx and RECOMMENDED PATTERN YYYY-MM-DD_Asset_DocumentType_v##.extension
- Example: 2026-09-15_Conveyor12_SafetyReport_v01.docx. Read across each row so that the details remain connected rather than treating individual cells as isolated facts.

RECOMMENDED PATTERN

- YYYY-MM-DD_Asset_DocumentType_v##.extension

Example: 2026-09-15_Conveyor12_SafetyReport_v01.docx

Part	Rule	Example
Date	Use year-month-day so files sort chronologically.	2026-09-15
Asset	Use the agreed machine or equipment identifier.	Conveyor12
Document type	Use a clear controlled term without unnecessary words.	Safety Report
Version	Use v01, v02, v03 when a document is revised.	v02
Extension	Keep the correct extension; do not change it to convert a file.	docx.

The examples table shows how vague names are improved. Compare the problem in the middle column with the added information in the improved name.

Poor name	Problem	Improved name
report.docx	No date, asset or purpose.	2026-09-15_Press3_ MaintenanceReport_v01.docx
final FINAL v2 (3).docx	Unclear version history.	2026-10-02_Motor7_InspectionReport_ v02.docx
photo1.jpg	No information about the image.	2026-09-20_Robot2_GripperFault_ Photo01.jpg
15-9-26 pump. pdf	Date order and wording are inconsistent.	2026-09-15_Pump4_Datasheet.pdf

FILE-NAME RULES

- Use only approved asset names and document types.
- Use underscores between major parts; avoid long sentences.
- Avoid characters Windows does not allow in file names: \ / : *? " < > |
- Do not use “final” repeatedly. Use version numbers and follow the teacher’s approval process.
- Keep the original extension. Renaming photo.jpg to photo.pdf does not convert the file.

Rename and Place the Practice Files

Use the scenario information to build each name, then place the file in the folder that matches its purpose. Complete one file at a time and check the result before continuing. This controlled sequence reduces the risk of losing a file or assigning the wrong name.

1. Copy the eight practice files supplied by the teacher into your main folder. Keep the teacher's original set unchanged.
2. Rename each file using the convention. Use the scenario information supplied with the files.
3. Move each file to the correct folder: reports, datasheets, photos or checklists.
4. Switch to Details view and sort by Name. Check that files appear in a logical order.
5. Ask your partner to find one named file in 15 seconds.

✓ **CHECKPOINT P3**

- All eight files have meaningful names.
- Dates use YYYY-MM-DD.
- Each file is in the correct folder.
- No extension was accidentally removed or changed.

Practical 4 Copy, Move, Search and Recover

Practical 4 compares actions that are often confused. Copying creates another instance, moving changes the location, searching locates an item within a selected scope, and recovery restores a deleted file when possible. Understanding the result of each action is more important than memorizing a shortcut.

Introduction

Practical 4 begins with a short purpose statement so that each command is connected to a professional reason. Before following the steps, identify the final result you must produce and the safety boundary that applies. Use the routine: the teacher demonstrates a small group of actions, you repeat them, then the result is checked.



Copy and Move are not the same. Copy creates another item while keeping the original. Move changes the location of the item. Before using either command, check the source location and destination location. After the action, verify where the file exists.

The table compares copy, move, search, and recovery by outcome, method, and risk. Use it to predict what will happen to the original file before performing an action.

Action	What happens	Useful method	Common risk
Copy	Original remains; a second copy is created.	Ctrl+C then Ctrl+V, or Copy/Paste.	Editing the wrong copy later.
Move	File leaves the first location and appears in the new location.	Ctrl+X then Ctrl+V, or drag carefully.	Moving to the wrong folder.
Rename	Name changes; content and location stay the same.	F2 or right-click -> Rename.	Removing the extension.
Delete	File normally goes to Recycle Bin first.	Delete key or right-click -> Delete.	Deleting the wrong file.
Restore	Returns a deleted file from the Recycle Bin to its original location.	Open Recycle Bin -> Restore.	Assuming every deleted file can be recovered.
Search	Finds matching names or properties in the chosen location.	Type a keyword in the search box.	Searching the folder or drive.

TEAM CHALLENGE (10 minutes): The 30-Second File Race

- **Step 1:** Form groups of three or four. One learner controls the computer; others may guide verbally.
- **Step 2:** The teacher announces a file request, such as "the safety report for Conveyor 12 from September".
- **Step 3:** Find and open the correct file. Rotate the searcher each round.
- **Step 4:** After four rounds, identify what made searching faster: folder, file name, sorting or search keyword.

Recovery Practice

Recovery is a safety net, not a substitute for careful work. The Recycle Bin may help with local files, but some network, cloud, or removable-drive deletions behave differently. Use only the practice file and verify the restored location before continuing.

1. Create a temporary text file named DELETE_PRACTICE.txt in your main folder.
2. Delete the file. Open the Recycle Bin and locate it.
3. Restore the file and check that it returned to the original folder.
4. Delete the practice file again only when the teacher gives permission.

SAFETY - Delete Carefully

- Never practice deletion on system files, real reports or another learner's work.
- Some network, cloud or removable-drive deletions may not go to the Recycle Bin.
- When uncertain, stop and ask before deleting.

Practical 5 Back Up and Verify Your Work

Practical 5 protects the completed folder system against loss or damage. A backup must be stored in a separate authorised location and must be tested. Simply seeing a copied folder is not enough; opening files from the backup proves that the copy is usable.

Introduction

Practical 5 begins with a short purpose statement so that each command is connected to a professional reason. Before following the steps, identify the final result you must produce and the safety boundary that applies. Use the routine: the teacher demonstrates a small group of actions, you repeat them, then the result is checked.

A backup is not simply another file in the same folder. It is a separate copy kept in a different authorised storage location so work can be recovered if the original is lost, corrupted, deleted, or the device fails. The backup is useful only if it is complete and can be opened.

The backup diagram distinguishes the working copy from copies stored in separate authorised locations. Follow the arrows away from the original folder and then read the verification message. A backup becomes useful only after the copied files can be opened.

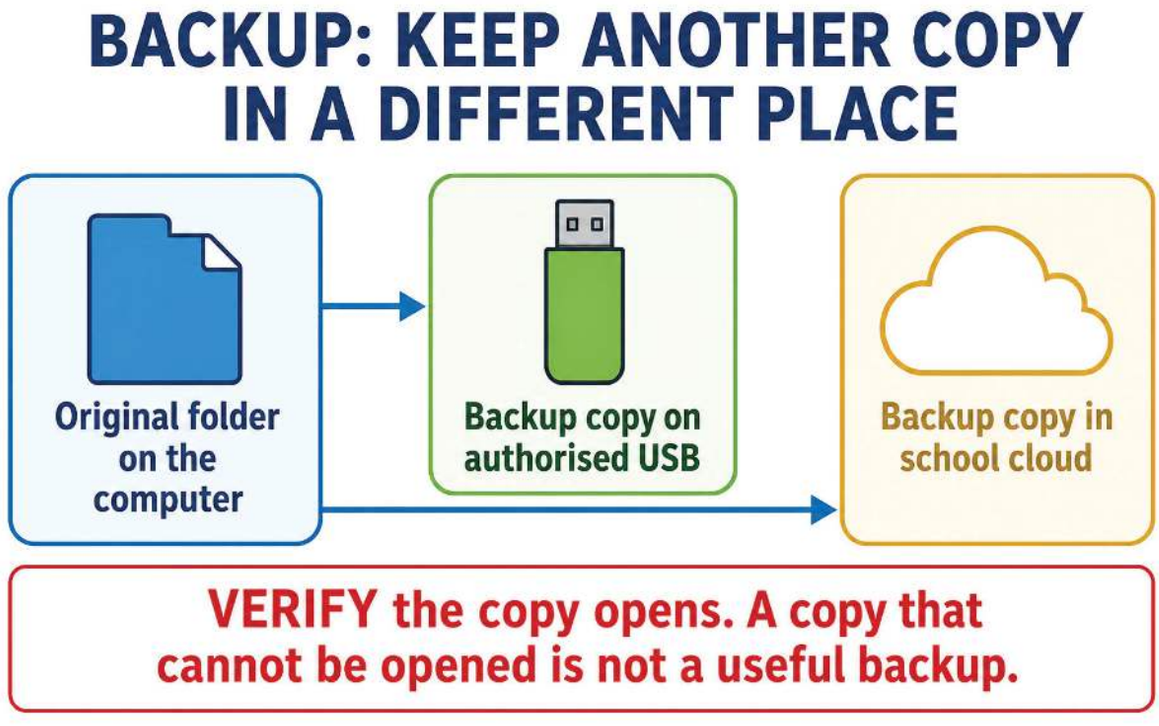


Figure 2.9: A backup copy is stored separately and verified.

Create the Backup

Copy the complete main folder so that its internal structure is preserved. Give the backup folder a date and wait until the operation is complete before disconnecting the device. The next step—verification—is part of the backup process, not an optional extra.

1. Close any practice files that are still open.
2. Select your main folder DT_Year1_YourName and choose Copy.
3. Open the authorised USB drive or school cloud backup location supplied by the teacher.
4. Create a folder named Backup_YYYY-MM-DD and paste your main folder inside it.
5. Wait until copying is complete. Open the backup and check that all four main folders are present.
6. Open one report and one photograph from the backup. Confirm both files work.
7. If using USB, close files and use the system Eject command before removing the device.

PROFESSIONAL EXTENSION: THE 3-2-1 IDEA

For important organisational data, a common resilience approach is to keep three copies, on two different types of storage, with one copy separated from the main system. Your school or employer may use a different approved backup policy. Always follow that policy rather than creating personal copies of confidential information.

Restore Test

A restore test answers the question: could the backup actually be used after a loss? Copy one file from the backup into a temporary restore folder, open it, and compare it with the expected content. This small test demonstrates the same principle used in professional backup plans.

1. Inside your main folder, create a new folder named Test_Restore.
2. From the backup, copy one practice file into Test_Restore.
3. Open it and confirm it matches the backup file.
4. Delete Test_Restore after the teacher checks your work.



CHECKPOINT P5

- The backup is in a different authorised location.
- The backup folder includes the date.
- You verified at least two file types open from the backup.
- You safely ejected the USB if one was used.

Practical 6 Troubleshoot a Simple File Problem

The final practical combines file management with the four-step problem-solving method. Your task is to gather evidence and locate the missing datasheet without changing system settings or using unsafe actions. The quality of your explanation matters as much as finding the file.

Introduction

This begins with a short purpose statement so that each command is connected to a professional reason. Before following the steps, identify the final result you must produce and the safety boundary that applies. Use the routine: the teacher demonstrates a small group of actions, you repeat them, then the result is checked.

The purpose of this final practical is not to create a difficult computer problem. It is to practice calm, evidence-based troubleshooting. Work within File Explorer only, and stop when the next action would need permission.

Use this pair task to apply the four-step method to a realistic file-management problem. One student acts as the operator and the other as the navigator; both must be able to explain the evidence used at each step.

PAIR PROBLEM CARD: The Missing Datasheet

- **Your partner says:** "I saved the motor datasheet, but I cannot find it." Begin with questions rather than random clicking.
- **UNDERSTAND:** identify the expected name, file type, and last known location.
- **FIND:** choose an approved search method and search location.
- **TRY:** use one safe action, such as searching the main folder for .pdf or the word Motor.
- **CHECK:** open the result, verify the correct motor model, and move it to 02_Datasheets if needed.
- Write one sentence of evidence for each step.

If the file is still not located, the pair should report the problem rather than continue with uncontrolled searches. The report must show what was expected, where the search took place, and which safe actions were already attempted.

WHAT TO REPORT IF IT IS NOT SOLVED

- What you expected to find.
- Where you searched.
- The exact search terms used.
- What results appeared.
- What safe action you already tried.

Lab Completion Checklist and Assessment

The checklist collects evidence from the entire laboratory. Complete it only after inspecting the actual folder structure, filenames, and backup. A tick should represent a verified result, not an assumption that the step was probably completed.

The completion table turns the lab outcomes into visible evidence. Inspect the real files and folders before marking the self-check column.

Criterion	Evidence	Self-check	
Folder structure	Four main folders and two report subfolders with exact names.	Achieved ☐	Not yet ☐
Navigation	Can read a path and use Back, address bar and search.	Achieved ☐	Not yet ☐
File naming	Eight files follow the agreed pattern and keep correct extensions.	Achieved ☐	Not yet ☐
Organisation	Files are placed in the correct folders.	Achieved ☐	Not yet ☐
Retrieval	Finds a requested file within 30 seconds.	Achieved ☐	Not yet ☐
Backup	Creates and verifies a separate authorised copy.	Achieved ☐	Not yet ☐
Problem solving	Explains one UNDERSTAND-FIND-TRY-CHECK example.	Achieved ☐	Not yet ☐

Lab Rubric

The rubric describes different levels of performance so that assessment is transparent. Read the secure-performance column first to understand the target, then use the developing and beginning columns to identify specific improvements. The same criteria can guide peer feedback before final submission.

The rubric table describes four performance levels for each assessed area. Begin with the secure column to identify the target and use the lower levels to locate precise improvements.

Area	3 - Secure	2 - Developing	1 - Beginning	0 - Not demonstrated
Folder tree	Complete, exact and independent.	One minor error corrected with prompt.	Several errors or heavy support.	Not created.
File naming & placement	All names clear and files correctly placed.	Most correct; one or two errors.	Inconsistent names or several misplaced files.	Files unchanged or missing.
Navigation & retrieval	Reads path and finds files quickly.	Finds files with some support.	Slow or confused navigation.	Cannot locate files.

Area	3 - Secure	2 - Developing	1 - Beginning	0 - Not demonstrated
Backup & verification	Separate backup created and files verified.	Backup created but verification incomplete.	Copy made in unsuitable location or needs major support.	No backup.

Lesson/Session Summary

- Hardware is the physical system; software is the set of instructions that makes the hardware useful.
- The CPU processes instructions, RAM holds active work temporarily, and storage keeps files after power is removed.
- Computers and PLC systems can be understood through Input-Process-Store-Output.
- The operating system manages programmes, memory, devices, files, users, and security.
- Extensions identify file types; size units help you judge storage and sharing requirements.
- Use UNDERSTAND-FIND-TRY-CHECK; change one thing at a time and stop when the action is unsafe or outside your authority.

Class Discussion

1. Explain the difference between RAM and storage using the workbench and cabinet analogy.
2. Describe Input-Process-Store-Output for a sensor-controlled conveyor.
3. Name three jobs of an operating system.
4. Explain the difference between Save and Save As.
5. A programme controls a live machine and stops responding. What should a trainee do?

Assessment

Part A: Choose the Correct Answer

1. The component that carries out programme instructions is:

a. CPU **b.** monitor **c.** folder **d.** printer

2. RAM is best described as:

a. permanent archive **b.** temporary working memory
c. operating system **d.** file extension

3. Which is application software?

a.motherboard **b.** Microsoft Word **c.** SSD **d.** keyboard

4. The main role of an operating system is to:

a. manage the computer's resources and provide services to applications
b. replace all applications
c. manufacture hardware
d. create electricity

5. Which extension normally identifies a PDF document?

a. pdf. **b.** xlsx. **c.** jpg. **d.** zip.

6. Which action creates a new version while keeping the original?

a. Save As **b.** Delete **c.** Cut **d.** Close

7. A live-machine control programme freezes. The trainee should:

a. end it immediately **b.** follow the authorised stop/escalation procedure

c. remove the power cable **d.** install new software

8. A table header row tells the reader:

a. what each column contains **b.** the computer password

c. the file size **d.** the processor speed

Part B: True or False

1. Data in RAM normally remains when the computer is switched off. **True** ☐ **False** ☐

2. A file extension helps the operating system identify the file type. **True** ☐ **False** ☐

3. Save As can be used to create v02 while keeping v01. **True** ☐ **False** ☐

4. A floating image that covers report text is acceptable if the picture is attractive. **True** ☐ **False** ☐

Part C: Short Answer

A student says: "The computer is broken." Write a professional four-part problem report using WHAT, WHERE, WHEN/CONTEXT, and WHAT WAS TRIED.

Self-Assessment Checklist

When I finish this lesson, I can	Yes	Not yet
Identify the main computer components and explain their jobs.	☐	☐
Use Input-Process-Store-Output to explain a PC or PLC system.	☐	☐
Explain the difference between hardware, software, OS, driver, and application.	☐	☐
Interpret common file extensions and size units.	☐	☐
Use Save, Save As, and professional version names correctly.	☐	☐
Apply safe first-aid troubleshooting and respect authority boundaries.	☐	☐

Reflect and Connect

Choose one "hidden computer" at home, such as a washing machine, television, microwave, or car display. In 3-4 sentences, identify one input, one processing function, one stored item, and one output. Then explain one action you should not attempt without authority or training.

Glossary قاموس المصطلحات

The dictionary table supports technical vocabulary development. Use the English term in class and practical work, use the simple meaning to check understanding, and use the Arabic equivalent as a learning support.

English term	Meaning	المصطلح بالعربية
Computer system	Hardware and software working together	نظام الحاسوب
Hardware	Physical parts that can be touched	المكونات المادية
Software	programme and instructions	البرمجيات
Firmware	Low-level software stored inside a device	البرامج الثابتة
CPU / processor	Component that executes instructions	وحدة معالجة المركبة / المعالج
RAM	Temporary working memory	ذاكرة الوصول العشوائي
Storage	Long-term place for files and programmes	وحدة التخزين
Motherboard	Main board connecting internal components	اللوحة الأم
Peripheral	Connected input, output, or storage device	جهاز طرفي / ملحق خارجي
Operating system	Main software that manages the computer	نظام التشغيل
Device driver	Software that links the OS to hardware	برنامج تشغيل الجهاز
Application	Programme for a specific user task	تطبيق
Boot/start-up	Process of loading and starting the operating system	الإقلاع /بدء التشغيل

English term	Meaning	المصطلح بالعربية
File extension	Letters after the dot showing the file type	امتداد الملف
Version	A numbered revision of a document	إصدار / نسخة منقحة
Troubleshooting	Finding and resolving the cause of a problem	استكشاف الأعطال واصلاحها
Symptom	What can be directly observed	غرض / علامة ظاهرة
Finding	A factual observation or measurement	نتيجة / ملاحظة فنية
Recommended action	Proposed next step based on evidence	إجراء موصى به

Abbreviations

The abbreviation table expands the shortened forms used in specifications, diagrams and instructions. Knowing the full form helps you read technical documents accurately.

Abbreviation	Full form
CPU	Central Processing Unit
RAM	Random Access Memory
SSD	Solid State Drive
HDD	Hard Disk Drive
OS	Operating System
PSU	Power Supply Unit
USB	Universal Serial Bus
PLC	Programmable Logic Controller
CNC	Computer Numerical Control
KB / MB / GB / TB	Kilobyte / Megabyte / Gigabyte / Terabyte

LESSON 3

Word Processing for Technical Documentation

Learning Outcomes

Look carefully at the two documents below. They contain the same information about the same machine. One was written in a hurry, with no plan and no formatting. The other was planned first and then built using the tools you will learn today. Ask yourself one question as you compare them: if you were a workshop manager with thirty seconds to decide whether a machine is safe to run, which one would you rather be holding?

By the end of this lesson, you will be able to:

- identify the parts of a professional technical document — header and footer, title, numbered headings, body text, tables, figures with captions, and lists — and explain the job each part does for the reader.
- plan a document before writing it using the Purpose–Reader–Sections method, and justify the order of the sections you choose.
- apply advanced Word tools with confidence: paragraph styles, headers and footers, automatic tables of contents, and templates.
- apply safe troubleshooting and create a professional inspection report.

Introduction:

In this lesson, you will use the operating system to start Word and manage a file, use storage correctly, work with structured information, insert a table and image, save regularly, and apply troubleshooting if something goes wrong.

In the laboratory session, you will integrate hardware, operating-system, file-management, and document-production skills into one professional task. Work through the practicals in order because

each stage creates the foundation for the next: save, structure, format, insert evidence, review and verify. This is not a classroom exercise invented for school. It is, almost exactly, the task a junior technician is given in their first week of work.

WARM-UP

Spot the Differences!

Your teacher will show you two versions of the same maintenance report. Version A is plain, with no formatting at all. Version B has been formatted professionally. Both contain identical information.

Step 1: Working in pairs, find FIVE differences between the two documents. You have two minutes.

Step 2: Answer this together: you are a busy workshop manager, and you have thirty seconds. Which version tells you the machine status faster, and why?

The five differences we found:

.....

.....

.....

.....

Before You Begin: The Four Jobs of Technical Documentation

A technical document is more than a page of information. It creates a chain of evidence between the person who observed a condition, the person who must decide what to do, and the people who may need to check the decision later. In a workshop, a good document normally performs four jobs at the same time: it records what happened, communicates it to the right reader, guides the next action, and preserves evidence for future checking.

Imagine that the night shift measures unusual vibration on a motor but leaves only a verbal message. By the morning, the exact value, time, and operating condition have been forgotten. A short written record with the measurement, unit, time, and machine ID turns a memory into evidence. The table below separates the four jobs so you can see how each one changes what you write.

Job of the document	Question it answers	Workshop example
Record	What was observed, measured or completed?	Vibration measured at 7.2 mm/s at 08:15.
Communicate	Who needs to know, and how quickly?	The supervisor and the next maintenance shift.
Guide action	What decision or next step is required?	Isolate the motor and inspect the bearing.
Preserve evidence	What may need to be checked later?	The signed report, issue date and approved version.

A single report may perform all four jobs. Planning and structure prevent one job from hiding another: the reader can find the evidence quickly, understand the required action and identify which version of the document is valid.

Topic 1: The Anatomy of a Professional Technical Document

Every professional document you will ever read — a maintenance report, a machine manual, a safety procedure, a university thesis — is built from the same small set of parts. Engineers call this repeated arrangement the document skeleton. In this topic, you will learn to recognize each bone of that skeleton, understand the specific job it does, and see why getting one of them wrong can have consequences far beyond an untidy page.

1.1 Why Structure Matters — the Thirty-Second Test

Start with what you have just experienced. In the warm-up, two documents held identical information, yet one of them answered the manager's question in seconds, and the other did not. Nothing was added to Version B except structure. No new facts, no extra pages, no better writing — only structure. That is the whole argument of this topic in a single observation.

Here is a useful way to think about it. Reading a document is not like watching a film, where you start at the beginning and continue to the end. A professional reader hunts. The supervisor does not want to read your report; the supervisor wants to find one fact inside your report and then get back to work. Structure is what makes hunting fast. Headings tell the reader where to look. A table lets the reader compare five values without reading five sentences. A caption under a photograph explains what the photograph proves without the reader having to guess.

Real-World Scenario: The Report Nobody Read

- A technician at a steel plant spent an entire afternoon writing a careful three-page report about a vibration problem in a rolling mill. He wrote it as continuous text: no headings, no table, no summary. His supervisor opened it, scrolled for eight seconds, could not find the recommendation, and put it aside "to read later". Later never came. Three weeks after, the bearing failed and the mill stopped for two days.

  The technician's analysis was correct. His conclusion was correct. His document was unreadable, so his correct work had no effect on anything. **Structure is not politeness — it is what makes your work count.**

1.2 The Document Skeleton — the Seven Parts

The table below lists the seven parts that appear, in some form, in almost every professional technical document. Read it as a checklist rather than as a list of facts to memorize: before you submit any document from today onwards, you can run down this table and ask whether each part is present and doing its job. The third column is the important one, because it explains the purpose of each part rather than just naming it.

Part of the document	What it is	Why it matters to the reader
Header and footer	Text repeated at the top and bottom of EVERY page: company or school name, document title, page number, sometimes a date or document code.	Pages get separated, printed, photocopied, and mixed with other documents. A header means a loose page can always be traced back to the document it belongs to.
Title	The name of the whole document, at the top of the first page, formatted larger than everything else.	It answers the reader's first question — "What am I holding?" — before they read a single sentence.
Headings (numbered)	The names of the sections and subsections, formatted consistently and usually numbered (1, 1.1, 1.2, 2 …).	They act as signposts. A reader who wants only the recommendation can jump straight to it without reading the rest.
Body text	The main writing: short, factual, precise sentences that state what was observed, measured and concluded.	This is where the actual information lives. Short factual sentences remove ambiguity. — and ambiguity in a workshop is a safety risk.

Part of the document	What it is	Why it matters to the reader
Tables	Numerical or comparative data arranged in rows and columns with a header row.	The eye can compare eight values in a table in a few seconds. The same eight values buried in a paragraph would take a minute and invite mistakes.
Figures with captions	Photographs, drawings or diagrams, each with a numbered label such as "Figure 1: Cracked bearing housing".	A figure is evidence. A photograph of a crack proves what a hundred words can only claim. The caption tells the reader what to look at and why.
Lists	Bullet lists for items with no required order, and numbered lists for steps that must happen in sequence.	Lists break long instructions into single actions the reader can follow — and tick off — one at a time.

Now look at how those seven parts appear together on a single real page. In the diagram below, an ordinary one-page maintenance report is shown in the centre, and every part of the skeleton is labelled with an arrow pointing at where it actually sits on the page. Notice particularly how the parts are arranged from general to specific as you move down the page: first what the document is, then what the machine is, then what was found, then what must be done.

Figure 3.1 — The Skeleton of a Professional Document

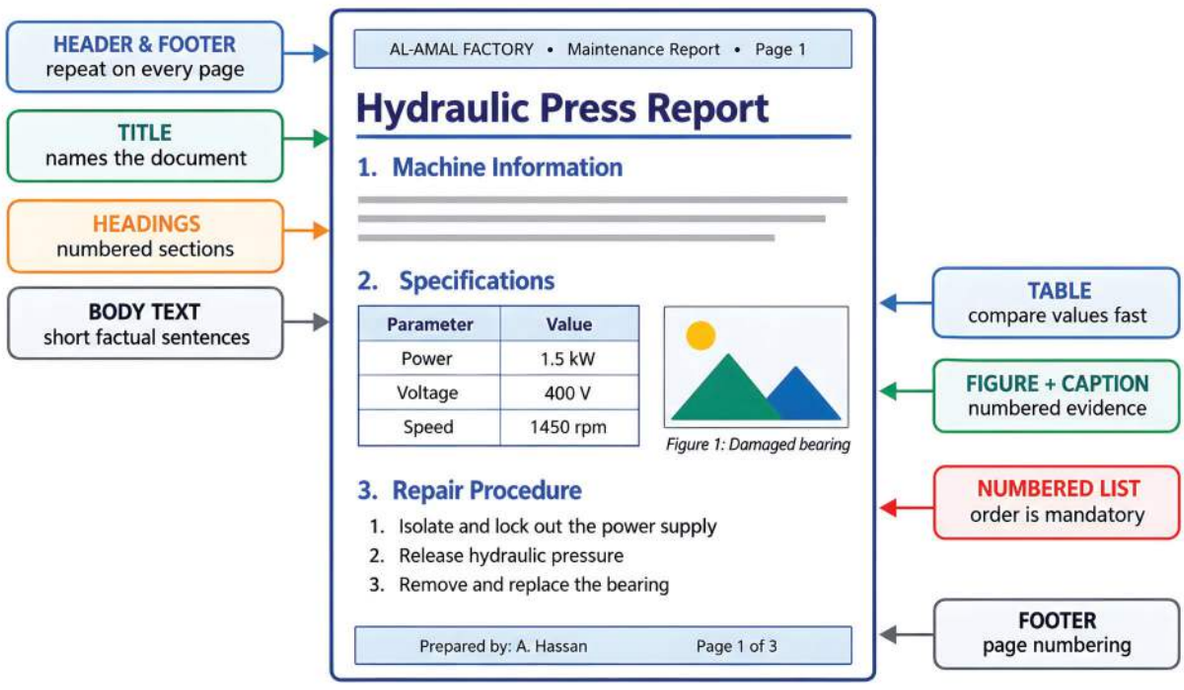


Figure 3.1: The document skeleton — every part labelled on a real report page.

1.3 Bullets or Numbers? A Safety Rule, Not a Style Choice

Most students think choosing between bullet points and numbers is a matter of taste, like choosing a font. It is not. In technical writing, the choice carries meaning, and the reader relies on that meaning.

A bullet list tells the reader: these items belong together, but their order does not matter. A list of the tools you need for a job is a bullet list — it makes no difference whether you pick up the spanner or the gloves first. **A numbered** list tells the reader something completely different: these are steps, and they must be performed in exactly this sequence. When you number a list you are making a promise to the reader that the order is deliberate.

Consider a lock-out procedure — the sequence a technician follows to make a machine electrically dead before working inside it. Step 1 is to switch the machine off. Step 2 is to isolate the main power. Step 3 is to attach a personal lock and warning tag. Step 4 is to test that the machine really is dead. Now imagine those four actions written as bullet points, and a nervous first-week apprentice reading them. If that apprentice performs the fourth item before the second, they will test a machine that is still live. That is not an untidy document. That is an injury.

This is why the rule below is worth memorising exactly. It will be assessed at the end of this lesson, and it will still be true in your final year and in your first job.

The diagram below places the two kinds of list side by side using two examples from your own workshop, so you can see at a glance what each one is promising the reader.

Figure 3.2 — Bullet List or Numbered List? A Safety Decision

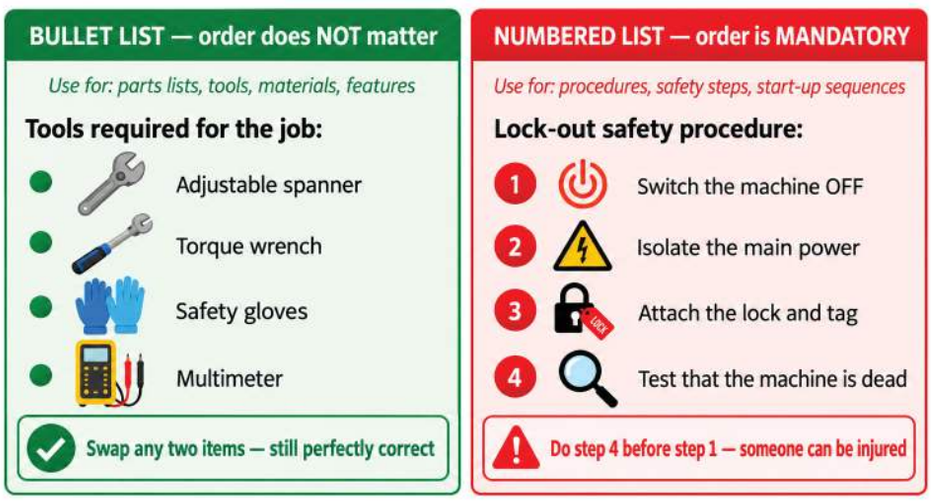


Figure 3.2: Bullet list or numbered list — the decision is about order, and order can be about safety.

The Rule

If the wrong order could cause harm, damage or failure, you must use a NUMBERED list. If the order genuinely does not matter, use bullets. When you are unsure, ask yourself: "Could someone be hurt by doing these in the wrong sequence?"

Use the short check below to test whether you can apply the list rule in a new situation, not merely repeat the wording.

Pop-Up Quiz 1

Answer from memory, then check with your partner before your teacher gives the answers.

1. A safety procedure with six steps should be written as:

a. bullet points **b.** a numbered list **c.** one long paragraph **d.** only images

2. Page numbers belong in the:

a. title **b.** table **c.** header or footer **d.** figure caption

3. A caption is:

a. the page number **b.** a numbered label describing a figure **c.** a font **d.**the footer

Write your answers: 1. ..... **2.** ..... **3.** .....

1.4 Document Control and Traceability

Professional documents change. A draft may be corrected after a measurement is checked, and a reviewed copy may later be approved for use. If the copies are not controlled, two technicians can act on two different instructions while both believe they are using the "latest" document. Document control is the small set of fields that prevents this confusion.



Definition - Traceability

Traceability means being able to identify who created or changed a document, when the change was made, which version is current and who approved it. It allows a later reader to reconstruct the history of the information.

The table below explains the control fields normally placed in a header, footer or document-information block. Read the example column as one connected identity: together, the fields point to one exact approved copy rather than to "the report" in general

Control field	What it tells the reader	Example
Document ID	The unique code for this document.	MTR-RPT-014
Version/revision	Which edition of the document this is.	v03
Issue date	When this version became available.	17-09-2026
Prepared by	Who created or revised the content.	A. Hassan
Reviewed/approved by	Who checked and authorised it.	M. Karim
Status	Whether it is Draft, Under Review or Approved.	APPROVED

The diagram below follows one report from draft to review and approval. Notice that earlier versions are not erased; they remain part of the history. However, only the copy marked APPROVED should be issued for work.

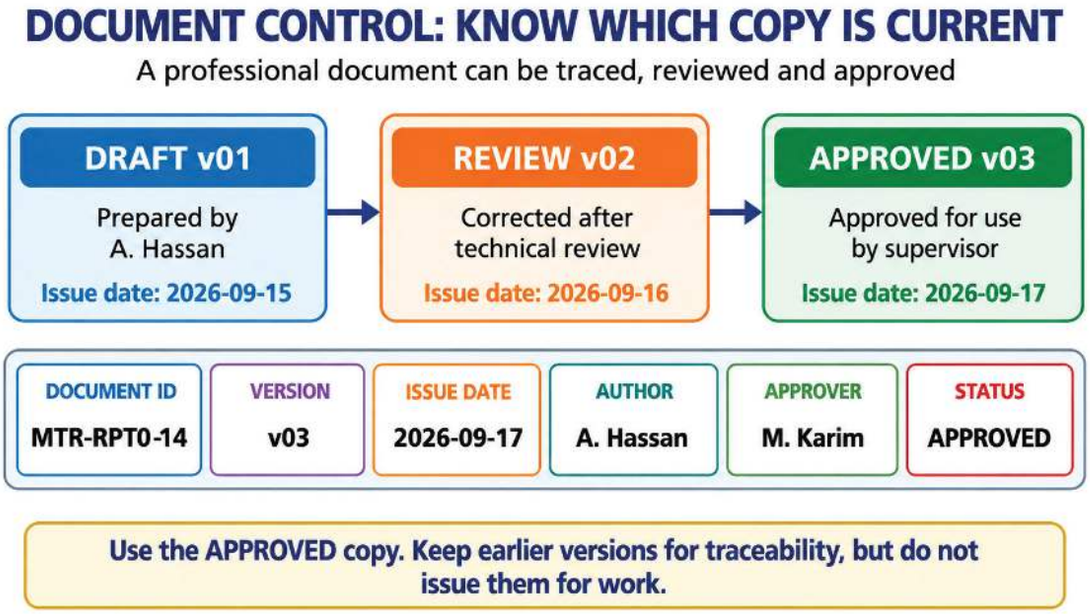


Figure 3.3: Document control makes the current approved copy visible and keeps earlier versions traceable.

Quick Check

A folder contains v01 Draft, v02 Review, and v03 Approved. Which copy should be used for work? Which copies should be kept as history? Explain your answer in one sentence.

Topic 2: Plan Before You Type — the Purpose–Reader–Sections Method

Topic 1 showed you what a finished professional document looks like. This topic answers the harder question: how do you get there? The answer is that professionals do their thinking before they do their typing. This topic gives you the exact method they use — three questions, asked on paper, in about two minutes.

2.1 The Problem With "Just Start Typing"

Two students are given the same task: write a report about a pump that has failed in the workshop. Student A opens Word immediately and types everything he remembers, in the order he happens to remember it. He starts with the noise he heard, then jumps to the model number, then back to the noise, then mentions a measurement, then remembers he should have described the machine first. Forty minutes later, he has three pages of true statements in no useful order. Everything in his report is correct. Almost none of it can be found.

Student B spends two minutes with a pencil first. She writes three short answers on a scrap of paper, then opens Word and types for thirty-five minutes. Her report is shorter than his, took less total time, and her supervisor finds the recommendation in eight seconds.

The difference is not talent, and it is not typing speed. The difference is that Student B answered three questions before she began. Those three questions are a professional planning method used in engineering companies everywhere, and you can learn them in the next five minutes.

2.2 The Three Planning Questions

The method has a name you can remember easily: Purpose–Reader–Sections. You ask them in that order, because each answer shapes the next. The purpose tells you what the document must achieve; knowing the reader tells you how to achieve it; and only then can you decide which sections to write and in what order to put them.

The table below sets out each question, what you should be asking yourself, and how Student B answered it for her pump report. Copy this table into your notebook — you will use it for every document you write from today until you graduate.

Question	Ask yourself	Student B's pump-report answer
1. PURPOSE	Why does this document exist at all? What decision or action does it need to support? What must the reader be able to do after reading it?	"So that my supervisor can decide whether to repair this pump or replace it." Every sentence in the report must serve that decision.
2. READER	Who will actually read this? What do they already know? How much time do they have? What one thing do they most need?	"A busy supervisor. He knows pumps better than I do, so I need no explanations of basic parts. He has two minutes, so I must use a table and clear headings."
3. SECTIONS	Which sections do I need, and in what logical order? Does the order move sensibly from general information to specific action?	"Machine information → Checks performed → Findings → Recommendation." General to specific, ending with the action she wants him to take.

The diagram below shows the same three questions drawn as a flow, with the pump example carried through each stage so you can see how one answer shapes the next. Notice the yellow bar along the bottom: it marks the moment when you are finally allowed to open Word. Everything above that bar happens on paper, with a pencil, in about two minutes.

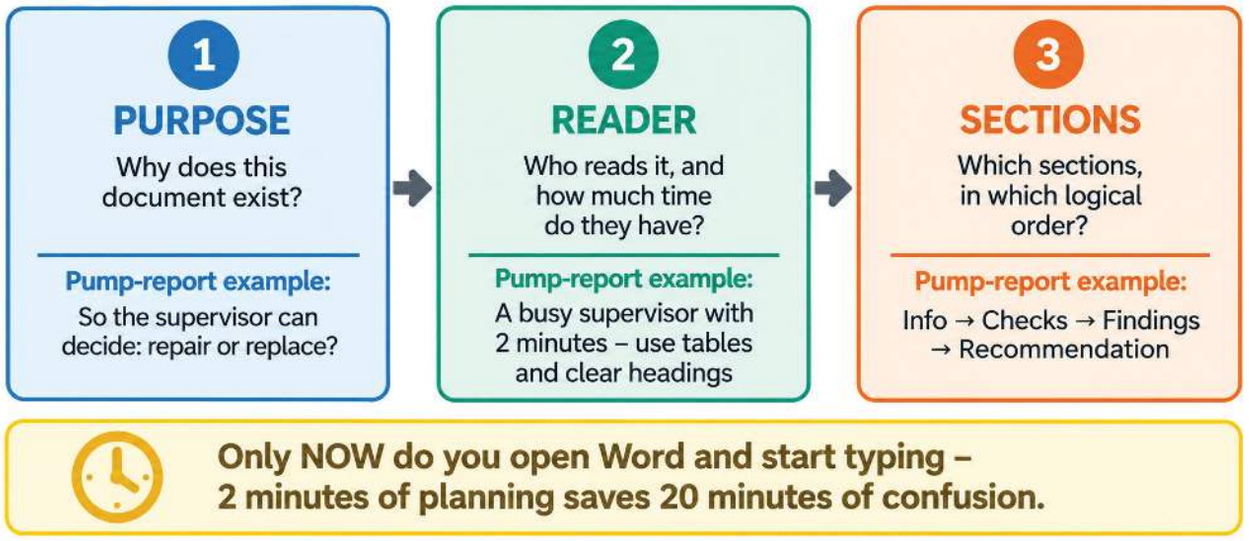


Figure 3.4: Purpose–Reader–Sections Planning Method—the two minutes of thinking that save twenty minutes of confusion.

2.3 From a Plan to a Working Outline

Purpose–Reader–Sections gives you the architecture of the document, but a professional outline goes one step further: it gives every section a specific job and identifies the form of evidence that belongs there. This prevents a common first-year problem in which a student creates four good headings but then places the same kind of paragraph under every heading.

The table below turns the pump-report plan into a working outline. Read across each row. The section name tells the reader where they are; the section job tells the writer what information belongs there; and the evidence form tells you whether a table, list or short paragraph will communicate it most efficiently.

Planned section	Specific job	Best evidence or format
Machine Information	Identify the equipment and the context.	Short fields or a compact table.
Checks Performed	Show exactly what was done and in what order.	Numbered list when sequence matters.
Findings	Record objective observations and measurements.	Short factual sentences with units.
Recommendation	State the proposed next action and reason.	Concise action paragraph.

The diagram below shows the full planning path. Purpose and reader decisions shape the sections; each section then receives suitable evidence; only after those choices are clear do you open Word and build the document.

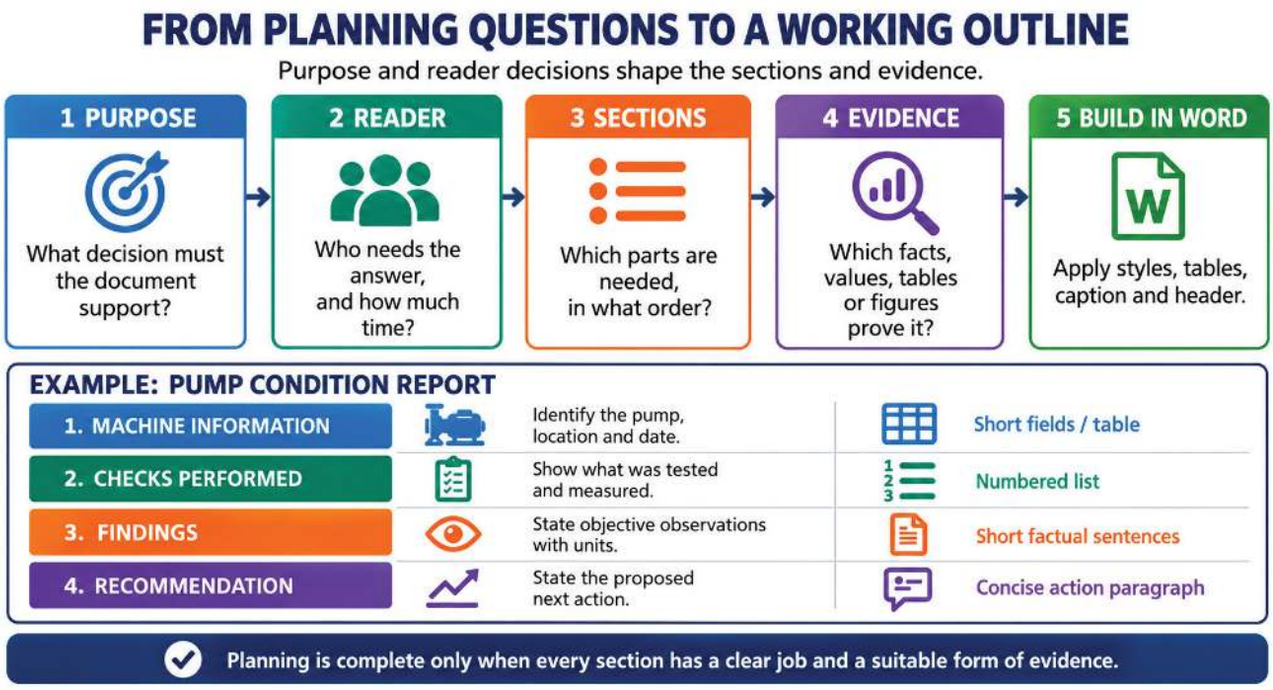


Figure 3.5: A professional outline connects purpose, reader, sections and evidence before Word is opened.

2.4 Common Patterns for Ordering Sections

There is no single correct order for every technical document. The correct order is the one that helps the intended reader reach the required decision. The table below introduces four common patterns. Choose one pattern deliberately instead of arranging sections in the order that ideas occur to you.

Ordering pattern	Use it when	Example section order
General to specific	The reader needs identification and context before detail.	Machine → location → checks → findings
Chronological	The sequence of events explains the result.	Before fault → event → response → current status
Problem–evidence–action	The document must justify a decision.	Problem → measurements → analysis → recommendation
Compare and recommend	Two or more options must be evaluated.	Criteria → Option A → Option B → recommendation

A clear document normally uses one main ordering logic. Mixing patterns without a reason makes the reader jump backwards and forwards, which increases the chance of missing an important condition or action.

Look back at the report you wrote in Lesson 2. It already followed a planned structure, because the structure was given to you. From today, you know why those sections were in that order, and

you can build the structure yourself for a document nobody has planned for you. This is the same problem-solving habit you met in Lesson 1 — understand the problem before you try to solve it — applied to writing.



Activity 1.1 *Plan It — Don't Write It!*

With your partner, PLAN the following document. Do not write a single sentence of the document itself

— planning is the skill being practiced here, and your teacher will stop you if you start writing.

The document: "Instructions for new students: how to shut down the workshop computers safely at the end of the day."

Step 1: PURPOSE — write one sentence explaining why this document needs to exist.

Step 2: READER — who are they, what do they already know, and what do they NOT know?

Step 3: SECTIONS — list three or four sections in a logical order. Then decide: will the shutdown steps be bullets or numbers, and why?

Our plan:

Topic 3: Power Tools in Word — Styles, Headers and Templates

You now know what a professional document looks like and how to plan one. This topic gives you the tools that let you build it quickly and consistently. Word contains dozens of features, but three of them do most of the professional work: styles, headers and footers, and templates. Learn these three properly, and you will never again format a heading by hand.

3.1 The One-Click Transformation

Your teacher is about to show you something that looks like a trick. A plain line of typed text, indistinguishable from every other line on the page, is selected. One click in the ribbon. The line becomes a heading: larger, bold, coloured, correctly spaced above and below. No font was chosen, no size was set, no colour was picked.

That single click applied a style. Understanding what just happened is the difference between someone who uses Word and someone who is good at Word.

3.2 Styles — What They Are and Why Professionals Depend on Them

A style is a saved package of formatting instructions — font, size, colour, bold or not, spacing before and after — stored under a name and applied to a whole paragraph with one click. Word

comes with several built-in styles, and the ones you will use constantly are called Title, Heading 1, Heading 2, and Normal.

The obvious benefit is speed. The real benefit is consistency. Imagine formatting the headings of a two-hundred-page machine manual by hand: you would set size 16, bold, dark blue for the first heading, and by heading forty you would have forgotten and used size 15. The document would look subtly wrong everywhere, and you would never find out why. With styles, every Heading 1 in the document is identical by definition, because they all point at the same saved instructions.

There is a third benefit, and it is the one that surprises students most. Because Word knows which paragraphs are headings — not merely which paragraphs look big and bold, but which ones ARE headings

— it can do intelligent things with them. It can build a table of contents automatically. It can show you a navigation pane you can click through. It can renumber your sections when you insert a new one. None of this works if you simply made the text bold by hand, because bold text carries no meaning for the software. This is worth stating plainly, because it is the single most common misunderstanding in this lesson:

Making text bold and big is not the same as applying a Heading style. It looks the same to your eye and means nothing at all to Word.

The diagram below compares the two approaches directly. On the left is the manual method, together with the five separate actions it demands for every single heading in the document. On the right is exactly the same visual result produced by one click — plus the bonus you receive for free, which the manual method can never give you.

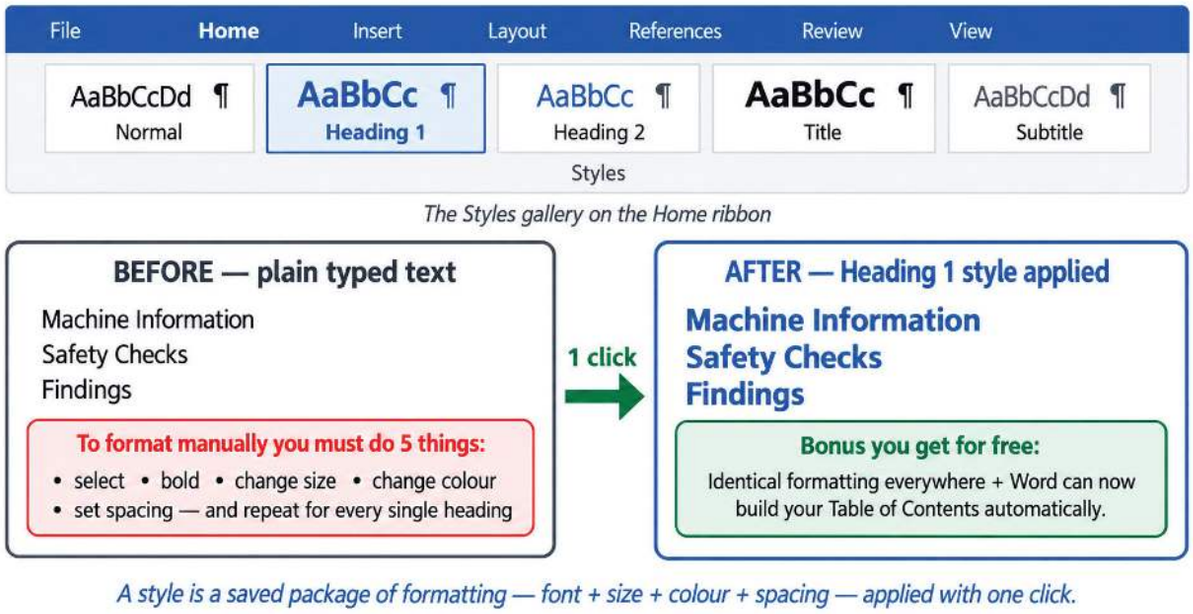


Figure 3.6: Styles replace five manual formatting actions with one click — and unlock the automatic table of contents.

How to apply a style — the three steps

Applying a style is genuinely this simple, and you will use these three steps in the lab:

1. Click anywhere inside the paragraph you want to format. You do not need to select the whole line
2. a style applies to the entire paragraph automatically.
3. Look at the Styles gallery on the Home tab of the ribbon, at the top of your screen.
4. Click the style you want: Title for the document name, Heading 1 for main sections, Heading 2 for subsections.

Headers and footers

A header is text that Word repeats automatically at the top of every page; a footer does the same at the bottom. You type the content once, and Word places it on page one, page seven, and page two hundred without being asked again. Headers usually carry the company or school name and the document title. Footers usually carry the page number and sometimes the author's name.

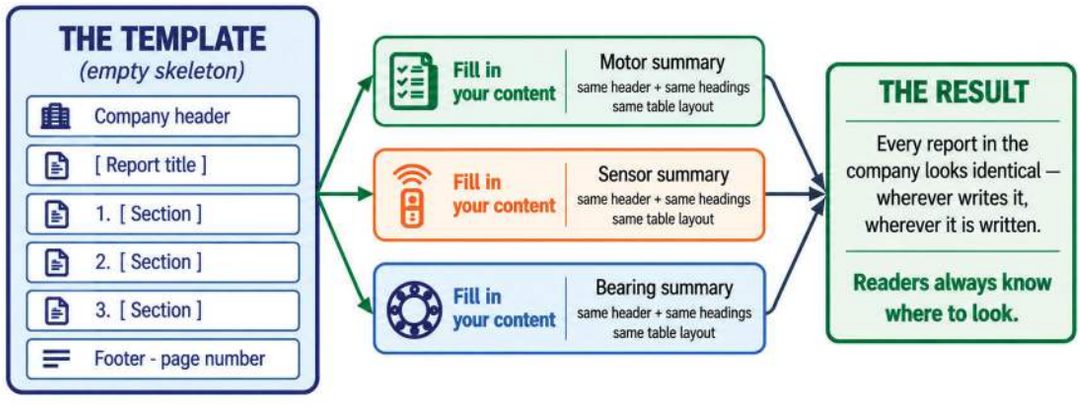
To insert one in Word: go to the Insert tab, click Header (or Footer), choose the Blank layout, type your text, then double-click anywhere in the main body of the page to close it. In the lab you will add a header containing your own name — the marker will be looking for it.

3.3 Templates — Build It Once, Reuse It Forever

A template is a ready-made document skeleton, saved so that you can create new documents from it instead of starting from an empty page. It already contains the header, the title placeholder, the section headings, the table layout, and the footer — everything except the content, which changes each time.

Companies rely on templates for a reason that connects directly back to Topic 1. If every maintenance report from a factory uses the same template, then every supervisor in that factory knows exactly where to look for the recommendation — it is always in the same place, in every report, written by every technician. The template turns a personal habit into a company standard.

The diagram below shows this idea in action: a single empty template on the left, filled three different times with three different components, producing three documents that are different in content but identical in structure.



In today's lab you build YOUR OWN datasheet-summary template — and reuse it for three years.

Figure 3.7: One template, many documents — different content, identical structure.

3.4 Modify a Style Once, Not Twenty Times

A style can be applied with one click, and it can also be changed from one place. Suppose a company decides that every Heading 1 must be dark blue, 15 pt, and followed by 6 pt of spacing. If the document uses real styles, you modify Heading 1 once and every Heading 1 update. If each heading was formatted manually, you must find and repair every heading separately.

Use this method only when you are authorised to change the document design: select a correctly styled heading, right-click the style in the gallery, choose Modify, change the required settings and confirm. In a school or company template, follow the approved style guide rather than inventing a personal colour scheme.

3.5 Captions, Cross-References and Automatic Numbering

A caption is not merely a sentence under an image. It is a formal numbered label that gives the figure an identity. Once a figure is called "Figure 1", the writer can refer to it precisely from another paragraph.

This reference is called a cross-reference. Precise references are especially important when a document is printed, when a figure moves to another page, or when the reader is using a screen reader.

The table below separates four related Word features that students often confuse. Notice that each feature performs a different structural job even though all four help navigation.

Feature	What it does	Professional example
Caption	Numbers and describes a table, photo or diagram.	Figure 1: Motor nameplate and rating data.
Cross-reference	Points the reader to a numbered item.	"See Figure 1 for the rated voltage."
Table of contents	Lists styled headings and page numbers.	1 Machine Information … 2
Navigation pane	Lets the writer move between styled headings on screen.	Click "3 Findings" to jump to that section.

Avoid vague phrases such as "the picture below" or "the table above". Layout changes can move an item to another page. A numbered caption and cross-reference remain meaningful even when the document is edited.

DID YOU KNOW? The Automatic Table of Contents

Word can build a complete table of contents by itself — with every section name and its correct page number — from the References tab, in two clicks. It can also update every page number in that contents list after you edit the document, again in two clicks.
But it can only do this if you used real Heading styles. If you formatted your headings by hand, Word finds nothing and builds an empty list. Companies producing two-hundred-page manuals would need days to renumber a contents page by hand; with styles it takes them seconds. This is the practical reason styles exist.

Answer the questions below after the demonstration. They check whether you understand the structural meaning of styles and templates.

Pop-Up Quiz 2

1. To create an automatic table of contents, you must first have used:

a. bold text **b.** Heading styles **c.** bright colours **d.** a large font

2. A template is best described as:

a. a finished report setting **b.** a ready-made document skeleton

c. a type of table **d.** a printer

Write your answers:

1. ..... **2.** .....

Micro-Practice — Style or Manual Formatting?

For each action, write STYLE or MANUAL and explain why: (1) make every main heading identical; (2) emphasize one warning word inside a sentence; (3) build an automatic table of contents; (4) change all Heading 1 paragraphs from 14 pt to 15 pt. Then apply Title and Heading 1 to a three-line practice document and confirm that the Navigation Pane recognizes the headings.

3 SESSION Producing a Formatted Safety Inspection Report in Word

The lab integrates hardware, operating-system, file-management, and document-production skills into one professional task. Work through the practicals in order because each stage creates the foundation for the next: save, structure, format, insert evidence, review and verify.

This lab combines the lesson's ideas. You will use the operating system to start Word and manage a file, use storage correctly, work with structured information, insert a table and image, save regularly, and apply troubleshooting if something goes wrong.

This workflow previews the practical sequence. Each box produces a visible result that becomes the starting point for the next stage, so do not skip directly to formatting or image insertion. The checkpoint question asks whether another person can find, read, and understand the work after every stage.

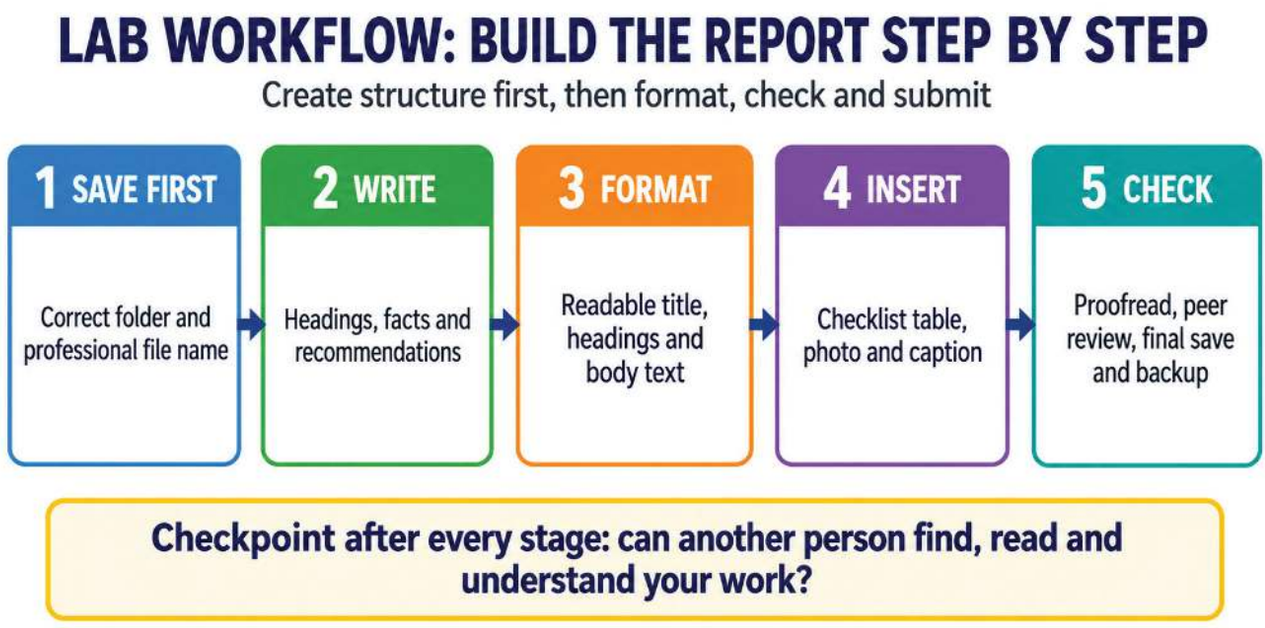


Figure 3.8 : The laboratory workflow for building the report.

Lab Purpose and Professional Standard

Before learning Word commands, identify the purpose and reader of the report. A supervisor needs accurate facts, visible status, and safe recommended actions, so the document must be easy to locate, scan, understand, and verify.

A safety inspection report is an official technical record of what was inspected, what was observed, and what action is required. It supports safe decisions, creates evidence of completed work, and helps maintenance teams prioritize repairs. A professional report must be accurate, objective, readable, traceable, and saved in the correct location.

Key Lab Definitions

These definitions describe the language of the report and the Word features used to build it. Read them before the practicals so that terms such as finding, status, caption, cell, and proofreading have a clear technical meaning.

The lab vocabulary table links each technical-writing or Word-processing term to the hydraulic-press task. Refer to it whenever an instruction uses a word that is new.

Term	Definition	Example in this lab
Finding	A factual observation or measurement, without guessing.	"A small oil leak was observed near the main cylinder."
Status	A short judgement against a check requirement.	OK, Not OK, Not Checked or Not Applicable.
Recommended action	A clear next step based on the finding.	"Isolate the press and arrange inspection of the cylinder seal."
Formatting	How text and objects are organised visually to guide the reader.	Title hierarchy, headings, font size, spacing and table style.
Cell/row/ column	The parts of a table.	A status is entered in a cell at the meeting of a row and column.
Caption	A short label that identifies an image or table.	"Figure 1: Oil leak near the main cylinder."
Proofreading	Checking language, facts, numbers and layout before submission.	Confirm that "180 bar" is copied correctly.

Equipment and Files Required

Preparation reduces wasted lab time. Check the list before opening Word so that the correct folder tree, situation card and authorised image are available and no student needs to search online or use unapproved data.

- A computer with Microsoft Word or an approved compatible word processor.
- Your DT_Year1_YourName folder tree from Lesson 2.
- The teacher's workshop situation card and an authorised image of the hydraulic press.
- An approved USB or school cloud location if the backup extension is completed.

Lab safety and data rule

Do not photograph people, confidential displays or real incidents without permission. Use only the training image supplied by the teacher. Never troubleshoot a live machine through this lab.

Workshop Situation: Hydraulic Press 3

The situation is the evidence source for the whole report. Every status, finding, and recommendation must be traceable to these facts; do not add repairs, measurements, or safety decisions that the scenario does not provide, but you can add a recommended actions list that also shows exactly what must happen next.

You are the duty trainee technician in the training workshop. During a supervised visual and functional check of Hydraulic Press 3, the following information was recorded:

- A small oil leak is visible near the main cylinder.
- The emergency stop button operates correctly during the authorised test.
- One warning label is damaged and difficult to read.
- The pressure gauge reads 180 bar, which is within the stated training range.
- No repair has been carried out. The press remains subject to the supervisor's decision.

Lab Working Routine: Prepare - Perform - Check

Each practical uses the same learning routine. First, prepare by reading the purpose, definitions, and success criteria. Next, perform only the demonstrated actions. Finally, check the visible result before moving on. This routine reduces memory load and makes errors easier to correct.

Pair roles can improve accuracy: the operator uses the keyboard and mouse, the navigator reads the instructions and checks names or paths, and both students confirm the checkpoint. Change roles between practicals so every student performs the key actions.

Practical 1: Create the File and Save Before Typing

Practical 1 establishes the file's identity and location before content is created. The professional habit is to confirm both the path and the file name early, then save regularly so that the document moves from temporary RAM to persistent storage.

Why this practical matters: Word can create a blank document immediately, but a blank document has no professional name or confirmed location. Saving first creates the digital container for your work and reduces the risk of losing it.

Definitions: Save updates the current file. Save As creates a new file name, location, or format. A file path shows exactly where the file is stored.

Before you begin, check that the Lesson 2 folder tree exists and that you can identify the Safety folder in File Explorer. Do not begin typing into an unsaved document because the location and version would still be uncertain.

Successful evidence for this stage is simple and visible: the file is in the correct path, the professional name appears in the title bar, and the document can be found again from File Explorer.

1. Open Microsoft Word and choose Blank document.

2. Press Ctrl + S before typing.

3. Navigate to DT_Year1_YourName → 01_Reports → Safety.

4. Name the file using today's date: YYYY-MM-DD_Press3_SafetyReport_v01.docx.

5. Select Save. Read the file name in the Word title bar and the folder path in the Save dialog.

6. Tell your partner the full folder path. Ask them to confirm that it is the Safety folder.

Checkpoint 1

The teacher or partner can locate your file in under 20 seconds. The file name includes date, machine, document type and version.

Practical 2: Build the Report Structure

A predictable structure helps a busy reader find information quickly. In this practical, headings are used as signposts, findings remain factual, and recommendations state what should happen next without claiming that unauthorised work has been completed.

Why this practical matters: structure comes before decoration. Readers should know where to find general information, inspection status, detailed findings, actions, and the responsible person. A consistent structure also makes reports easier to compare.

Before you begin, read the hydraulic-press situation once without typing. Identify the factual observations, the status of each check, and the actions that still require a supervisor's decision.

A successful structure allows a reader to predict where information will appear. The five headings should be complete and in order, and every finding or recommendation should be placed under the correct heading.

The figure shows the five sections in the order a reader needs them. General information identifies the record, the checklist summarizes status, findings present objective evidence, recommended actions state the next steps, and the signature section records responsibility.

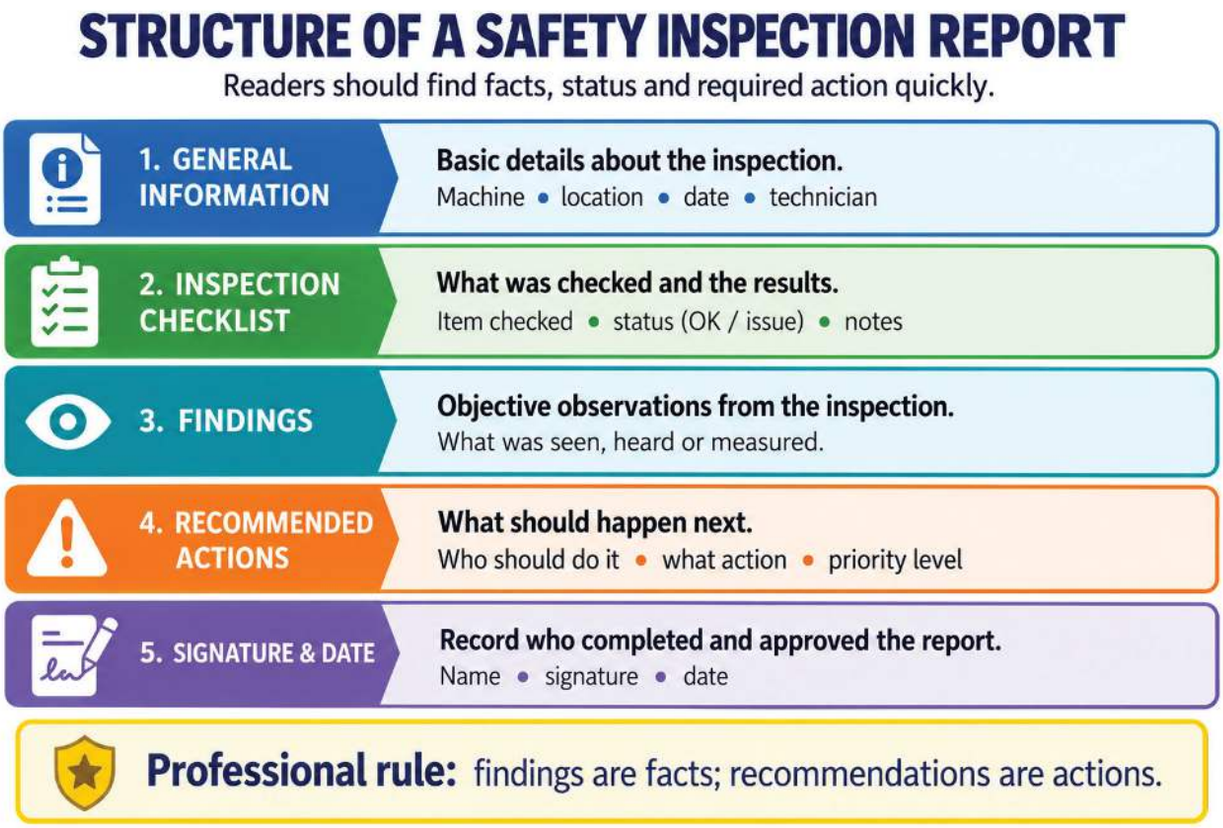


Figure 3.9: Required structure of the safety inspection report.

1. Type the main title: Safety Inspection Report.
2. Type five section headings: 1. General Information; 2. Inspection Checklist; 3. Findings; 4. Recommended Actions; 5. Technician Signature.
3. Under General Information, type separate lines for Machine, Location, Date, Technician, and Report Version.
4. Under Findings, write four factual sentences based on the situation. Use past tense and objective language.
5. Under Recommended Actions, write two or three safe next steps. Do not claim that a repair has been completed.
6. Under Technician Signature, type your name and leave space for supervisor review.
7. Press Ctrl + S.

Factual Language: Objective vs Unclear

Technical language should allow another person to understand what was seen, tested, or measured. The comparison shows how objective wording improves traceability and reduces disagreement about what the report actually proves.

The language table compares vague or unsupported wording with objective alternatives. The

final column explains how the improved sentence supports accuracy, traceability, or safe decision-making.

Avoid	Use instead	Why
"The press is dangerous."	"A small oil leak was observed near the main cylinder."	The second sentence describes evidence rather than making an unsupported judgement.
"The label is bad."	"One warning label is damaged and difficult to read."	It identifies the problem clearly.
"Fix it soon."	"Replace the damaged warning label before the press returns to normal service."	It gives a specific action and timing condition.

Practical 3: Format for Readability

Formatting creates a visual hierarchy that guides the reader from title to sections and then to supporting detail. Use consistent choices rather than decoration so the report remains professional on screen, in print, and when exported to PDF.

Why this practical matters: formatting is not decoration. It creates visual hierarchy. The title shows the document purpose, headings divide the content, and body text presents details without distracting the reader.

Before you begin, finish the report content. Formatting incomplete text repeatedly wastes time and increases inconsistency. Work from the main title to headings and then to body text.

A successful layout uses a clear visual hierarchy, consistent body text, sufficient white space, and no decorative effect that competes with technical information.

Use this figure as a map rather than trying to memorize every button. First select the correct ribbon tab, then select the text or object you want to change. Headings guide the reader, and frequent saving protects the document while you work.

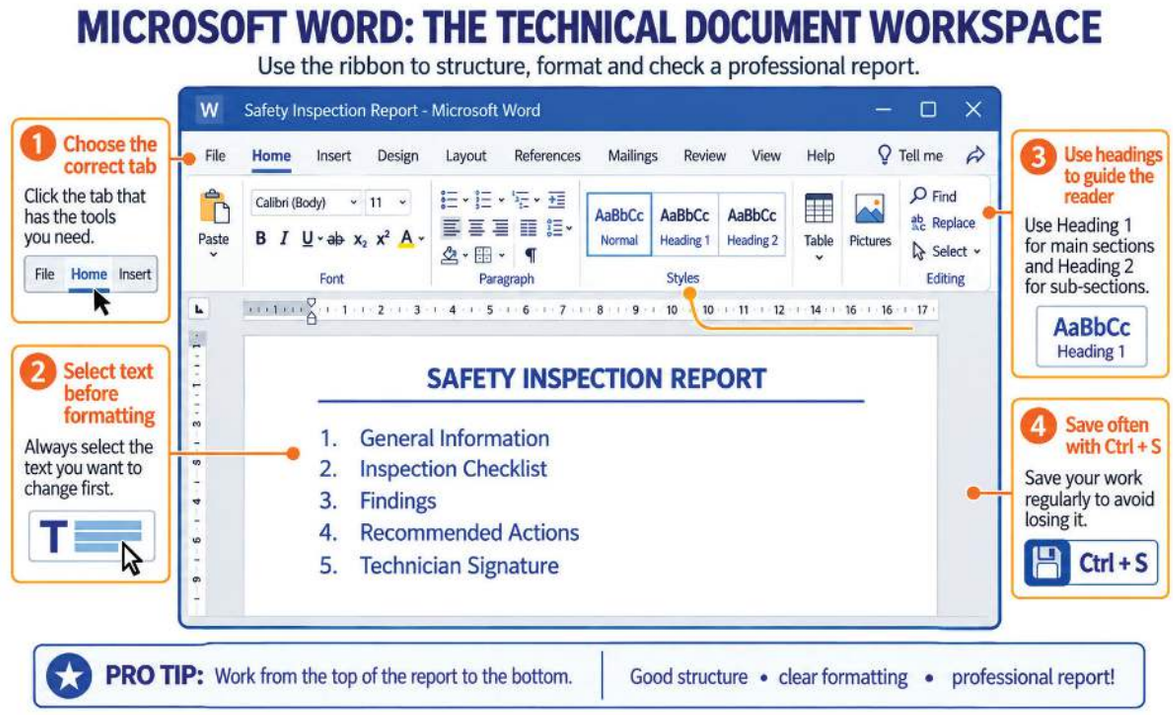


Figure 3.10: Simplified Word workspace and the tools used in this lab.

1. Select the main title. Apply bold, 18 pt, and centre alignment.

2. Select each numbered section heading. Apply bold, 14 pt, and a dark blue font colour.

3. Select the body text. Use an easy-to-read font such as Aptos, Calibri, or Arial at 11-12 pt.

4. Use left alignment for body text and consistent spacing after paragraphs.

5. Insert the date through Insert → Date & Time if this option is available in your version.

6. Insert a page number through Insert → Page Number. The exact menu may vary slightly by Word version.

7. Press Ctrl + S and check that the file is still v01 in the correct folder.

Formatting rule

Use no more than one main body font and one consistent heading system. Too many colours, fonts or effects make a technical report harder to scan.

Practical 4: Insert and Complete the Inspection Checklist Table

A table makes repeated information easier to compare. This practical develops both Word skills and information discipline because every row must agree with the written findings and every value must be copied accurately.

Why this practical matters: a supervisor may need to scan many checks quickly. A table places each inspection item, status, and note in a predictable position. The header row names the information in each column.

Definitions: a column runs vertically, a row runs horizontally, and a cell is the box where a row and column meet. The header row explains what each column contains.

Before you begin, compare the four checklist items with the written findings. The table is a summary, so it must not introduce a different status, measurement, or wording that changes the meaning.

A successful table can be scanned in a few seconds: the header row is clear, the notes column is wide enough, all text remains inside the page margins, and the numerical reading is copied exactly.

1. Place the cursor below 2. Inspection Checklist.

2. Select Insert → Table and create 3 columns × 5 rows.

3. In the header row, type: Item Checked | Status | Notes.

4. Complete four data rows using the information in the situation card.

5. Make the header row bold and apply a simple table style with high contrast.

6. Adjust column widths so the notes column is widest. Do not make the table wider than the page.

7. Press Ctrl + S.

Checklist Data to Enter

The table below is the approved data set for the checklist. Use it to check spelling, status, and numerical accuracy before formatting the table or moving to the photograph.

Item Checked	Status	Notes
Main cylinder area	Not OK	Small oil leak observed near the main cylinder.
Emergency stop	OK	Authorised test completed; button operated correctly.
Warning labels	Not OK	One label damaged and difficult to read.
Pressure gauge	OK	Reading recorded as 180 bar; within the stated training range.

Practical 5: Insert a Photo Without Covering Text

Line with Text treats it like a large character in the document, which prevents unexpected overlap and movement.

Why this practical matters: photographs can provide valuable visual evidence, but a large or floating image can cover text, move unexpectedly, or make a report difficult to print. In this course,

keep images In Line with Text unless the teacher gives another instruction.

Before you begin, confirm that you are using the authorised local image supplied for training. The photograph supports the report; it does not replace the written finding or provide permission to make a diagnosis.

A successful image remains proportional, stays within the margins, has an informative caption, and does not cover or split the surrounding text.

Compare the two sides of the figure. The good layout keeps the table, image and caption inside the normal text flow, so they move together and remain readable. The unsafe layout allows a floating image to overlap text and makes the report unstable when content is edited.

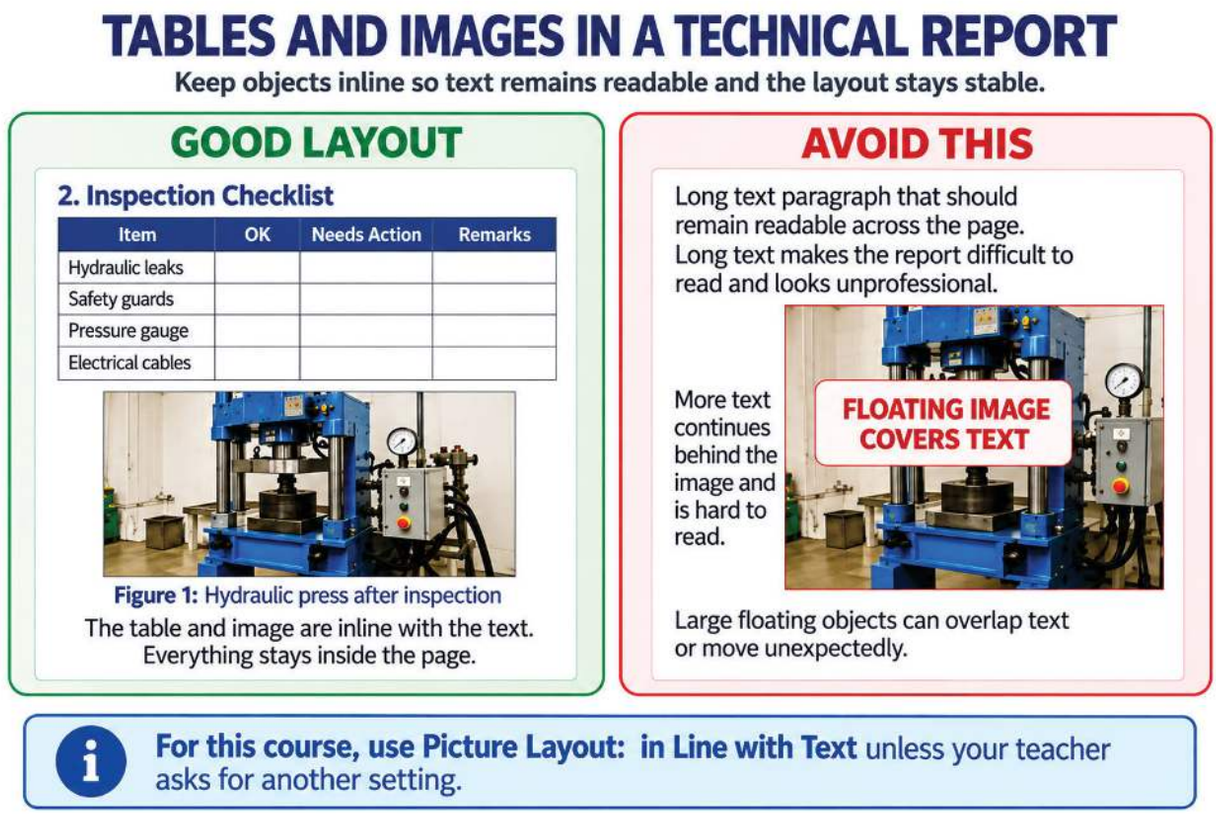


Figure 3.11: Stable inline image layout compared with a floating image that covers text.

1. Place the cursor under the Findings section after the factual sentences.

2. Select Insert → Pictures and choose the authorised training photo from 03_Photos.

3. Select the image and choose Picture Layout or Wrap Text → In Line with Text.

4. Resize using a corner handle so the image stays proportional. Keep it inside the page margins.

5. centre the image if appropriate.

6. Type a caption underneath: Figure 1: Hydraulic Press 3 inspection image.

7. Add alternative text if your Word version provides the command: "Training image of Hydraulic Press 3 used in the inspection report."

8. Press Ctrl + S.

Practical 6: Quality Check, Peer Review and Final Save

The final practical treats checking as part of production, not an optional extra. The reviewer compares the report with requirements, the writer corrects meaningful issues, and the final reopen test proves that the saved file can actually be used.

Why practical 6 matters: technical documents should be checked before submission. Peer review is a form of quality assurance. The reviewer checks the document against requirements; the author decides and applies the corrections.

Before reviewing, save the current file and agree that feedback will refer to the task requirements rather than personal preference. Read the report as if you are the supervisor receiving it for the first time.

Successful verification requires more than seeing the document on screen. The final file must be saved, closed, located through File Explorer, and reopened with the text, table, image, and caption still in the correct positions.

A. Content Quality Check

This checklist focuses on meaning: whether the report is complete, factual, internally consistent and safe. A visually attractive document should not pass if its content contains unsupported claims.

Use this checklist one row at a time rather than giving a general impression of the report. A "Not yet" response identifies a specific correction that must be completed before submission.

Check	Yes / Not yet
All findings match the situation card and contain no invented repairs or measurements.	
The checklist status agrees with the written findings.	
Recommended actions are clear, safe and within trainee scope.	
Machine, date, technician and version are present.	

B. Layout and File Quality Check

This checklist focuses on usability: whether the file is correctly named and stored, the table and image are stable, and the document can be reopened without missing or overlapping content.

Use this checklist one row at a time rather than giving a general impression of the report. A "Not yet" response identifies a specific correction that must be completed before submission.

Check	Yes / Not yet
The file is saved in 01_Reports\Safety with a professional v01 name.	
The title and section headings are clear and consistent.	
The table fits on the page and the header row is easy to read.	
The image is clear, proportional and does not cover text.	
Spelling, numbers and units have been checked.	
The final file opens after saving.	

1. Swap reports with a partner. Give one specific strength and one specific improvement.

2. Return to your own document and make the required corrections.

3. Use Save As to create v02 only if you make a meaningful revision after feedback.

4. Close Word, locate the file through File Explorer, and reopen it to verify that the saved copy works.

5. Optional extension: export a PDF copy and place it in the same Safety folder.

6. Optional extension: copy the completed folder to the approved backup location.

Lesson/Session Summary

This summary gathers the key ideas of Lesson 3 and its Word-processing practical. Use it as a quick revision map before you attempt the assessment.

- Word processing produces clear, consistent technical documents – the CREATE stage of the digital-literacy cycle.
- Styles (Title, Heading 1, Heading 2) build structure and let Word generate a table of contents automatically; they are not the same as bold text.
- A professional report has a predictable skeleton: title, headings, body, tables and a controlled file name.
- Inline images stay stable in the layout; a professional inserts and captions figures rather than floating them freely.
- Document control – a clear version and file name – makes the current approved copy easy to identify.
- The safety inspection report is saved with a professional name in the correct folder, ready to share.

Class Discussion

These questions require explanation rather than one-word answers. Support each response with an example from your report, a formatting reason, or evidence from the lab.

1. Why are heading styles better than bold text for a long technical document?

2. What can Word build automatically once heading styles are applied?

3. Why does a professional use inline images instead of freely floating ones?

4. How does a clear version and file name help a maintenance team?

5. Which part of the report was hardest to format correctly, and why?

Assessment

The assessment checks knowledge, interpretation, and practical judgement. Read each question carefully and use the terminology from the lesson.

Part A: Multiple Choice

Choose the best answer. Eliminate options that are unsafe, incomplete, or inconsistent with the definitions. Each item is worth one mark.

1. A table of contents can be generated automatically from:

(a) bold text **(b)** heading styles **(c)** coloured text **(d)** manual page numbers

2. The correct way to make a section title in a professional document is to:

(a) type it bigger **(b)** apply a heading style **(c)** underline it **(d)** centre it

3. An inline image is preferred because it:

(a) looks nicer **(b)** stays stable in the layout

(c) prints faster **(d)** needs no caption

4. A professional file name for a report should include:

(a) only the word final **(b)** date, asset, type and version

(c) the author's nickname **(d)** random numbers

5. Word processing mainly supports which stage of the cycle?

(a) FIND **(b)** EVALUATE **(c)** CREATE **(d)** SHARE

Part B: True or False

A sentence may contain familiar words but still be false because one important condition is missing.

1. Applying a heading style is the same as making text bold. _____

2. A table of contents can update automatically when headings change. _____

3. A clear file name and version help identify the current approved report. _____

Part C: Short Answer

Use concise technical sentences that answer every part of the question.

1. Give two elements that make a Word document professional rather than plain.

2. Write a professional file name for a safety inspection report on Press 2, produced today, first version.

Self-Assessment Checklist

I can	Yes	Not yet
Apply Title and Heading styles to structure a document		
Generate a table of contents automatically		
Insert and caption an inline image correctly		
Build a professional table that fits the page		
Save the report with a professional file name and version		

Reflect and Connect (At Home)

Write two or three sentences: think of a document you have made before (a school report, a form, a letter). How would applying heading styles and a clear file name make it more professional and easier for someone else to use?

Glossary قاموس المصطلحات

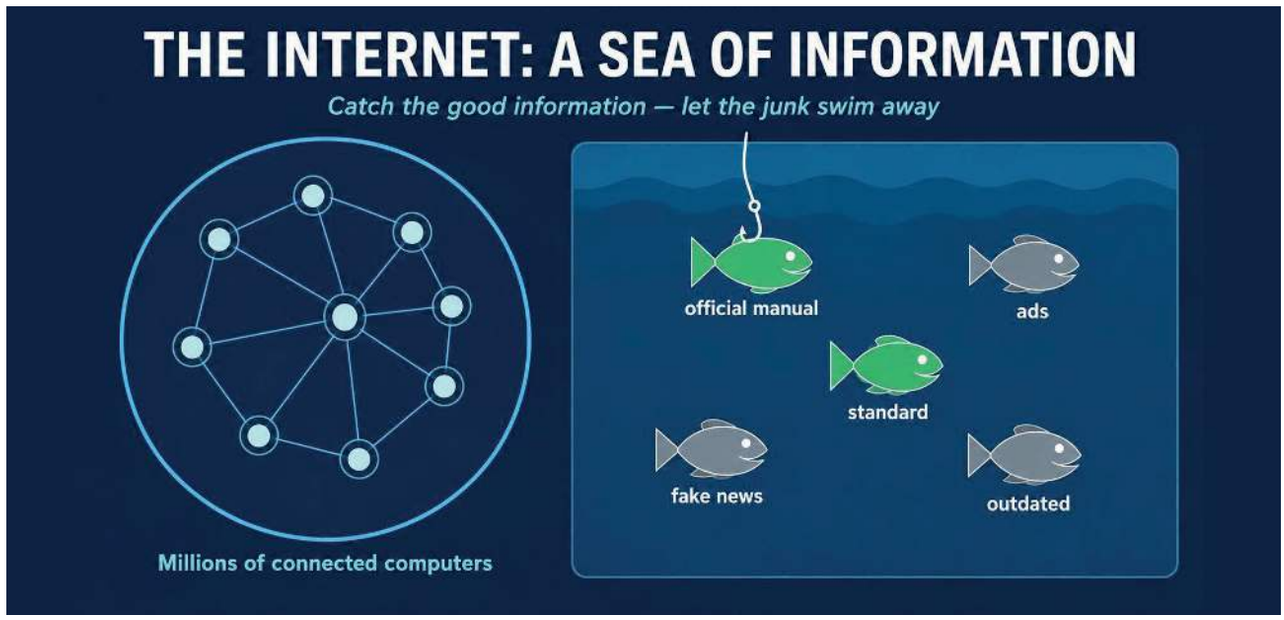
English Term	Meaning	المصطلح بالعربية
Word processing	Creating and formatting text documents on a computer	معالجة النصوص
Style	A saved set of formatting applied with one click	نمط / تنسيق
Heading style	A style that marks a section title and builds structure	نمط العنوان
Table of contents	An automatic list of headings and their pages	جدول المحتويات
Inline image	A picture fixed within the text flow	صورة مضمّنة
Version	A numbered state of a document	إصدار / نسخة
Document control	Managing versions and names of a document	ضبط الوثائق

Abbreviations

The abbreviation table gives the complete wording behind shortened technical terms.

Abbreviation	Full Form
ToC	Table of Contents
docx	Microsoft Word document file format
PDF	Portable Document Format
v01	Version 01 of a document

LESSON 4 Internet Foundations & Safe, Effective Online Search



The internet is a vast sea of information. This lesson trains you to catch the good information and let the junk swim away.

This lesson has one big goal: to turn you from someone who simply uses the internet into someone who commands it. By the end, you will understand what happens when you open a web page, search with the precision of a professional technician, judge whether a website deserves your trust, diagnose common connection problems, and use built-in, read-only network tools to trace and report a workshop connection safely.

Learning Outcomes

By the end of this lesson, you will be able to:

- explain how the internet works (IP address, DNS, server, router).
- identify the parts of a URL and recognize HTTPS security.
- apply advanced search operators (quotation marks, minus, site:, filetype:, OR).
- evaluate website credibility using WHO – WHAT – WHEN – WHY (4-w).
- explain what a datasheet is, name its five key sections, and decode an IP protection code digit by digit.
- complete a professional motor-datasheet search task by searching, evaluating, saving, and documenting specifications from a trustworthy source.

Introduction

Nothing in this unit stands alone. In Lesson 1, you learned Understand → Find → Try → Check. In Lesson 2, you applied lesson 1 concepts and methods in computer/file management system. In Lesson 3, you created a safety inspection report. In this Lesson 4 now connects internet foundations, precise search, source evaluation, and connection first aid through **a motor datasheet search/evaluation.**

WARM-UP

True or Fake?

Your teacher will show you TWO short technology 'news' statements — one is true, one is invented. Working in pairs:

Step 1: Vote: Decide which statement is fake. Raise your hand for A or B.

Step 2: Discuss: HoW could you actually check which is true? What clue or source would you look for?

Step 3: Remember: Write down your checking method. You will use the very same instinct later today when you learn the 4-W credibility test.

Topic 1 How the Internet Works

Before you can search the internet well or fix it when it breaks, it helps to picture what the internet actually is. You do not need to be an engineer; you need a clear mental model. This topic builds that model with

one simple story (the one-second journey of a web page) and four easy ideas. Everything else in networking, right up to the smart factories you will study in Year 3, is built on these same foundations.

1.1 Words That Are Often Confused

Students often use the words internet, web, browser and search engine as though they mean the same thing. They are related, but each has a different job. Understanding the distinction makes later explanations and troubleshooting much clearer. Read the table from left to right, then identify which item you use when you type a URL and which item you use when you type a search question.

Term	Meaning in simple words	Example
Internet	he worldwide network that connects devices and networks.	School network connected through an ISP to other networks.
Web / World Wide Web	A service on the internet made of linked pages and resources.	A manufacturer website containing motor datasheets.
Browser	The application used to request, display and interact with web pages.	Microsoft Edge, Google Chrome, Mozilla Firefox or Safari.
Search engine	A service that indexes online content and helps users find it.	Google, Bing or another approved search service.

Term	Meaning in simple words	Example
Website	A collection of related pages under one domain.	abb.com or skf.com.
Web page	One specific document or screen within a website.	A motor product page or safety manual page.
ISP	The organisation that connects a home, school, or company network to the wider internet.	A local telecommunications or internet service provider.

Example: when you open a browser and type a manufacturer URL, the browser requests a page directly. When you type "1.5 kW motor datasheet" into a search engine, the search service helps you find possible pages. The browser displays both, but the browser and search engine are not the same thing.

1.2 The One-Second Miracle

Let us start with something you do every day without thinking. You open a browser, type **abb.com**, and press Enter. In less than one second, a full page of text and images appears on your screen. Pause and consider what just happened. Your request left your laptop, crossed your school's Wi-Fi, travelled through cables, some of them running along the bottom of the ocean between continents — reached a powerful computer possibly on the other side of the planet, and that computer sent the whole page back to you. All of that, faster than you can blink. It sounds like magic, but it is not. It is four simple ideas working together, and once you know them, the 'magic' becomes something you understand and can rely on.

DID YOU KNOW?

Real-World Scale

The undersea cables that carry the internet between continents are real, physical fibre-optic lines — some more than 20,000 kilometres long — resting on the ocean floor. When you load a page hosted abroad, your data very likely travelled through one of them and back in the time it took you to lift your finger off the Enter key.

Request, Response and Data Packets

A web page does not normally travel as one giant object. Your device sends a request, and the returned text, images and other resources are divided into smaller units of data that networks can route efficiently.

At first-year level, you only need three ideas: the request asks, the response answers, and packets carry the data across the network.

Term	Meaning	Workshop-style analogy
Request	A message asking a server for a page, file, or service.	A technician submits a parts request.
Response	The server's answer, including the requested data or an error message.	The store issues the part or explains why it is unavailable.
Packet	A small labelled unit of data sent across a network.	A large delivery divided into numbered boxes and reassembled at the destination.
Protocol	An agreed set of rules for formatting and exchanging data.	A standard form that both departments understand.

Packets may travel through several devices and routes, but the receiving system reassembles them into the page or file you requested. This is why network communication can continue even when the route is complex.

1.3 The Four Key Ideas

The whole journey depends on four building blocks: the **IP address**, the **DNS**, the **router**, and the **server**. Rather than memorizing technical definitions, we will attach each one to a familiar real-world analogy. Before you read the table, here is the single most important definition to hold in your mind:



Definition - The Internet

The **internet** is millions of computers connected together all over the world by cables, fibre-optic lines and wireless links, all agreeing to speak the same 'language' — a set of rules called **protocols** — so that they can exchange data reliably with one another.

The table below introduces the four key ideas. Read each row as a pair: the technical term on the left, and on the right the everyday **analogy** that makes it stick. The analogy is not just a memory trick — it genuinely captures what each part does. As you read, imagine sending a physical letter through the post, because that is exactly how these four pieces cooperate.

Term	What it really is	Everyday analogy
IP address	The unique number of every device on the network, e.g. 192.168.1.10	Like the **postal address** of a house — it tells data exactly where to go
DNS	The service that converts a name (google.com) into an IP address	Like a **phone book**: you know the name, it gives you the number
Router	A device that directs data between networks	Like a **post office** sorting letters and sending each toward its destination
Server	A powerful computer that stores websites and files and answers requests	Like a **warehouse** that sends you exactly what you ask for

Notice how the analogies fit together. Every device has an address (IP). When you know a website's

name but not its number, the phone book (DNS) looks it up for you. The post office (router) then sorts and forwards your request, and the warehouse (server) sends back what you asked for. Keep this picture ready — it explains the journey diagram you are about to meet.

1.4 The Journey of a Web Request

Now we can watch the whole trip happen, step by step. The diagram below traces a single web request from the moment you press Enter to the moment the page appears. Read the five numbered stages from left to right, then follow the green arrow underneath, which shows the finished page travelling all the way back to your screen. The small red badge is a reminder that this entire round trip usually takes less than one second.

Figure 4.1 — The Journey of a Web Request

You type a web address and press Enter. Follow the numbered path — it all happens in under one second.



Figure 4.1.The journey of a web request: laptop → router → ISP → DNS lookup → server, and back again.

Walk through 4.1 in your own words: (1) your **laptop** creates the request when you type the address; (2) your **Wi-Fi router** sends that request out of the building; (3) your **Internet Service Provider** (ISP) carries it onto the wider internet; (4) the **DNS** — our phone book — turns the website's name into its numeric IP address so the request knows where to go; and (5) the **web server** at that address sends the finished page back along the same path. If you can narrate these five stages to a classmate, you understand how the internet works.

Why a Connection Can Feel Slow

A slow page does not automatically mean the computer is faulty. The delay may be caused by the capacity of the connection, the time taken for data to travel, heavy traffic, a weak wireless signal, or a slow server. The table separates four commonly confused ideas. Use it to describe what you observe instead of saying only "the internet is bad."

Factor	What it means	What a student may notice
Bandwidth	How much data the connection can carry in a period of time.	Large files or video are slow when many users share a limited connection.
Latency	The delay before data begins to return.	A page pauses before responding even though the final download is small.
Congestion	Too many users or devices competing for network capacity.	The network is slower at a busy time than early in the day.
Signal quality	The strength and stability of a wireless connection.	The Wi-Fi icon is weak or the connection drops near machinery or far from the access point.
Server response	How quickly the remote service processes the request.	One website is slow while other websites work normally.

A professional diagnosis begins by comparing: Is one page slow or every page? Is one device affected or the whole room? Does moving closer to the access point change the result? These observations narrow the problem before any action is taken.

1.5 Reading a Web Address — and the Padlock

Every web address, properly called a URL (Uniform Resource Locator), is made of readable parts. Consider https://www.abb.com/motors. The https at the front means the connection between your browser and that domain uses encryption. The domain www.abb.com identifies the site you are contacting, while/ motors is the path to a particular location within the site. Browsers may display a padlock or another security-controls icon, but the important check is the HTTPS connection and the exact domain name. HTTPS protects data in transit; it does not prove that every statement on the site is accurate or that the site deserves your trust.

Figure 4.2 — The Parts of a URL (Web Address)

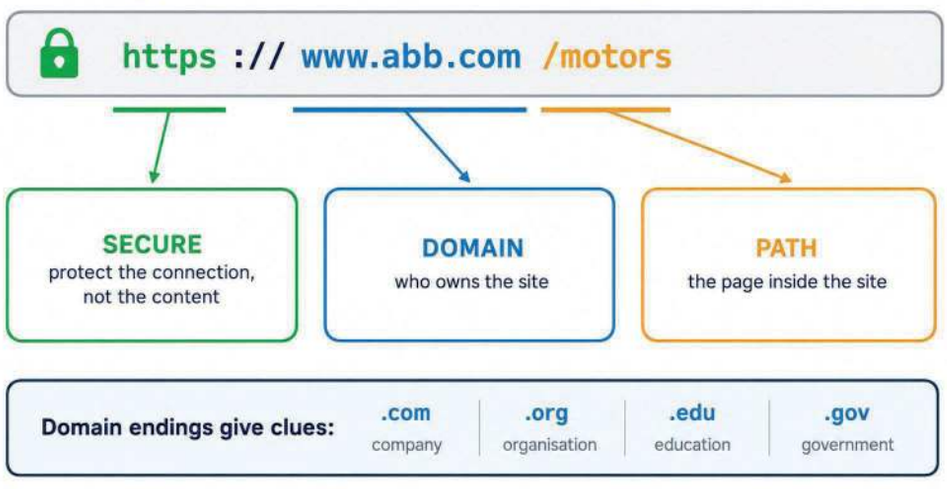


Figure 4.2.The parts of a URL. Learn to read 'who owns it' and 'is it secure?'before you click or type anything sensitive.

Figure 4.2 also shows the domain ending, sometimes called a top-level domain. Endings such as .com, .org, .edu and .gov can provide context, but they are not a complete credibility test. Many .com and .org names can be registered by almost anyone, and country rules for government or educational domains differ. Read the full domain carefully and then apply the 4-W test, technical-fit checks and cross-checking before relying on the content.

PADLOCK / HTTPS RULE

- HTTPS means the connection to the displayed domain is encrypted. Before entering a password or sensitive information, check HTTPS and read the full domain carefully. Then ask a second question: is this the correct and trustworthy site? Encryption protects the journey; source evaluation checks the content.

Mechatronics Scenario — Wi-Fi on the Factory Floor

- Modern factories run on industrial Wi-Fi networks. A technician's tablet connects to the maintenance server to open work orders; smart temperature sensors send readings across the same network to a live dashboard; and automated machines report their status in real time. Every one of those devices has an IP address and relies on routers and servers — exactly the four ideas you have just learned. Understanding how today's web page reaches your laptop is the very same understanding you will need to work with tomorrow's connected machines. This is a core **Industry 4.0** skill, and you are starting it now.

Pop-Up Quiz 1

1 **DNS converts**:

a. images to text
b. website names to IP addresses
c. Word to PDF
d. Wi-Fi to cables

2 **The padlock and 'https' in the browser mean:**

a. the site is free
b. the connection is encrypted / secure
c. the site has videos
d. the site is fast

Write your answers here:

1: .....

2: .....

1.6 Checkpoint: Connection Security Is Not Source Credibility

A secure connection and a credible source solve different problems. HTTPS reduces the risk of someone reading or changing data while it travels. Credibility checks whether the publisher, evidence, date and purpose deserve trust. A misleading or fraudulent site can still use HTTPS, so technicians must perform both checks.

THINK Quick scenario

A result uses HTTPS and displays a professional design, but it has no author, no date, and gives a torque value for a different motor model. Decision: do not use the value. The connection may be encrypted, but the source and technical fit are inadequate.

Topic 2: Search Like a Professional: Advanced Operators

You already know how to type words into a search box. This topic is about the difference between typing words and giving commands. Professionals do not scroll through more results than everyone else — they get better results in the first place by telling the search engine precisely what to include and what to ignore. These commands are called **search operators**, and just five of them will transform how quickly you find reliable technical information.

2.1 Plan the Search Before You Type

Good searching starts before the search box. A technician first defines the information need, identifies the exact asset and decides what kind of evidence is acceptable. This two-minute plan prevents vague searches and reduces the risk of selecting information for the wrong model. The table gives a reusable planning checklist.

Planning question	What to record	Hydraulic-press example
What decision or task must the information support?	Inspection, maintenance, troubleshooting, compliance or purchasing.	Confirm the approved inspection interval.
Which asset is involved?	Manufacturer, model, serial family and component.	Hydraulic Press 3, model HP-250.
What document type is needed?	Manual, datasheet, standard, checklist, drawing or service bulletin.	Manufacturer maintenance manual in PDF.
Which limits will improve the search?	Official domain, file type, exact phrase, excluded words, language or date.	site:manufacturer.example filetype:pdf "HP-250" maintenance interval.
What evidence must be saved?	URL, publisher, document title, revision/date, and relevant page.	Record the manual revision and page number used.

Professional habit: write the asset model exactly as it appears on the nameplate or approved record. One missing letter or digit can lead to a different machine, rating, or procedure.

2.2 From Scrolling to Precision

Here is the difference in a single example. In Lesson 3, you improved a search by adding more specific keywords — a good first step. Now go further. Compare two searches for a machine manual. Searching for **hydraulic press manual** returns a messy mix of online shops, videos, blogs, and advertisements. But searching for **hydraulic press manual filetype:pdf** returns almost nothing except actual PDF manuals. One extra command completely changed the result. That command, filetype: is a search operator, and it is only one of five you will learn today.



Definition - Search Operator

A **search operator** is a special symbol or word that you add to a search to control it — telling the search engine to lock a phrase together, remove unwanted results, look inside one website only, or return one type of file. Operators are typed directly in the search box alongside your normal words.

2.3 The Operator Toolbox

The table below is your toolbox. Each row shows the operator, what it does, and a realistic technician example you could type today. Read across each row and picture the situation in which you would reach for that tool. After the table, you will find a matching diagram with a memory hook for each operator, so you never forget which is which.

Operator	What it does	Example you can type
"quotation marks"	Finds the EXACT phrase, in that exact order	"three-phase induction motor"
minus (–)	Removes unwanted results containing a word	bearing installation –video
site:	Searches only inside one chosen website	site:abb.com 1.5kW motor
filetype:pdf	Finds only PDF documents — manuals, datasheets	hydraulic press manual filetype:pdf
OR	Accepts either one word or the other	motor OR gearbox maintenance checklist

To make these five operators stick, attach each to a simple picture, shown in the diagram below. Quotation marks are **handcuffs** — they lock the words together so they cannot drift apart. The minus sign is a **broom**
— it sweeps unwanted junk out of your results. The site: operator is a **flashlight** shone into one single building. The **filetype**: operator is a **fishing net** that catches only one kind of fish. And **OR** is a **pair of doors** — either one lets the result through.

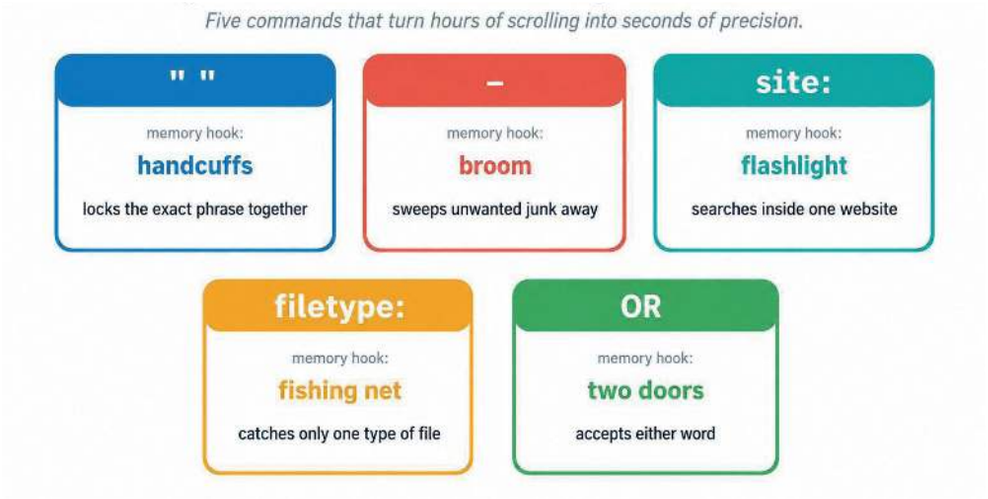


Figure 4.3. The five search operators, each paired with a memory hook. Invent a hand gesture for each one to remember it even faster.

DID YOU KNOW?

Search engines process billions of searches every single day, and studies repeatedly show that most people never look past the very first page of results. Professionals do not scroll further — they search **smarter** with operators, so the answer they need appears at the top. The reference below is the official guide to these operators.

Reference — Google Search Help (official operator guide

2.4 Combining Operators for Real Technical Tasks

Operators are most powerful when they are combined with precise technical words. The examples below show how the same toolbox supports different goals. Read the purpose column first, then identify which part of the search phrase performs each job.

Task	Possible search phrase	Why it is efficient
Find an official PDF manual.	site:abb.com "M3AA 90L-4" manual filetype:pdf.	Limits the result to one domain, one exact model and PDF documents.
Find a checklist without shopping pages.	proximity sensor inspection checklist -shop -price.	Uses the required topic and excludes common commercial results.
Find either of two accepted terms.	"lockout tagout" OR "energy isolation" procedure.	Accepts either recognised phrase while keeping each phrase together.
Find a recent service notice.	site:manufacturer.example service bulletin model-name after: 2024-01-01.	Limits the domain and adds a date boundary where the search service supports it.

The diagram below presents a complete professional search process. Follow it from left to right: define the need, build a query, scan results, verify the source and only then use and record the information. The final step is essential because another technician should be able to trace the evidence you used.

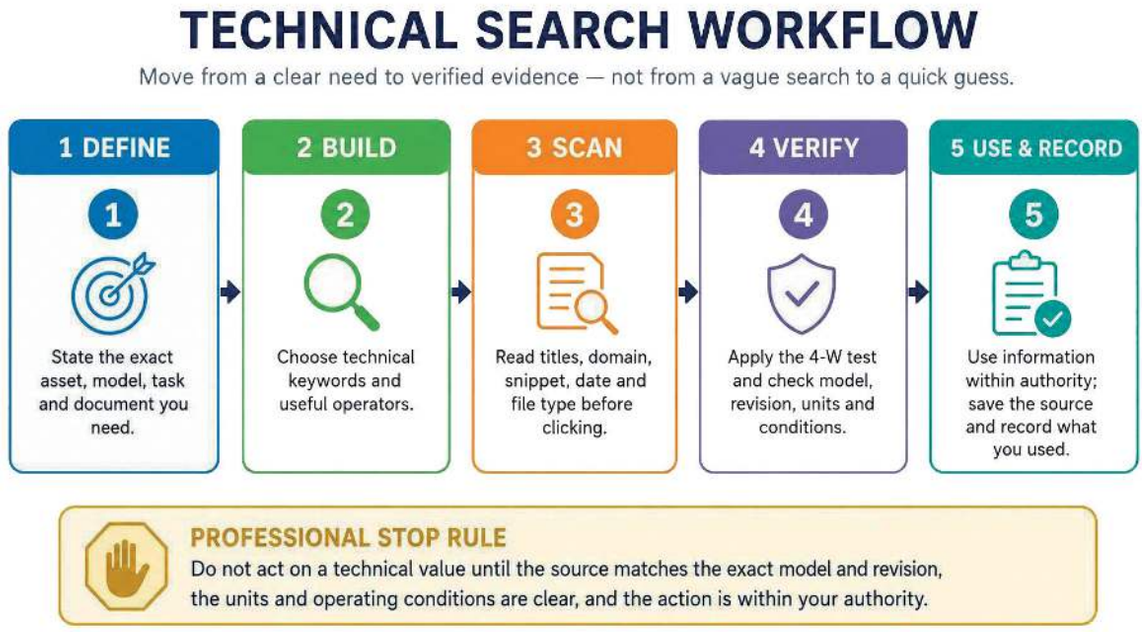


Figure 4.4. Technical search workflow: define → build → scan → verify → use and record.

2.5 Read the Results Page Before You Click

A results page provides useful clues before a site opens. Scanning these clues saves time and reduces unnecessary exposure to unreliable or irrelevant pages. The table explains what to inspect and what question each element answers.

Result element	Question to ask	Example of a useful clue
Title	Does the title match the exact task and asset?	"M3AA Low Voltage Motors — Technical Catalogue".
Displayed domain	Who appears to own or publish the page?	A manufacturer, government agency, university or standards body.
Snippet	Do the summary words match the required model, units or procedure?	The snippet includes the model number and "installation manual".
Date or revision	Is recency shown and appropriate for the task?	Updated 2025 or Revision 4.
File type	Is the result likely to be a manual, drawing or data table?	A PDF result for an approved manual.
Warnings / advertisements	Is the result clearly labelled as sponsored, unsafe or blocked?	Avoid clicking through a warning or confusing an advertisement with official guidance.

2.6 Safe Downloading and File Handling

Technical searches often lead to PDF manuals, spreadsheets or compressed files. Downloading adds a second safety decision: not only "Is the information correct?" but also "Is the file safe and authorised?" Follow the steps below before opening or sharing a downloaded file.

STOP Safe download routine

1. Prefer the official publisher or an approved shared repository.
2. Check the file type and name before opening it.
3. Do not run unknown .exe, .msi, macro-enabled or compressed files.
4. Save to the approved folder, keep the original name or add a traceable name, and record the source.
5. If the browser, antivirus or school system displays a warning, stop and ask the authorised person.

Topic 3: Can You Trust This Website? The 4-W Credibility Test

Finding information quickly is only half the skill. The other half — and in a technical trade, the more important half — is deciding whether that information is true and safe to act on. A confident-looking web page can still be wrong, out of date, or written by someone with no idea what they are talking about. This topic gives you a fast, reliable inspection method that professionals use every day: the 4-W test.

3.1 The Problem: The Internet Has No Inspector

Think about how a machine part reaches a factory. It is designed, manufactured, and then inspected before anyone trusts it. The internet has no such inspector. Anyone — a curious twelve-year-old, a joker, or a company that only wants your money — can publish a professional-looking website in a single afternoon. Nobody checks it for truth before it goes online. That means you must be the inspector. And just as a good inspection follows a checklist, a good credibility check follows four simple questions.

Why This Matters to a Technician

- Imagine you search for the correct torque setting to tighten a critical bolt, and you follow a random blog that got the number wrong. The bolt fails, the machine is damaged, and someone could be hurt. In a workshop, trusting the wrong source is not just embarrassing — it can be dangerous and expensive. The 4-W test is your safeguard.

3.2 The 4-W Test

The 4-W test asks four questions about any source: **WHO** published it, **WHAT** it actually says, **WHEN** it was published, and **WHY** it exists. For each question, there are green flags that build trust and red flags that destroy it. Read the table below carefully — you will apply it in the very next activity. Then keep the pocket-card figure that follows; many students photograph it or print it and keep it in their notebook for the rest of the programme.

W	The question	Green flags — trust it	Red flags — be careful
WHO?	Who published it?	Manufacturer, government, university, standards body	No author, unknown or anonymous site
WHAT?	Is it accurate and complete?	Data with units, references, matches other sources	Errors, no sources, contradictions
WHEN?	Is it up to date?	Recent date, current product versions	No date, obsolete models
WHY?	Why was it published?	To inform, or as official documentation	Only to sell, shock, or get clicks

The card below arranges the same four questions as four quadrants, each with an example green flag and red flag. Its purpose is to be small enough to keep with you: a quick reference you can glance at whenever you are about to rely on something you found online.

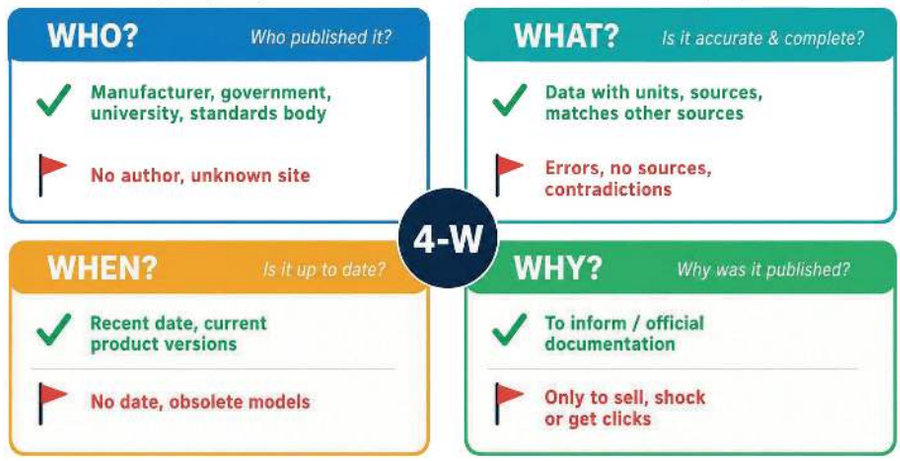


Figure 4.5. The 4-W test card. Print it, keep it in your notebook, and run these four checks before trusting any source.

Activity 1 — Judge These Two Sources!

Your supervisor asks about **safe hydraulic oil disposal**. Your search returns two sources:

Source A:

a page on a government environment agency website (.gov), updated this year, quoting the official regulation number.

Source B:

a nine-year-old forum post by user 'oilking99' saying 'just pour it in the ground, no problem lol'.

Step 1: In pairs, apply the 4-W test to both sources. Score each W as a green flag or a red flag.

Step 2: Decide which source you follow. Then discuss: what could happen if a technician followed Source B? Think about the environment, the law, and the company's reputation.

Our 4-W verdict: .....

.....

3.3 Technical Fit: Credible Is Not Always Applicable

A source can be credible and still be wrong for your task. For example, a manufacturer manual may be genuine but written for another model, voltage, revision or operating environment. After the 4-W test, complete a technical-fit check. The table shows the minimum comparisons a first-year technician should make before using a value or instruction.

Fit check	What to compare	Why it matters
Asset identity	Manufacturer, model, type code and component.	Similar products may use different ratings, wiring or procedures.
Revision and date	Document revision, issue date and product generation.	An older procedure may not include later safety changes.
Units and conditions	Units, supply, temperature, load, environment and test method.	A correct number can be unsafe when conditions differ.
Scope and authority	Installation, operator, service or specialist instruction.	Some actions require isolation, permits or qualified personnel.
Internal approval	Company procedure, approved list or supervisor instruction.	An organisation may require a controlled document even when public information exists.

3.4 Cross-Checking and Recording Evidence

When a technical decision matters, do not rely on one isolated statement. Cross-check the value against another authoritative source or the approved internal document. Cross-checking does not mean collecting many random pages; it means comparing independent, relevant evidence and resolving differences before action.

EVIDENCE Worked example

A forum states that a bearing should be tightened to 75 N·m. The manufacturer manual for the exact assembly gives 52 N·m and the controlled workshop instruction also gives 52 N·m. The forum is not used. The technician records the manual title, revision and page, then follows the authorised procedure.

Topic 4: When the Internet Fails: Connection First Aid

Sooner or later, right in the middle of important work, your screen will announce: 'No internet connection.' The unprofessional reaction is to panic or immediately call for help. The professional reaction is to diagnose calmly, in a logical order, and only then report the problem clearly. This topic gives you that logical order
— and it is the exact same **Understand** → **Find** → **Try** → **Check** thinking you learned in Lesson 1, now applied to networks.

4.1 Key Terms for Connection Troubleshooting

Troubleshooting becomes safer when students separate what they observe from what they think caused it. The table introduces five terms used in the diagnostic method. Read the symptom first, collect evidence, consider possible causes, and choose only an authorised action.

Term	Meaning	Example
Symptom	The visible or measurable problem.	"The browser displays No internet."
Possible cause	An explanation that has not yet been proved.	Wi-Fi is off, the router is unavailable, or the website is down.
Evidence	An observation or test result that supports or rejects a cause.	A second website works, so the internet connection is available.
Action	A safe step taken to test or solve the problem.	Turn Wi-Fi on or reconnect to the approved network.
Escalation	Passing the problem and evidence to an authorised person.	Report that all classroom devices lost connection at 10:15.
Outage	A service is unavailable to multiple users or systems.	The shared internet connection is unavailable, not only one website.

4.2 The Most Common 'Emergency'

The golden rule of connection troubleshooting is to check from **nearest to farthest**: start with the thing closest to you (your own device), then move outward step by step (your connection, then the shared router), and finally report what you found. Along the way, change only one thing at a time — otherwise, if the problem disappears, you will not know which change fixed it. The table below lists the four checks in the correct order, and what each one tells you.

Check (in order)	How to do it	What it tells you
Your device	Is Wi-Fi turned ON? Is Airplane mode OFF?	The problem may be only your own settings
Your connection	Does the Wi-Fi icon show connected? Try a different website	One site being down is not the same as the whole internet being down
The router	Do other devices work? Ask a classmate near you	If nobody has internet, the problem is the router or the ISP — not you
Report	Tell the teacher or IT exactly what you checked	Your checks make the fix faster for everyone

The diagram below shows the same four checks as a staircase descending from 'nearest' to 'farthest'. Picture yourself starting at the top step — the device in your own hands — and only moving down a step once you have ruled out the one above it. This simple discipline solves the majority of everyday connection problems before anyone else even needs to get involved.

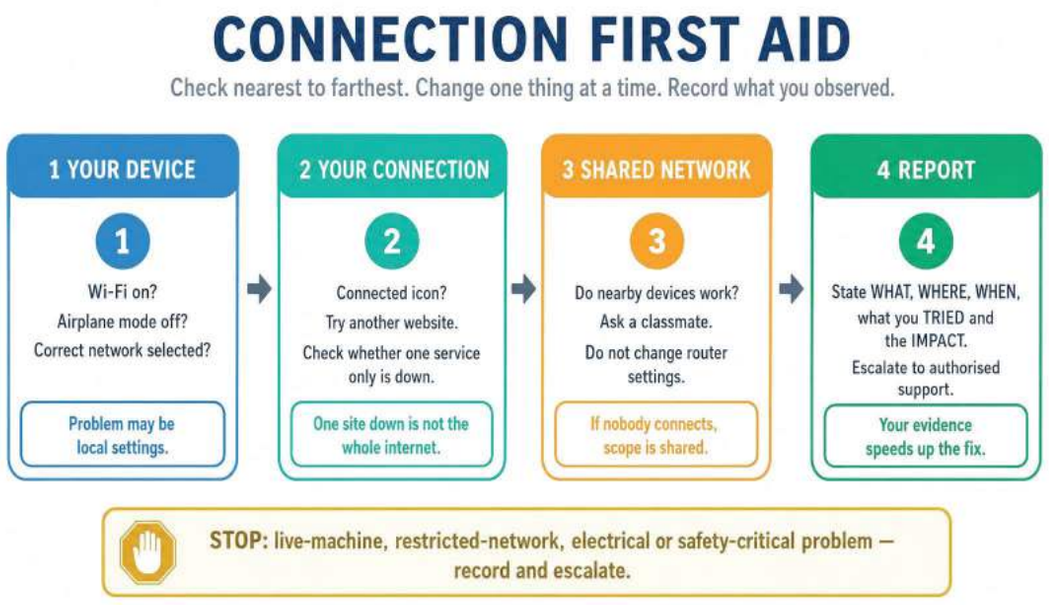


Figure 4.6. Connection first aid: always check nearest to farthest, and change only one thing at a time.

Notice the professional habit built into this method: you check from nearest to farthest, you change one thing at a time, and you report what you observed. Saying 'I checked my own Wi-Fi, I tried another website, and my neighbor's computer has no internet either' gives the IT team a head start. The very same logic will help you troubleshoot machine networks and automated systems in Year 3 — the scale grows, but the thinking stays identical.

4.3 Diagnose the Scope Before You Escalate

The quickest useful question is often "Who else is affected?" Comparing one site, one account, one device, and the whole room helps locate the problem. The table below is not a list of final diagnoses; it is a guide to the safest first check.

Observed pattern	Likely scope	Safe first check
One website fails; other websites open.	The remote site, URL, or access permission.	Recheck the address and try an official alternative page.
One browser fails; another browser works.	Browser session, extension, or cached data.	Save work and reopen the browser; report if the problem continues.
One device has no connection; nearby devices work.	Device Wi-Fi, airplane mode, network selection or local settings.	Check the approved network and the device connection icon.

Observed pattern	Likely scope	Safe first check
All nearby devices lose connection.	Shared access point, router, ISP or scheduled outage.	Do not change network equipment; report the time and affected area.
Internet works, but one account cannot sign in.	Account, password, permission or service issue.	Check the username and authorised sign-in page; do not share passwords.
Connection works but is unusually slow.	Congestion, weak signal, large transfer or slow server.	Compare another site/device and note the time, location and task.

4.4 Report the Problem Professionally

A useful problem report gives the next support person a head start. Avoid "It does not work." Use the five-part formula below and report only facts you observed or safe actions you completed.

Report element	Question	Example
WHAT	What exact symptom or message appeared?	"No internet" appeared in Edge.
WHERE	Which device, room, network or service?	Workshop PC 07 on the Student-Workshop Wi-Fi.
WHEN / CONTEXT	When did it start and what were you doing?	10:15 while opening the approved SKF page.
WHAT WAS TRIED	Which safe checks were completed, in order?	Wi-Fi on; airplane mode off; two sites tested; neighbour also affected.
IMPACT	What task is delayed or at risk?	The search challenge and report submission cannot continue.

SAFETY Stop and escalate immediately

Do not restart, disconnect or alter a computer that controls a live machine or safety-critical process. Do not change router, firewall, proxy, DNS or industrial-network settings. Record the symptom and follow the authorised escalation procedure.

Pop-Up Quiz 2

1 **To find PDF manuals only, add:**

a. more words
b. filetype:pdf
c. CAPITAL LETTERS
d. emojis

2 **'No internet' appears. Your FIRST check should be:**

a. the school's main server
b. your own device's Wi-Fi setting
c. the national network
d. buy a new laptop

Write your answers here:

1 : .....

2 : .....

Topic 5: The Datasheet, a Technician's Best Friend

This topic is about how to read a document— and specifically about the one document that will matter most in your working life. Before we define it, consider what happens when someone ignores it.

5.1 A Costly Mistake

Mechatronics Scenario: The Wrong Replacement

A motor on a production line burns out at eight in the morning. The line stops. A technician is sent to the storeroom and finds a motor that looks right: same size, same shape, same mounting holes, same shaft diameter. It fits perfectly. He installs it in forty minutes and presses start.

Before you read on, predict what happens next.

The burned motor was rated 400 V. The replacement was rated 230 V. It is destroyed in under five seconds, and the fault trips a distribution board, which takes the maintenance electrician a further two hours to reset and inspect. Total production loss: one full day. Total cost: far more than the motor.

Everything the technician checked — size, shape, mounting, shaft — was visible from the outside. The one thing that mattered was not. It was written in a document he did not open. One minute reading that document would have saved a day of production. That document is the datasheet.

5.2 What a Datasheet Is

A datasheet is an official document, published by the manufacturer of a component, that lists all of that component's technical specifications. Motors have them. So do sensors, bearings, hydraulic valves, programmable controllers, cables, and even screws. It is written by the people who designed and tested the part, which is exactly why it carries authority.

Two words in that definition deserve attention. "Official" means it is the manufacturer's own statement, not a summary written by a shop or a blogger — copies drift, get edited, and go out of date. "Specifications" means exact values with units: not "powerful", but 1.5 kW; not "fast", but 1450 rpm. A specification is a number you can check a replacement against.

The practical way to think about a datasheet is this: it is the source of truth for that component. When your memory disagrees with the datasheet, the datasheet is right. When the storeroom label disagrees with the datasheet, trust the datasheet. When a supplier's website disagrees with the manufacturer's datasheet, trust the manufacturer.

5.3 Reading a Motor Datasheet — the Five Key Sections

A datasheet can be twenty pages long and can look intimidating the first time you open one. It becomes manageable as soon as you realize that almost all of them are organised into the same five sections. Once you know the five, you stop reading datasheets from the beginning and start jumping straight to the section you need.

The table below names each section, tells you what you will find inside it, gives a typical example from a small industrial motor, and — most usefully — states the practical question that section answers for you.

Section	What you find there	Typical example	The question it answers
Ratings	Power, voltage, current, speed, frequency, number of poles.	1.5 kW · 400 V · 3.2 A · 1450 rpm	Will it do the job, and will my supply drive it safely?
Dimensions	Physical sizes, frame size, shaft diameter, mounting hole pattern, drawing.	Frame 90L · shaft Ø 24 mm	Will it physically fit where the old one was?
Performance curves	Graphs of efficiency, torque and current plotted against speed or load.	Efficiency 85.3 % at full load	How much will it cost to run, and how will it behave under load?
Operating environment	Ambient temperature range, altitude limit, protection class (IP code), cooling method.	IP55 · −15 °C to +40 °C	Can it survive the place where it will actually live?
Standards and approvals	The international standards and certifications the component meets.	IEC 60034 · CE · IE3	Is it legal, safe and acceptable to install here?

The diagram below shows a realistic datasheet page with all five sections marked and labelled.

Spend a moment tracing each labelled area back to its row in the table above. Notice in particular that the ratings.

section is placed first and given the most space on the page — manufacturers know that this is what most readers came for.

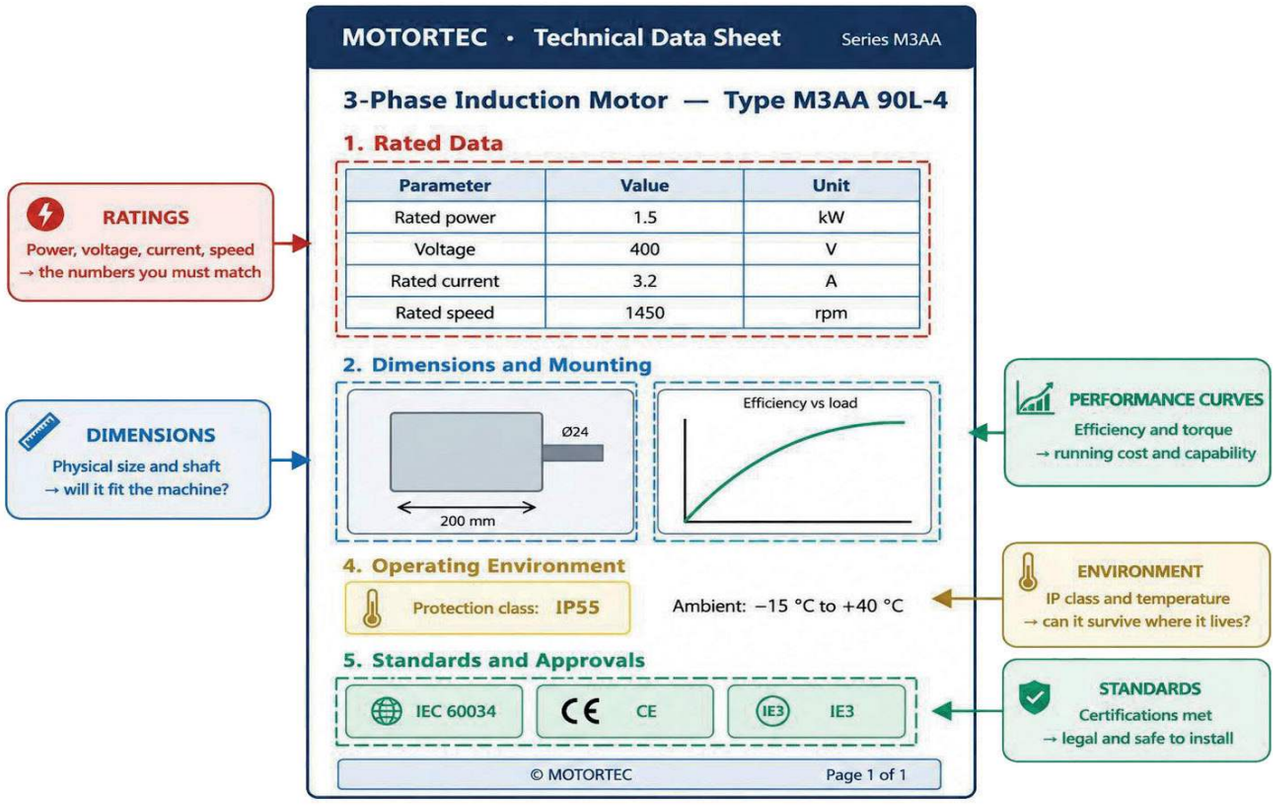


Figure 4.7. A motor datasheet page with its five key sections identified.

5.4 Decoding IP — the Protection Class

Of everything in the environment section, one code appears more often than any other and causes more confusion than any other: the IP code. You will see it printed on the nameplate of almost every industrial component you ever touch, written as two letters followed by two digits — IP55, IP20, IP65, IP67.

IP stands for Ingress Protection. “Ingress” means getting in, so the code describes how well the component's housing keeps unwanted things OUT. This matters enormously in a steel workshop or a wash-down food factory, where the air and the floor are full of things that destroy electrical equipment.

The code is simple once you know that the two digits mean two completely different things and are never interchangeable. The first digit rates protection against solid objects and dust, on a scale from 0 to 6. The second digit rates protection against water, on a scale from 0 to 9. So IP55 means level 5 dust protection and level 5 water protection — dust-protected, and safe against water jets from any direction.

The diagram below sets out both scales with the levels you are most likely to meet, and highlights the two digits of IP55 so you can trace exactly where they come from.

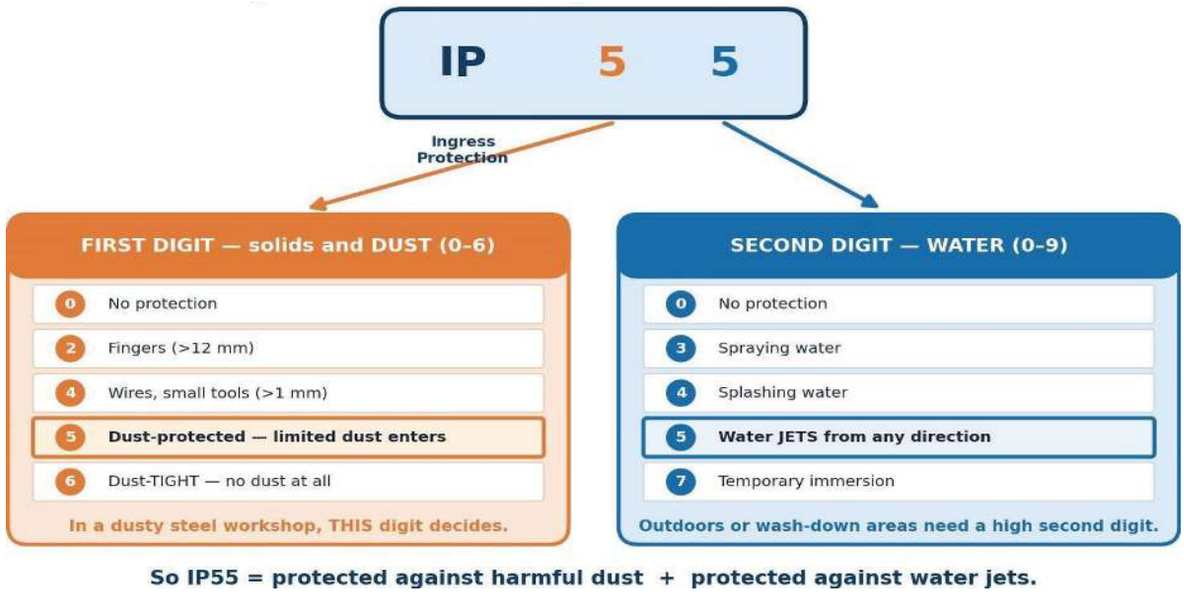


Figure 4.8. Decoding an IP code — first digit dust, second digit water.

Here is a question to hold in your mind until the laboratory session. Suppose you must choose between two motors for a dusty steel workshop. They have the same power and almost the same efficiency, but one is IP23 and the other is IP55, and the IP23 motor is slightly cheaper. Which digit should decide your recommendation, and what would happen to the cheaper motor after six months in that workshop? You will answer this properly in Part D of the lab.

5.5 Reading a Simple Performance Curve

Datasheets do not contain only tables. A performance curve shows how one value changes when another value changes—for example, how motor efficiency changes as load increases. You do not need advanced mathematics to read a basic graph, but you must read it in a disciplined order so that you do not confuse the axes or units.

Use four questions: What is on the horizontal axis? What is on the vertical axis? What units are used? Where is the intended operating point? Then trace from the operating point to the curve and across to the value. A curve is not a decoration; it is evidence about behaviour under conditions that a single rated number cannot show.

5.6 Nameplate, Datasheet, Manual and Standard — Not the Same Thing

A technician may consult several technical sources for the same component, but the sources do not answer the same questions. The nameplate is fast and local, the datasheet gives exact ratings, the manual explains procedures, and a standard defines the test or requirement behind a code. Treating them as interchangeable leads to incomplete decisions.

The table below shows the best use and the limitations of each source. Read it with a real task in mind: identifying a motor, checking whether it fits, installing it safely, and confirming that its protection class is acceptable.

Source	Best use	Important limitation
Machine nameplate	Fast identification and key ratings at the machine.	Limited space; cannot show full procedures or curves.
Official datasheet	Exact manufacturer specifications and dimensions.	May not explain installation or maintenance steps.
Installation/operating manual	Procedures, warnings, settings, and troubleshooting.	May cover several models; confirm the exact variant.
Standard/certification document	Definition of a requirement or test method.	Often technical and not specific to one product.

The diagram below turns the table into a source-selection guide. The professional habit is to begin with the model and variant, then use the datasheet and manual together, consulting the relevant standard when a code or legal requirement must be interpreted.

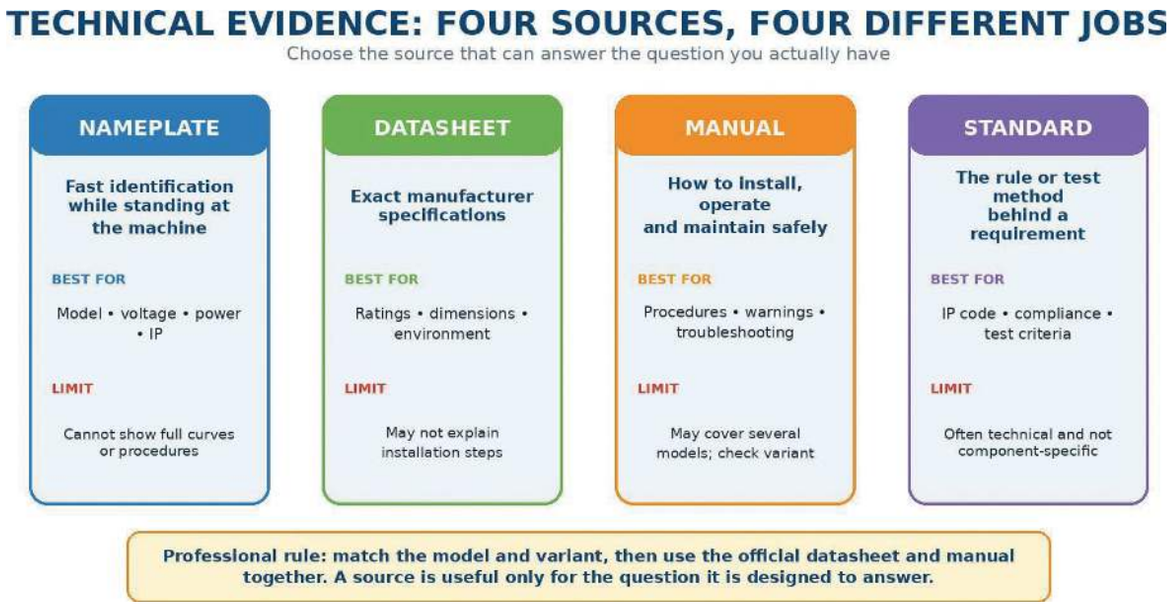


Figure 4.9. Nameplates, datasheets, manuals and standards are complementary sources, not substitutes.

5.7 Replacement Decisions Require a Match, Not a Guess

The wrong replacement motor in the opening scenario failed because only its appearance was compared. A safe replacement decision checks the whole system. Electrical compatibility answers whether the supply can drive it; mechanical compatibility answers whether it will fit; duty and performance answer whether it can do the work; environmental compatibility answers whether it can survive; and compliance or control features answer whether it can connect legally and functionally to the existing system.

The table below turns these five areas into questions. A replacement should not be recommended until every required area has acceptable evidence. “Almost the same” is not a technical conclusion.

Match area	Questions to answer	Example evidence
Electrical	Voltage, frequency, phase, current, and connection match?	400 V, 50 Hz, 3-phase, star/ delta data.
Mechanical	Frame, shaft, mounting and orientation match?	Frame 90L, Ø24 mm shaft, foot mount.
Duty and performance	Power, speed, torque and duty cycle suit the load?	1.5 kW, 1450 rpm, S1 duty.
Environment	IP code, temperature and cooling suit the location?	IP55, –15 °C to +40 °C, IC411.
Compliance and control	Standards, efficiency class and control features match?	IEC 60034, IE3, encoder/brake as required.

The diagram below shows the five evidence areas feeding one decision. It deliberately places "safe replacement decision" in the centre: no single rating is allowed to make the decision on its own.

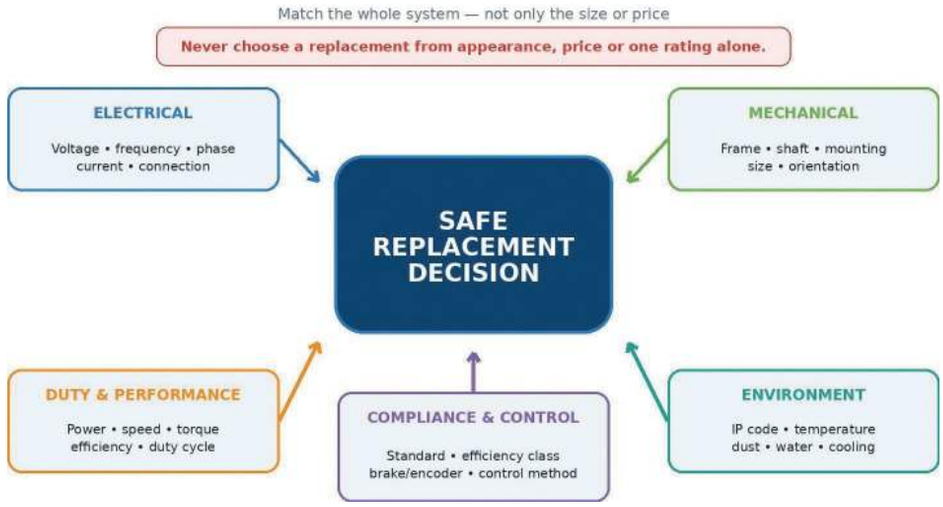


Figure 4.10. A replacement component must match the electrical, mechanical, duty, environmental, and compliance requirements.

Pop-Up Quiz 3

1. A datasheet is written by:

a. any blogger b. the component manufacturer

c. social media users d. a classmate

2. In IP55, the first digit describes protection against:

a. water b. heat c. dust and solid objects d. noise

3. Which datasheet section tells you whether a motor will fit the old mounting?

a. ratings b. dimensions c. standards d. performance curves

My answers:

1 : .....

2 : .....

3 : .....

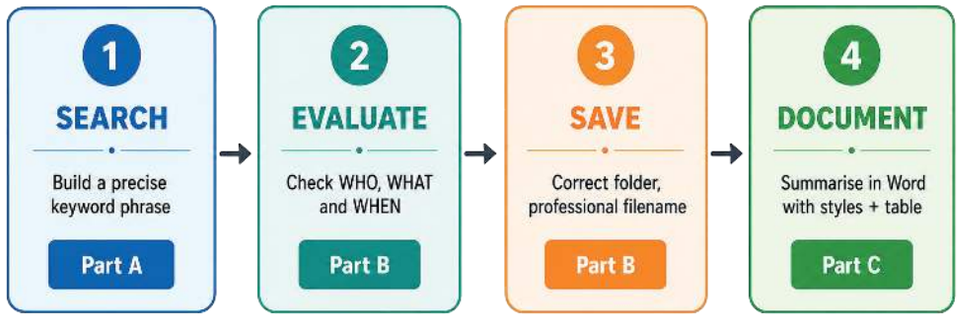
4.1 SESSION Searching, Locating & Evaluating a Motor Datasheet Online

Lab Introduction

"Find me the datasheet for this motor." This is one of the most common instructions a junior technician receives, often in the first week of work. It sounds like a small task. In fact, it contains four separate professional skills, and most people who do it badly fail at the second one.

Today you will run the complete cycle for the first time. You will search with precise keywords rather than vague ones. You will evaluate the source before trusting a single number from it. You will save the file where you can find it again in six months. And you will document what you found in a Word summary built with the styles and structure from Topics 1 to 3. Each of these four stages maps onto one part of this lab.

The diagram below shows the cycle and tells you which part of the lab covers each stage. Notice that it is drawn as a chain: skipping a link does not save time; it just means you end up documenting numbers you never verified.



SKip a step and you may document data from a source you never checked.

Figure 4.11. The professional information cycle — the four stages of today's lab.

The Evidence Chain

Every laboratory stage leaves evidence for the next stage. Your search phrase explains how you looked; the source check explains why you trusted a document; the professional filename preserves the original evidence; the Word summary records exact values with units; and the final recommendation points back to those values. If a number cannot be traced to a source, it is not ready to support a decision.

Key terms for this lab

Four words will be used repeatedly during the next hour. Make sure you can explain each one to your partner before you begin, because the instructions assume you know them.

Term	Meaning	Why it matters today
Search engine	A website such as Google or Bing that finds pages matching the words you type.	It cannot read your mind. It can only match the words you give it.
Keywords	The specific words you type into the search box.	Quality in, quality out. Vague keywords return millions of useless pages.
Source	Who published the information you found.	The same specification from the manufacturer and from an unknown blog are not equally trustworthy.
Specification (spec)	One exact technical value taken from the datasheet, with its unit.	"1.5 kW" is a specification. "Quite powerful" is not — you cannot check a replacement against it.

The equipment and files listed below are required because every stage of the task produces evidence that must be saved and reused. Check these items before the timed work begins.

What You Need

A computer with internet access · a web browser · Microsoft Word · your Unit 1 folder system from Lesson 1 (you will need the 02_Datasheets folder) · this book, open at the lab pages.

The mission box turns the laboratory into a workplace request. Read it as an instruction from a supervisor and identify the required component, ratings, output, and deadline before searching.

Your Mission

Your supervisor hands you a note:
"Find the datasheet for a 3-phase induction motor, around 1.5 kW, 4-pole (approximately 1450 rpm). I need the key specifications summarized on one page by the end of your shift."
Your teacher may give your group a different component — a proximity sensor, a gearbox or a bearing. The four steps are identical whatever the component.

Part A: The Hunt — Smart Searching

Why this practical matters: vague searching wastes time and increases the chance of selecting the wrong model. A junior technician who types only "motor" may find thousands of products, while a planned search phrase narrows the task to the required component and document type. The goal is not to collect many links; it is to build a small set of relevant candidate sources.

Search Plan Before Typing

Before opening the search engine, extract the important nouns and ratings from the supervisor's request. The table below shows how each part of the request becomes one part of the search phrase. If the manufacturer is unknown, leave that element blank at first and add a manufacturer after you identify a promising product family.

Search element	Example from the mission	Why it narrows the results
Manufacturer	ABB or Siemens, if known	Prioritizes the official product family.
Component	three-phase induction motor	Removes unrelated meanings of "motor".
Key rating	1.5 kW	Narrows to the required power range.
Type/variant	4-pole / approximately 1450 rpm	Separates similar models.
Document and format	datasheet filetype:pdf	Targets technical PDF documents.

Success evidence for Part A is a search phrase that includes the task's important identifiers and two or three candidate tabs that appear relevant. Opening ten weak results is not better than opening three strong ones.

A search engine is not intelligent in the way people imagine. It does not understand that you are a mechatronics student who needs an official manufacturer document. It matches words. This means the quality of what you get out is decided entirely by the quality of what you put in — and most people put in far too little.

The professional keyword formula has five parts, and you can build it like a sentence: manufacturer, then component, then the key rating, then the type or variant, then the word "datasheet", and optionally the file format. Each part you add cuts the results down and pushes the official document upwards.

The comparison below shows the same search performed weakly and professionally, with the professional phrase broken into its parts so you can see how it was constructed. You are about to type both of these yourself and compare the results with your own eyes — the contrast is the point of the exercise.

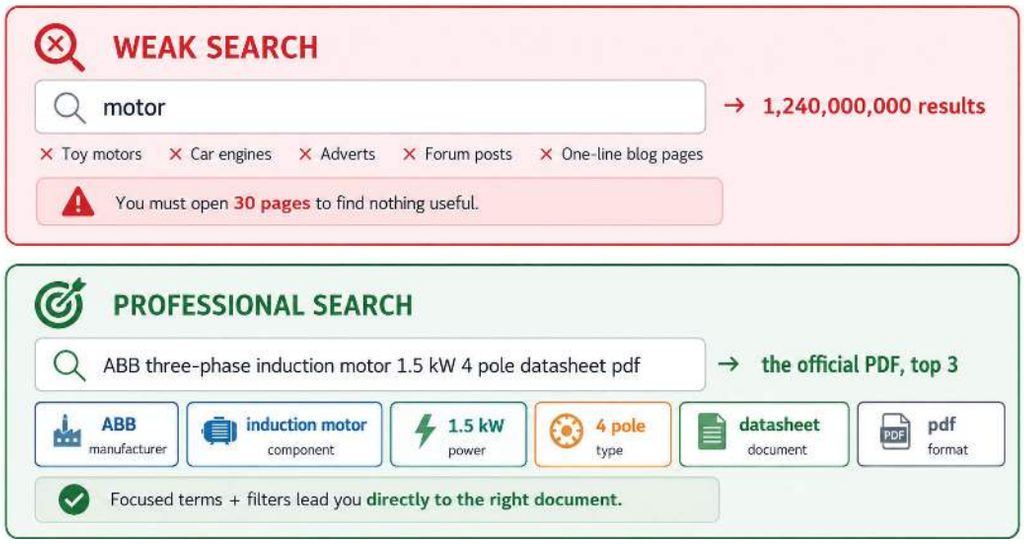


Figure 4.12. The same search, weak and professional — and how the professional phrase is built.

Step 1:

Open your browser and go to Google. First, deliberately try a WEAK search: type only the single word "motor". Look at what comes back — toy motors, car engines, advertisements, forum discussions. Millions of results, almost none of them useful to you.

Step 2:

Now try a PROFESSIONAL search: type "three-phase induction motor 1.5kW 4 pole datasheet pdf". Compare the first five results with your partner. What changed?

Step 3:

Improve it once more by adding a well-known manufacturer at the front: "ABB 1.5kW motor datasheet" or "Siemens 1.5kW motor datasheet". This usually pushes the official PDF into the top three results.

Step 4:

Open two or three promising results in new tabs (hold Ctrl and click the link). Do NOT download anything yet — evaluation comes first, in Part B.

Search Like a Professional

More specific words mean better results. Build your phrase from: manufacturer + component + power + type + the word "datasheet".
Adding the word "pdf" often leads straight to the official document, because datasheets are nearly always published as PDF files.
Prefer results from manufacturer websites — abb.com, siemens.com, sew-eurodrive.com, weg.net. In the next lesson you will learn the complete rules for judging websites.
The first results are sometimes marked "Sponsored" or "Ad". Those are paid placements, not the best answers. Look past them.

Part B: Evaluate Before You Trust

Why this practical matters: a document can be genuine yet still be the wrong fit for your task. It may describe an older product, a different frame size, or a similar model with another voltage. Evaluation therefore checks both credibility and fit.

Three Words That Protect the Decision

Use the three terms below while discussing candidate sources. They turn a vague reaction—"this site looks good"—into a technical explanation that another person can check.

Term	Meaning in this task	Question to ask
Authoritative	Published by the manufacturer or another recognized technical authority.	Who is responsible for this information?
Current	Recent enough for the product generation and application.	Is there a date, revision or current model range?
Correct variant	Matches the exact model, ratings and configuration required.	Does the code and specification match my component?

Here is the step that separates a professional from someone who merely knows how to use Google. Anyone can publish anything on the internet. A datasheet copied onto a random parts-reseller site may be five years out of date, may refer to a different variant of the motor, or may simply have been retyped with an error in it. If you copy a wrong voltage into your summary, the mistake is now yours — and it will be your recommendation that destroys a motor.

Evaluation does not need to be slow. Four quick questions, asked about every tab you have open, will catch almost every bad source. Ask them in this order: WHO published it, WHAT exactly is it, WHEN was it produced, and WHY it exists.

The diagram below sets out all four questions, and for each one gives you both a good sign to look for and a warning sign that should make you close the tab. Use it as a checklist as you work through your open tabs with your partner.

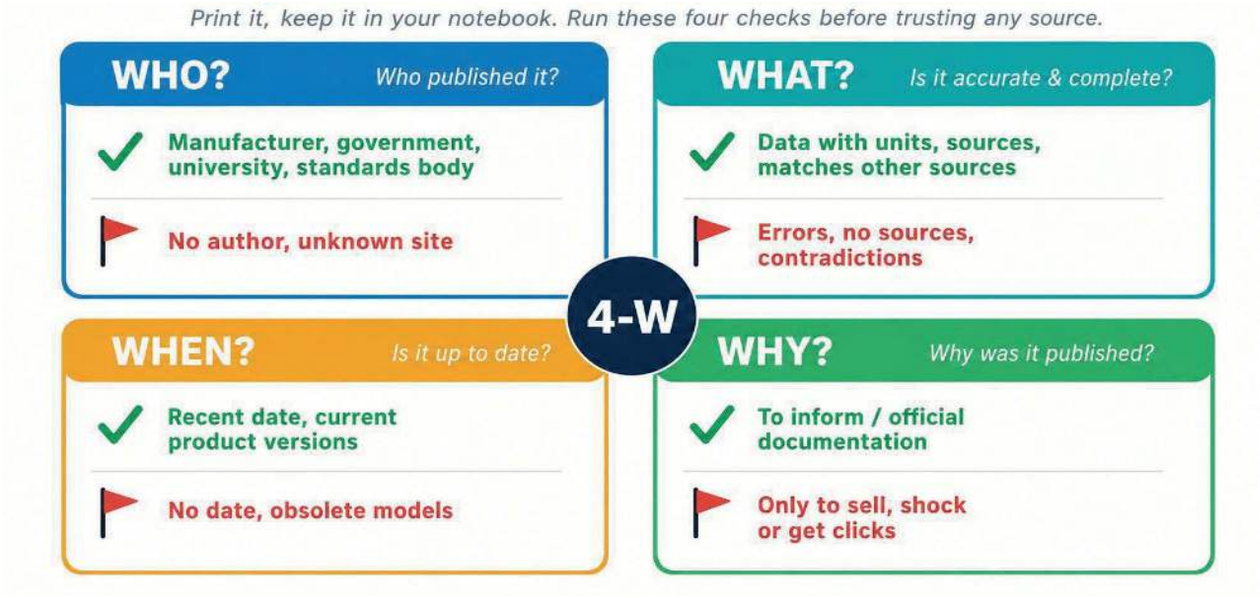


Figure 4.13. The four-question source check — apply it to every tab before downloading.

Record the comparison rather than relying on memory. The table below gives you space to compare up to three candidate sources. A short written decision also becomes useful evidence if another technician later asks why that PDF was selected.

Candidate source	WHO published it?	WHAT exactly is it?	WHEN / revision	WHY does it exist?	Decision
Source 1					
Source 2					
Source 3					

Step 1: Working with your partner, apply WHO, WHAT, WHEN, and WHY to each of your open tabs. Say your answers out loud — explaining them to another person is what catches lazy thinking.

Step 2: Choose the BEST source and state why in one complete sentence. You will write this sentence into your Word summary later, so make it a good one.

Step 3: Download the datasheet PDF. In the download dialog, choose Save As rather than accepting the default location, and navigate to your 02_Datasheets folder.

Step 4: Rename the file professionally before you save it: 2026-XX-XX_Motor1.5kW_Datasheet.pdf, using today's date.

A file that cannot be found again is a file you did not really save. The diagram below reminds you where this file belongs in your folder system, and then breaks the filename into its parts so you can see why it is built that way. Each part does a specific job, and the date is written first so that your files sort themselves into chronological order automatically.

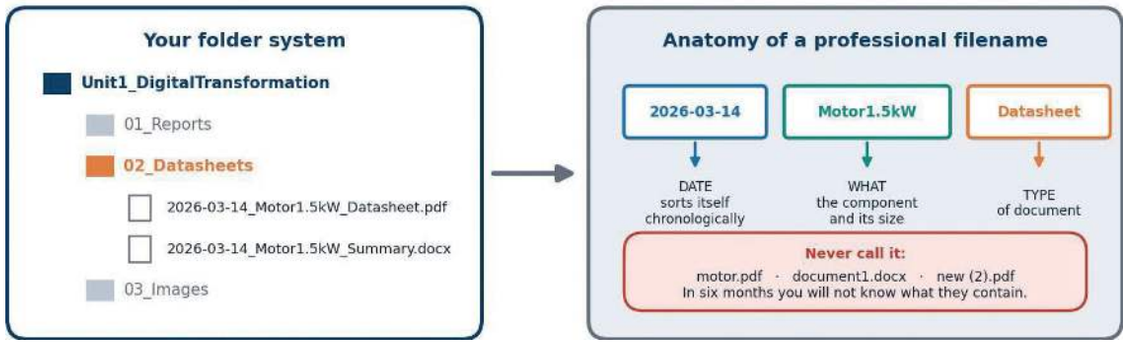


Figure 4.14. Where the file goes, and how a professional filename is constructed.

Part C: Document It — the One-Page Summary in Word

Why this practical matters: summarizing is not copying less text. It is selecting the facts the reader needs for a specific decision while preserving the exact meaning of the source. The supervisor needs a one-page technical view, but every value on that page must remain faithful to the manufacturer document.

Rules for Faithful Technical Summarizing

Copy the exact value and its unit. "400" is incomplete; "400 V" is evidence.

Keep the model or variant beside the values so they cannot be separated from the component they describe.

Write "not stated" when a value is genuinely absent. Never invent a likely number.

Do not change rounding, symbols or conditions unless you clearly state the conversion.

Keep the source address, access date and trust sentence with the summary.

The table below shows common first-year errors and the professional correction. Read it before building your page so that you prevent the error instead of repairing it during final checking.

Common error	Why it is risky	Professional correction
A value without a unit	The reader cannot interpret or compare it.	Record 400 V, 3.2 A, and 1450 rpm.
A decimal point copied incorrectly	The value may change by a factor of ten.	Check the PDF again and compare the unit.
Values from a similar model	The summary no longer represents one component.	Record manufacturer and exact model first.
Headings made bold manually	The page looks structured, but Word cannot navigate it.	Apply Title and Heading styles.
A floating image covering text	Evidence becomes unreadable or moves unexpectedly.	Use In Line with Text and resize from a corner.

You have a trustworthy datasheet. Your supervisor does not want the twenty-page PDF; he wants one page with the numbers he needs. This is where every skill from Topics 1 to 3 comes together in a single deliverable, and it is the part of the lab your teacher will assess.

Plan first — do not open Word yet

Apply Topic 2 before you type anything. Take thirty seconds and answer the three questions out loud with your partner:

- **PURPOSE:** to give the supervisor the key specifications of this motor at a glance, so he can decide whether it suits the application.
- **READER**: a busy supervisor with two minutes, who knows motors well and needs no explanations—so use a table, not paragraphs.
- **SECTIONS:** Key Specifications (as a table) → Source Information (as a numbered list).
  A Recommendation section will be added in Part D.

Now build it

Step 1: Open Word and create a new blank document. SAVE IT FIRST into 02_Datasheets, as 2026 - XX-XX_Motor1.5kW_Summary.docx. Saving first means a power cut costs you nothing.

Step 2: Insert a header: Insert tab → Header → Blank. Type "Component Datasheet Summary — [Your Name]". Double-click in the body of the page to close the header. It will now appear on every page automatically.

Step 3: Type the document title, "Motor Datasheet Summary", then click inside that line and click the Title style in the Styles gallery. One click.

Step 4: Type your two section headings — "Key Specifications" and "Source Information" — and give each of them the Heading 1 style. Do not make them bold by hand; use the style, for the reasons in Topic 3.2.

Step 5: Under Key Specifications, insert a table with 2 columns and 9 rows (Insert → Table). Put "Parameter" and "Value" in the top row, then fill the rest FROM YOUR REAL DATASHEET: manufacturer and model, power (kW), voltage (V), current (A), speed (rpm), efficiency (%), protection class (IP), weight (kg).

Step 6: Struggling to find a value in a long PDF? Press Ctrl + F inside the PDF viewer and search for "kW", "rpm" or "IP". This is a genuine professional trick and it will save you minutes. If a value is genuinely not given, write "not stated" — this is the correct professional answer, and inventing a number is not.

Step 7: Under Source Information, create a numbered list with three items: (1) the web address you downloaded from, copied from the browser address bar with Ctrl + C and Ctrl + V; (2) the date you downloaded it; (3) one sentence explaining why you trusted this source — the sentence you prepared in Part B.

Step 8: Optional bonus: capture the motor image from the datasheet (Windows key + Shift + S), paste it into your document, and add the caption "Figure 1: Motor overview (source: manufacturer datasheet)" underneath it.

Step 9: Press Ctrl + S. You now have a complete professional summary — and the personal template you will reuse for sensors, bearings and controllers in every unit that follows.

Before you call your teacher over, compare your page with the model below. Every labelled feature in it corresponds to something you were asked to do in the nine steps above, so use the labels as a checklist. Check especially that you applied real styles rather than manual bold, and that your table has a proper header row.

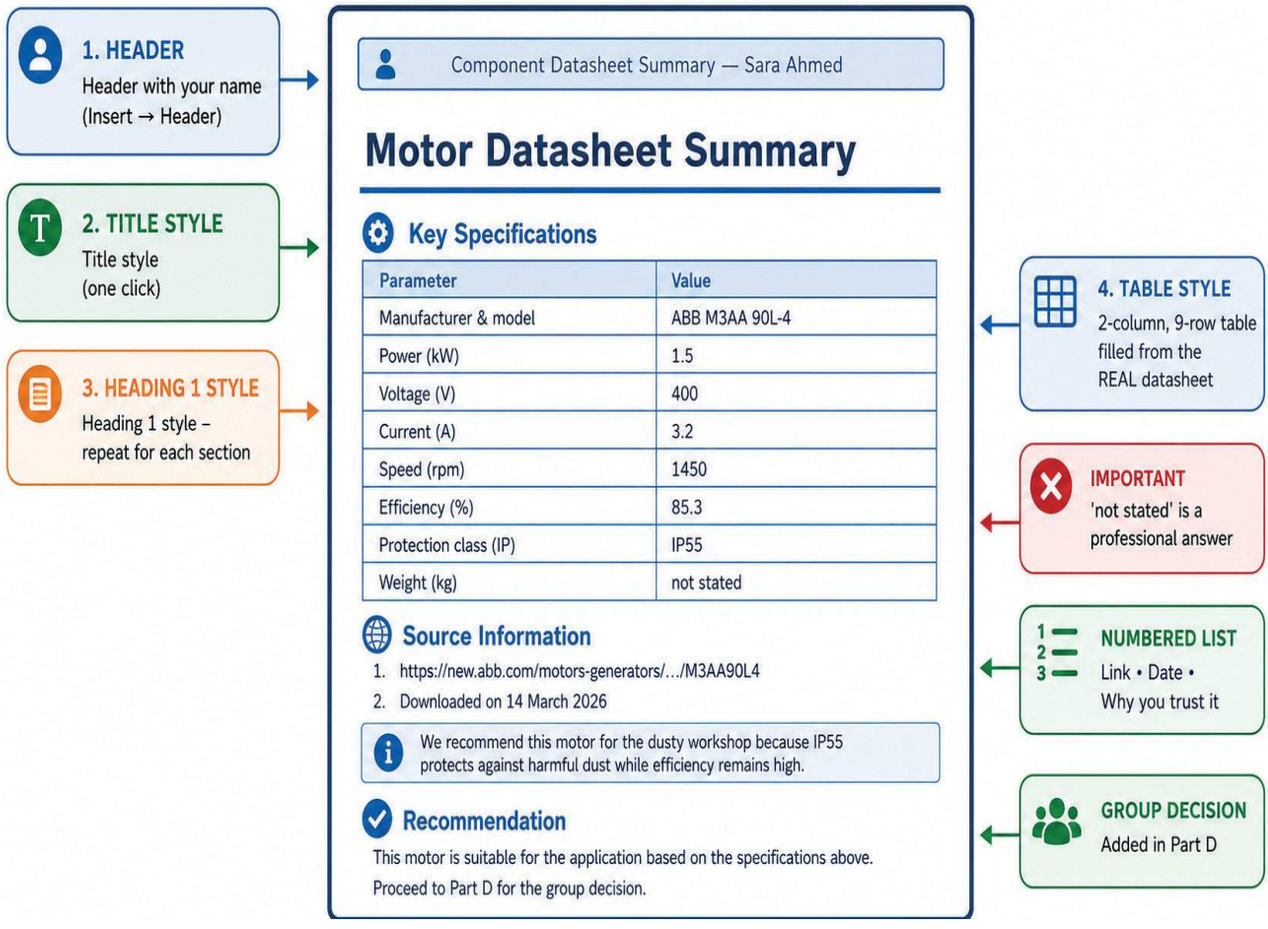


Figure 4.15. A completed one-page datasheet summary, with every required feature labelled.

Self-Check Before You Submit

1. Does the header show my name, and does it appear on the page automatically?
2. Did I use the Title and Heading 1 STYLES, not manual bold formatting?
3. Does my table have a header row and all eight specification rows filled in — with "not stated" where the datasheet is silent?
4. Does every value carry its correct unit (kW, V, A, rpm, %)?
5. Is my source list numbered, and does it include the link, the date and my reason for trusting it?
6. Is the file saved in 02_Datasheets with the professional filename?

Final checking has two different purposes, so use two separate checklists rather than mixing them. First check data accuracy; then check document presentation. A document can look excellent and still contain a wrong voltage, or contain correct data and still be impossible to scan.

The data-accuracy table below checks whether every technical value can be trusted.

Data-accuracy check	Yes / Fix
Exact manufacturer and model recorded	
Every value includes the correct unit	
Missing values shown as "not stated"	
Source address, date and trust reason included	
Values rechecked against the original PDF	

The document-presentation table below checks whether a busy supervisor can find and interpret the information quickly.

Document-presentation check	Yes / Fix
Header, title and Heading 1 styles applied correctly	
Specification table fits within the margins	
Lists use bullets or numbers for the correct reason	
Any image is inline and has a caption	
Final file is saved, closed and reopened successfully	

Part D: The Selection Meeting

Why this practical matters: a technical recommendation is a claim supported by evidence. In a selection meeting, the winning option is not the one presented most confidently; it is the one that best satisfies the stated conditions. Every speaker must point to a value, section or source rather than to personal preference.

Before the meeting, read the decision criteria below. The criteria should be agreed before the options are discussed; otherwise, a group may choose a favorite first and invent reasons afterwards.

Selection criterion	Evidence to compare	Decision question
Electrical suitability	Voltage, frequency, phase and current	Can the existing supply operate it safely?
Mechanical fit	Frame, shaft and mounting dimensions	Will it fit without unsafe modification?
Performance	Power, speed, efficiency and duty	Can it do the required work continuously?
Environment	IP code and temperature range	Can it survive the actual workshop conditions?
Source quality	Manufacturer, revision and exact variant	Can every comparison value be traced?

A short meeting works best when each person has a defined responsibility. The roles below keep all students involved and prevent one confident speaker from making the decision alone.

Meeting role	Responsibility
Presenter	Summarizes the option and points to two key specifications.
Evidence checker	Confirms that each claimed value appears in the source.
Environment checker	Tests the option against the workshop conditions and IP requirement.
Recorder/chair	Keeps time, records the decision and writes the recommendation.

Real engineering decisions are almost never made by one person reading one document. They are made in short meetings where people bring evidence and argue for a recommendation. Part D is a miniature version of that meeting, and it is where the IP code from Topic 5.4 finally earns its place.
A strong recommendation names the chosen option, the relevant conditions and the evidence.
Model: "We recommend Motor B because its 400 V, 1.5 kW rating matches the supply and load, and its IP55 enclosure provides stronger dust protection for the steel workshop. Motor A is not selected because its IP23 enclosure provides less protection against dust."
After feedback, save a revised copy as v02 only if you make a meaningful change. Keep v01 as the original and reopen v02 from File Explorer to confirm that the final document is readable and complete.

Group Activity: The Selection Meeting

In groups of four, each pair presents its motor summary to the other pair, then:

Step 1: Compare your two motors directly. Which has the higher efficiency? Which is lighter? Which has the better ingress protection?

Step 2: As a group, DECIDE: which motor would you recommend to the supervisor for a dusty steel workshop, and why? Think carefully about the FIRST digit of the IP code before you answer.

Step 3: Write your group's decision as one clear sentence at the bottom of your Word summary, under a new Heading 1 called "Recommendation". Save the file again.

Our group recommendation:

.....

.....

.....

.....

Lesson/Session Summary

This summary gathers the key ideas of Lesson 4 and its datasheet-search practical. Use it as a quick revision map alongside the assessment.

- Finding information online well is the FIND stage of the cycle: precise search terms beat vague ones.
- Every source must be evaluated before it is trusted – the EVALUATE stage – using the 4-W test: who, what, when, why.
- A datasheet for the wrong model gives the wrong ratings, so confirming the exact model is a safety step.
- A found document is saved with a professional name in the correct folder so it can be found again.
- Reading and recording a key rating from a datasheet turns raw information into usable evidence.
- When the internet fails, connection first aid works from nearest to farthest.

Class Discussion

1. Why is searching for "motor" less effective than searching for the exact motor type, rating, and document needed?
2. Can a website use HTTPS and still contain incorrect information? Why?
3. Which of the 4-W questions is easiest to answer when you find a technical website? Which is the most difficult? Explain.
4. Why should you check the exact model and revision before copying a specification from a datasheet?
5. A website is published by a well-known manufacturer, but the document is for an older motor model. Would you use it? Why or why not?
6. You are working in the workshop, and one computer cannot connect to the internet, but all the other computers are working. What would you check first?
7. Why is "It doesn't work" a poor technical problem report? What information should you provide instead?
8. Which is more important when selecting technical information: finding it quickly or making sure it is correct? Can a professional technician do both?

9 How can poor online information lead to a real problem in a mechatronics workshop? Give one example.

10 What is one search habit from today's lesson that you can use in your next technical task?

Reflect and Connect (At Home)

Write two or three sentences: think of the last time you searched for something important online. Did you check who published it and when? How would the 4-W test change the way you decide what to trust?

.....

.....

Assessment

Part A. Choose the correct answer:

1. A figure caption is:

a. the page number **b.** a numbered label describing an image

c. a type of font **d.** a footer

2. To create an automatic table of contents, you must first use:

a. bold text **b.** Heading styles **c.** colours **d.** big fonts

3. In the planning method, READER asks:

a. how many pages to write **b.** who will read it and what they need

c. which font to use **d.** when to print

4. The official document listing a component's full specifications is the:

a. invoice **b.** datasheet **c.** poster **d.** email

5. In IP55, the second digit describes protection against:

a. dust **b.** water **c.** heat **d.** electricity

6. Which datasheet section tells you whether a motor will fit the existing mounting?

a. Ratings **b.** Dimensions **c.** Standards **d.** Performance curves

Part B. True **(T) or False (F):**

- A header repeats on every page of the document.
- Any website is an equally good source for a datasheet.
- The steps of a procedure should be written as a numbered list.
- Writing "not stated" for a missing specification is unprofessional.

Part C. Practical:

Write the professional search phrase you would use to find the datasheet of a Siemens 0.75 kW three-phase motor .....

Self-Assessment Checklist

When I Finish This Lesson, I can:	Yes	Not yet
1. Explain the basic roles of an IP address, DNS, router, and server.		
2. Read a URL and explain what HTTPS means.		
3. Use search operators to find a relevant technical datasheet or manual.		
4. Use the 4-W test to check whether an online source is trustworthy and suitable.		
5. Find, save, and record information from a motor datasheet accurately.		

Glossary قاموس المصطلحات

Use this table whenever a term in the lesson is unclear. The Arabic column is there to help you connect the English technical vocabulary to words you already know — but in industry, the English terms are the ones written on the datasheets you will read.

English term	Meaning	المصطلح بالعربية
Datasheet	The manufacturer's official specification document for a component.	ورقة بيانات فنية
Specification	One exact technical value or requirement, written with its unit.	مواصفة
Document skeleton	The standard set of parts every professional document is built from.	الهيكل العام للوثيقة
Header / Footer	Text repeated automatically at the top / bottom of every page.	رأس /تذييل الصفحة
Style (Word)	A saved package of formatting applied to a paragraph with one click.	نمط تنسيق
Template	A ready-made document skeleton that you fill with new content.	قالب
Table of contents	An automatic list of the document's sections with their page numbers.	جدول المحتويات

English term	Meaning	المصطلح بالعربية
Caption	A numbered label placed under a figure or table to explain it.	تعليق توضيحي
Planning	Deciding purpose, reader and sections before writing anything.	التخطيط
Keyword	A specific word typed into a search engine to find information.	كلمة مفتاحية
Source	The person or organisation that published a piece of information.	المصدر
Induction motor	The most common type of industrial AC electric motor.	محرك حثي
Rated power	The power a machine is designed to deliver continuously.	القدرة المقننة
Efficiency	How much of the electrical power supplied becomes useful output.	الكفاءة
Protection class (IP)	A two-digit code for protection against dust and water.	درجة الحماية
Ingress	The entry of something unwanted, such as dust or water, into a housing.	التسرب إلى الداخل
rpm	Revolutions per minute — a measure of rotational speed.	لغة في الدقيقة
Traceability	The ability to identify a document's author, version, date, approval and history.	إمكانية التتبع
Document ID	A unique code that identifies one document.	معرّف الوثيقة
Revision/version	A numbered edition of a document after a change.	مراجعة / إصدار
Approval	Formal authorisation that a document may be used.	اعتماد
Cross-reference	A precise reference from one part of a document to another numbered item.	إحالة مرجعية
Authoritative source	A source published by the responsible manufacturer or recognised authority.	مصدر موثوق / معتمد

English term	Meaning	المصطلح بالعربية
Variant	One exact version or configuration within a product family.	طراز فرعي / إصدار محدد
Performance curve	A graph showing how one technical value changes with another.	منحنى أداء
Operating point	The condition on a graph where the component is expected to work.	نقطة التشغيل
Compliance	Meeting a required standard, rule, or certification.	امتثال
Search engine	A tool that finds web pages from keywords	محرك بحث
Keyword	A precise word used to search	كلمة مفتاحية
Datasheet	A document giving a component's technical ratings	ورقة بيانات فنية
4-W test	Who, what, when, why – a source-credibility check	اختبار المصداقية
Source	Where information comes from	مصدر
URL	The address of a web page	عنوان الموقع
Revision	The version or edition of a document	مراجعة / إصدار

Abbreviations

Technical documents are full of abbreviations, and datasheets are among the worst offenders. These are the ones you will meet in this lesson and in the lab.

Abbreviation	Full form	Where you will see it
PDF	Portable Document Format	The file format almost every datasheet is published in.
IP	Ingress Protection	On the nameplate of nearly every industrial component.
rpm	Revolutions Per Minute	The speed row of a motor's ratings table.
kW	Kilowatt — unit of power	The rated power of a motor.
V	Volt — unit of voltage	The supply voltage a component is designed for.

Abbreviation	Full form	Where you will see it
A	Ampere — unit of current	The current a motor draws at rated load.
IEC	International Electrotechnical Commission	The standards section of a datasheet (e.g. IEC 60034).
CE	Conformité Européenne	A conformity mark shown among a component's approvals.
TOC	Table of Contents	• The automatic contents list Word builds from your styles. • A navigation list built automatically from Heading styles.
ID	Identifier	A unique code used for a document, machine or component.

4.2 SESSION Network Detective — Trace, Test and Report a Workshop Connection (Optional Network Extension)

This session is optional. You study it only if the available unit time allows for its completion without affecting the core lessons and learning outcomes of the unit.
If instructional time is limited, this session may be **skipped without affecting the achievement of the unit's essential learning outcomes**.

Lab Introduction

A workshop computer cannot open the manufacturer's support page for a network-connected HMI. Is the problem inside the computer, between the computer and the local router, in DNS, or at the remote website? A professional technician does not guess and does not immediately change settings. The technician collects evidence from nearest to farthest, checks an official support source, and reports the result clearly.
This lab uses only free, built-in Windows tools and an approved browser. You will inspect information, not reconfigure the network. The final product is a Workshop Connectivity Incident Brief that another technician or IT support person can understand and act on.
The following terms are the tools and evidence words used in the lab. Read the table before opening Command Prompt. The commands shown are read-only when used exactly as written.

SAFETY - Read-only network investigation

Use only the commands printed in this lab: ipconfig, ping, nslookup, and teacher-approved tracert. Do not type /release, /renew, /flushdns, or any command that changes settings. Do not reset equipment, inspect another person's device, or connect to an industrial control network. If a command is blocked, record "blocked by policy" and continue with the evidence provided by the teacher.

Term or command	Meaning
Command Prompt	A built-in Windows window for entering text commands.
ipconfig	Displays the current adapter, IPv4 address, and default gateway.
IPv4 address	The address currently identifying the device on its network.
Default gateway	The local router address used to reach other networks.
ping	Sends test messages and reports whether a target replies; a failed reply is evidence, not final proof.
nslookup	Asks DNS to translate a domain name into an IP address.
tracert	Shows the sequence of network hops toward a destination; some hops may not reply.

The lab produces four pieces of evidence. Read the following table first so you know what must be visible before your group moves to the next stage.

Evidence	Minimum success condition	Why it matters
Connection Identity Card	Active adapter, IPv4 address, default gateway, DNS server and approved HTTPS page.	Identifies the device and its normal route.
Web-request test record	Results from local, gateway, DNS and destination checks, with screenshots or copied output.	Separates observation from assumption.
Fault-card diagnosis	Likely problem scope, evidence, verified support source and safe next step.	Connects search, credibility and troubleshooting.
Connectivity Incident Brief	WHAT, WHERE, WHEN, TRIED, EVIDENCE, IMPACT and ESCALATION, saved with a traceable file name.	Allows authorised support to act efficiently.

The workflow diagram below shows the four stages. Follow it from left to right. Each stage ends with a checkpoint because later conclusions depend on the evidence collected earlier.

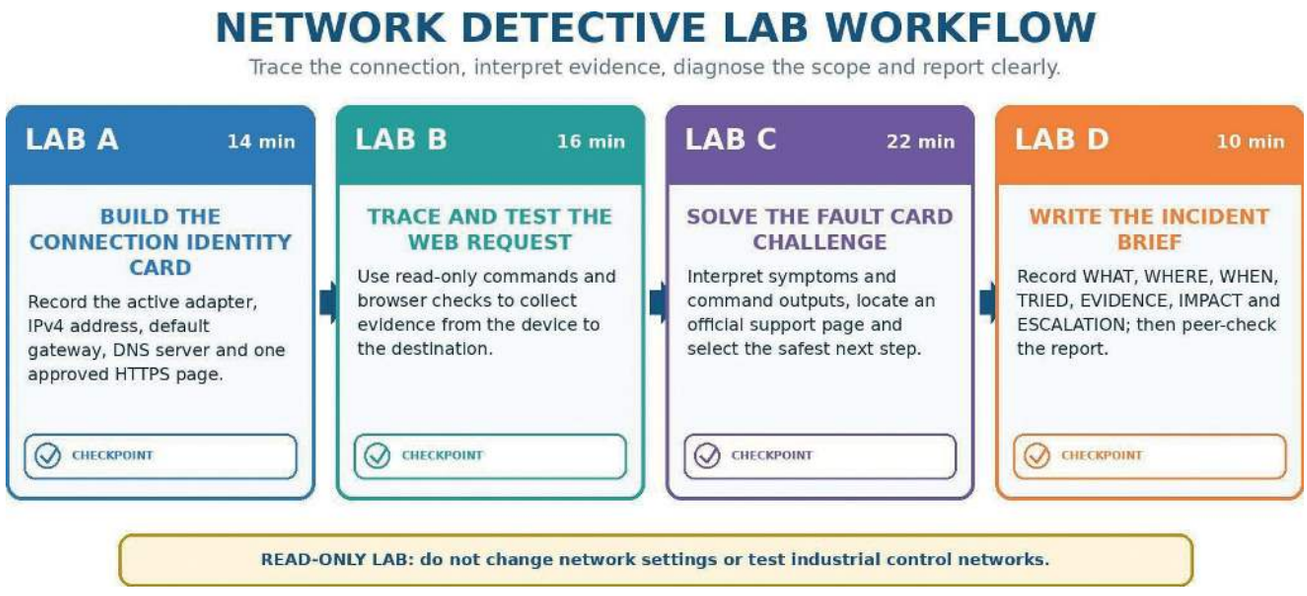


Figure 4.16. Network Detective lab workflow: identify → trace → diagnose → report.

Part 1: Lab A: Build the Connection Identity Card

Start with the device in front of you. The purpose is to record its current network identity without changing anything. The values will differ between computers, so copy only what appears on your own screen. If the school hides a value, write "not displayed by policy" rather than guessing.

1. Open Start, type Command Prompt or cmd, and open the approved Command Prompt application.
2. Type ipconfig and press Enter. Find the active Wi-Fi or Ethernet adapter. Record its IPv4 Address and Default Gateway.

3. Open the network-status screen approved by your teacher and record the active adapter name and whether it shows Connected.
4. Open the approved HTTPS page supplied by your teacher. Record the complete displayed domain and confirm that HTTPS is present.
5. Type nslookup followed by the approved domain, for example nslookup support.microsoft.com. Record one returned IP address.
6. Save one screenshot of the command output, or copy the relevant lines into your evidence file if screenshots are restricted.

Connection Identity Card	Your evidence
Active adapter	
IPv4 address	
Default gateway	
DNS server or nslookup server	
Approved HTTPS domain	
Resolved IP address	

Part 2: Lab B: Trace and Test the Web Request

Now follow the connection from nearest to farthest. Each test answers a different question. Do not treat one failed result as a complete diagnosis: some devices and websites are configured not to reply to ping or tracert. Record the result, compare it with the other checks and keep the conclusion proportionate to the evidence.

Stage	Command or check	Question answered	Result
Local TCP/IP	ping 127.0.0.1.	Does the computer's own TCP/IP system reply?	
Local route	ping [your default gateway].	Can the device reach the local router?	
Name resolution	nslookup [approved domain].	Can DNS translate the name into an IP address?	
Web service	Open the approved HTTPS page.	Does the actual web service open in the browser?	
Optional route trace	tracert -h 10 [approved domain].	Which network hops appear before the trace ends?	

Checkpoint: explain in one sentence what each result can and cannot prove. For example, a successful gateway ping supports the local path, but it does not prove that a remote website is available.

Part 3: Lab C: Solve the Fault Card Challenge

You will receive one simulated workshop fault card. The card contains symptoms and command results so nobody needs to disconnect a real computer. In groups, do the following:

1. Identify the likely scope: local device, local network, DNS/name resolution, or remote website/service.
2. Write one operator-based query for an official support page, for example: site:microsoft.com "DNS troubleshooting" or site:manufacturer-domain "network connection" HMI.
3. Apply the 4-W test and confirm that the page is relevant to the equipment or connection issue. Record the title, publisher/domain, date or revision, and the exact guidance used.
4. Choose the safest next step within a first-year trainee's authority. Do not invent a repair or change network settings.
5. Prepare a 45-second evidence defence: conclusion → two pieces of evidence → safe next step.

Fault card	Evidence supplied
Card 1 — Local device	PC-04 shows "Media disconnected" in ipconfig. The default gateway is blank. The computer beside it opens approved websites normally.
Card 2 — Name resolution	The gateway replies. An approved site opens when its IP address is used, but nslookup for the domain returns a DNS error.
Card 3 — Remote service	Other approved websites open. nslookup resolves the manufacturer domain. The manufacturer page displays "503 Service Unavailable".
Card 4 — Shared connection	Three nearby workshop PCs cannot reach the default gateway. Their Wi-Fi icons show connected, but no approved website opens.

Use the diagnosis table to keep the claim, evidence, and next step separate. A good diagnosis may still end with "escalate" when the required action is outside your authority.

Likely scope	Evidence supporting it	Verified support guidance	Safe next step

Part 4: Lab D: Write and Peer-Test the Connectivity Incident Brief

A diagnosis becomes professionally useful only when another person can understand what happened and what has already been checked. Create a one-page brief in Word or LibreOffice Writer. Keep it

factual: report observations, not guesses. Your classmate will read it as though they are the authorised support technician receiving the incident.

1. Save first as 2026-XX-XX_WorkshopConnectionIncident_v01.docx in the approved Reports folder.
2. Use the seven headings WHAT, WHERE, WHEN, TRIED, EVIDENCE, IMPACT, and ESCALATION.
3. Insert one small, readable screenshot or copied command extract with a caption. Use In Line with Text.
4. State one likely scope and one safe next step. If action is outside your authority, write who should receive the escalation.
5. Exchange files with another pair. The reviewer checks: Can I identify the affected device, reproduce the checks, and understand the requested action?
6. Correct one weakness, save v02, close the file and reopen it to confirm that the final version works.

Incident field	Required content
WHAT	Exact symptom or message.
WHERE	Device name, workshop area and connection type.
WHEN	Time first observed and whether it is continuous or intermittent.
TRIED	Checks completed in nearest-to-farthest order.
EVIDENCE	Key command results, browser result, and verified support source.
IMPACT	What work is delayed or unavailable.
ESCALATION	Who should act next and what evidence is attached.

Lesson/Session Summary

The internet connects devices and services; DNS translates names into addresses; HTTPS protects the connection, not the truth of the content. Search operators improve precision. The 4-W and technical-fit checks protect decisions.

Connection diagnosis should move from nearest to farthest, use several pieces of evidence and end with a clear, safe report.

Class Discussion
Explain DNS using the phone-book analogy.
Which search operator was most useful, and for what task?
Give one red flag that makes you distrust a website.
What are the first two checks when a connection fails?
Which read-only test helped you locate the problem scope, and what could it not prove?

Common Misconceptions Corrected

In pairs, discuss and explain each correction from the following table:

Misconception	Correct understanding
The browser and search engine are the same.	A browser displays web content; a search engine helps locate possible content.
A padlock proves the website is accurate.	HTTPS encrypts the connection; credibility and technical fit still require checking.
The first search result is the best result.	Ranking is not the same as authority, accuracy, or relevance.
Misconception	Correct understanding
If ping fails, the whole internet is down.	Ping can be blocked or ignored. Compare gateway, DNS, browser, and other-device evidence before deciding.
Any network command is safe because it only displays text.	Use only approved read-only commands. Some options change addresses, caches, or settings and are outside a first-year trainee's role.

Assessment

Part A — Choose the correct answer:

1. an ip address is:
a. a password **b.** the unique network number of a device
c. a type of cable **d.** a document format.

2. to search only inside siemens.com, use:
a. "siemens" **b.** -siemens **c.** site:siemens.com **d.** filetype:siemens.

3. in the 4-w test, when checks:
a. price **b.** recency **c.** colours **d.** number of images.

4. the professional connection-check order is:
a. random **b.** nearest to farthest
c. server first **d.** buy a new device.

5. the command used to ask dns for an ip address is:
a. ipconfig **b.** nslookup **c.** filetype:pdf **d.** ctrl + s.

Part B — True (T) or False (F)?

1 HTTPS means the connection is encrypted.
2 filetype:pdf can help locate manuals and datasheets.
3 One failed ping result proves that the whole internet is unavailable.

Part C — Practical search:

Write one search phrase using at least TWO operators to find a PDF maintenance manual for conveyor belts on a manufacturer site.

.....

.....

Part D — Network diagnosis and reporting:

PC-04 shows "Media disconnected" in ipconfig and no default gateway. The nearby PC opens approved websites normally. State the most likely problem scope, name one safe next step, and write a four-sentence incident report using WHAT, WHERE, TRIED, EVIDENCE, and IMPACT.

Lab Performance Checklist

Criterion	2 marks — secure	1 mark — developing	0 marks — not yet
Safe evidence collection	Uses approved read-only checks in order; records accurate results.	One check or result is incomplete.	Uses unsafe commands or invents evidence.
Search and support source	Precise query; official, relevant support source passes 4-W/fit.	Source is relevant, but checking is incomplete.	Random, promotional or unsuitable source.
Diagnosis reasoning	Scope and safe next step follow from several pieces of evidence.	Conclusion is plausible but weakly supported.	Conclusion ignores or contradicts evidence.
Incident brief and escalation	Traceable file; all core fields, readable evidence and clear escalation.	One field or presentation feature needs correction.	Brief is missing, unclear or unsafe.

Self-Assessment Checklist

When I Finish This Lesson, I can:	Yes	Not yet
Explain how a web request travels (device → DNS → server).	☐	☐
Read a URL and explain what HTTPS protects.	☐	☐
Use quotation marks, minus, site: and filetype: in searches.	☐	☐
Apply the 4-W and technical-fit checks.	☐	☐
Do connection checks from nearest to farthest.	☐	☐
Use ipconfig, ping, and nslookup safely and write a clear connectivity incident brief.	☐	☐

Reflect & Connect

Write 2–3 sentences. Think of a recent connection problem on a phone or computer. Which nearest-to-farthest checks would you use now, and what evidence would you report before asking for help?

Glossary قاموس المصطلحات

English term	Meaning	المصطلح بالعربية
Internet	Worldwide network of connected devices	الإنترنت / الشبكة العالمية
IP address	Number identifying a device on a network	عنوان بروتوكول الإنترنت
DNS	System converting names to IP addresses	نظام أسماء النطاقات
URL	Full web address of a page	رابط / عنوان الموقع
HTTPS	Encrypted web connection	بروتوكول نقل آمن
Router	Device directing data between networks	موجه / راوتر
Server	Computer providing pages, files or services	خادم
Search operator	Symbol or command controlling a search	معامل بحث
Credibility	How trustworthy a source is	المصداقية
Technical fit	Suitability for the exact model and conditions	الملاءمة الفنية
Command Prompt	Windows window used to enter text commands	موجه الأوامر
ipconfig	Command displaying IP and gateway information	أمر عرض إعدادات الشبكة
ping	Test checking whether a target replies	اختبار الاتصال
nslookup	Tool asking DNS for an IP address	أداة استعلام نظام أسماء النطاقات
Default gateway	Router address used to leave the local network	البوابة الافتراضية

Abbreviations

Abbreviation	Full form
IP	Internet Protocol
DNS	Domain Name System
URL	Uniform Resource Locator
HTTPS	HyperText Transfer Protocol Secure
ISP	Internet Service Provider
IPv4	Internet Protocol version 4
PDF	Portable Document Format
TLS	Transport Layer Security

Reflect and Connect (At Home)

Write two or three sentences: think of a time a device would not connect to the internet or a network. Using the 'nearest to farthest' idea, what would you check first next time, and why is a calm, ordered check better than random tries?

LESSON 5

Professional Email & Digital Workplace Etiquette

The same email buttons can produce a careless message or a professional one. This lesson is about always choosing the professional path.

Learning Outcomes

By the end of this lesson, you will be able to:

- Apply the rules of professional email tone and etiquette (netiquette).
- Choose correctly between the To, CC and BCC fields.
- Recognize phishing emails and respond to them safely.
- Handle difficult email situations — angry messages, your own mistakes and no reply — professionally.
- Plan, compose, attach, send, verify, reply to and organise a professional maintenance-request email.

WARM-UP

Warm-Up: Fix This Disaster!

Step 1: In pairs, look at the following email and find at least FOUR problems with this email:

"Hey,
Machine 5 is broken again ツ
Fix it or whatever.
bye
Sent from my phone"

Step 2: Rewrite ONE sentence of it professionally.

Step 3: Share: what impression does the original give the supervisor about this technician?

Problems we found:

Introduction:

Writing an email is a little like operating a machine: knowing where the buttons are is only the beginning. The professional skill is using them safely, accurately, and consistently, even under pressure. Think about how much of a modern technician's day passes through a screen. You report faults by email. You receive work orders, safety notices, and shift changes by email. Your supervisor forms an opinion of you — reliable or careless, calm or hot-headed — partly from the way you write. In many companies, your manager will read dozens of your messages long before you ever sit across a desk from them. Your writing arrives before you do.

Today you will master professional tone and recipient choice, how to detect and report phishing, and how to respond calmly when communication becomes difficult. In the lab, these ideas come together as you plan, send, and verify a maintenance-request email with an attachment, then practice professional follow-through.

Scenario: A Day in the Life — Why This Lesson Matters

- Imagine your first week as a junior maintenance technician at a steel-rolling plant. On Monday you report a worn conveyor bearing. On Tuesday a supplier emails a quotation. On Wednesday an urgent-looking message asks you to verify your login. On Thursday a colleague sends an angry email blaming you for a stoppage. On Friday you send the inspection report to the supervisor, verify the attachment, reply to the response and file the record. Every one of those moments appears in this lesson.

Topic 1: Email Etiquette - The 8 Golden Rules

Before we look at any rule, let us understand the big idea behind all of them. Every email you send is a small sample of your professionalism. Get the small things right consistently, and people trust you with big things. This topic gives you eight simple, memorable rules that together form the foundation of professional email.

Before writing, ask three questions: Who will read this? Why am I writing? What should the reader do next? A supervisor may need a decision-focused summary, while a teammate may need practical details. A clear purpose prevents long messages that contain information but no usable request.

Professional email is more than conversation. It creates a dated record of what was reported, requested, approved or corrected. This supports traceability: another person can later see who communicated, what information was available and what action was requested. Technical emails should therefore contain checkable facts, not guesses presented as certainty.

1.1 Your Email Is Your Uniform

In a workshop, your uniform and clean tools signal that you take the job seriously before you have even spoken. In digital communication, your email plays exactly that role. Consider a real situation you may soon face: two technicians apply for the same promotion. Their technical skills are almost identical. But one writes clear, polite, well-structured emails, while the other writes like the warm-up disaster above — no subject, slang, no name. Who looks ready for more responsibility? The manager barely has to think about it.

This is not about being formal for its own sake. It is about respect and clarity. A well-written email respects the reader's time and makes your meaning impossible to misunderstand. In a company, your emails ARE your professional image — so let us learn to keep that image sharp.



Definition - Netiquette

Netiquette (network + etiquette) means polite, professional behaviour in digital communication — the agreed 'good manners' of the digital workplace. Just as etiquette guides how we behave at a shared table, netiquette guides how we behave in a shared inbox.

1.2 The 8 Golden Rules

The table below sets out the eight rules that professionals follow without even thinking. Read each rule together with its reason in the right-hand column — the reason is what turns a rule you memorize into a habit you keep. As you read, try matching each rule to a problem in the warm-up disaster email; almost every rule is broken by that one short message.

	Rule	Why it matters
1	Specific subject line (Type – Machine – Problem).	Busy people decide in about two seconds whether to open your email. A clear subject gets it read.
2	Polite greeting and closing with your full name.	Respect, and the reader instantly knows who is writing and how to reply.
3	Short paragraphs, simple, clear sentences.	Nobody reads a wall of text on a busy shift. Short is professional.
4	Professional tone: no slang, no emojis, no CAPS.	CAPS reads as SHOUTING; emojis read as unserious at work.
5	Proofread before sending.	Spelling and number errors make even good work look careless.
6	Reply within one working day.	Silence blocks other people's work; a quick reply keeps the line moving.
7	Think before you Reply All.	Reply to everyone only if everyone truly needs your answer.
8	Never send when angry — wait, reread, then send.	Emails are permanent; anger is temporary. One angry message can follow you for years.

Notice a pattern across all eight rules: each one is really about respect — for the reader's time, for their inbox, and for your own professional reputation. The poster below gathers the rules into a single image you can picture in your mind while writing. Study it now; many students find they can recall the whole set just by remembering the layout.

The poster below condenses the eight rules into one visual checklist. Read it clockwise and connect each rule to a concrete action you can perform before pressing Send.



Figure 5.1: The 8 Golden Rules of Email — a poster to keep in your mind's eye whenever you write

1.3 Worked Example: The Same Message, Two Ways

Rules are easier to trust when you see them side by side. Below, the same fault report is written twice. Read the weak version first, then the strong version, and notice how each golden rule improves the message. The table places them next to each other so the difference is obvious at a glance.

Weak email ×	Strong, professional email ✓
Subject: (none)	Subject: Repair Request – Machine 5 – Conveyor Belt Fault
Hey boss	Dear Mr. Karim,
Machine 5 is broken again ツ	Machine 5 has stopped due to a conveyor belt fault. Could you approve a repair this morning?
Fix it or whatever.	I have attached two photos.
bye	Best regards,
Sent from my phone	Ahmed Hassan (Maintenance, Line 2)

What changed: a specific subject (Rule 1), a greeting and full name (Rule 2), short clear sentences (Rule 3), a professional tone with no slang (Rule 4), and a clear request the reader can act on. The words took ten extra seconds; the impression lasts far longer.

DID YOU KNOW?

Over 300 billion emails are sent worldwide every day, and studies show office workers spend around a quarter of their working day on email. Clear, professional email is not a small extra skill — it is one of the most-used job skills on Earth. Every hour you invest in writing well pays back for the rest of your career.

Reference: Statista — Daily number of emails worldwide

Topic 2: To, CC, BCC — Who Gets the Email?

Every email pogramme gives you three address fields: To, CC, and BCC. They look almost the same, but choosing the wrong one can expose private information to fifty strangers or bury an important request where nobody acts on it. This topic makes the choice simple and automatic.

Send information only to people who genuinely need it for the task. Do not forward photographs, reports, contact details or production information to personal accounts or public groups. When uncertain, ask the teacher or supervisor before sending.

Choosing recipients is part of information control. Sending too widely can expose personal or workplace information; sending too narrowly can leave an important person uninformed. The field you choose should reflect responsibility and the minimum information needed.

2.1 Small Choice with Big Consequences

Here is a true-to-life scenario. A trainee is asked to announce a training day to 50 colleagues. She types all 50 addresses into the To field and clicks send. Now every one of those 50 people can see every other person's private email address. Worse, one of them clicks Reply All with a joke — and 50 inboxes fill with a conversation none of them needed. Private addresses are exposed, time is wasted, and the trainee looks careless. A single different click — using BCC instead of To — would have prevented all of it.

That is the power hidden in these three little fields. Let us learn them properly so you never make that mistake.

2.2 The Three Fields Explained

The table below defines each field, tells you when to use it, and gives a workshop example so the idea sticks. Read across each row: the field, what it means, what it is FOR, and a concrete example from your own future work.

Field	Meaning	Use it for	Workshop example
To	Main recipient — must act.	The person you need something from.	The supervisor who must approve the repair.
CC	Carbon copy — should know; visible to all.	People who need the information only.	A teammate working on the same machine.
BCC	Blind copy — hidden from others.	Large lists; protecting addresses.	An announcement to 50 trainees — addresses stay private.

Memory line: **To = act. CC = know. BCC = hidden.**

The programme below shows the three fields in action for one message. Follow the arrows from you, the sender, to each type of recipient, and notice how BCC keeps the large list of trainees invisible to one another. Picture this flow whenever you are about to send to more than one person.

The flow diagram below shows that recipient fields assign responsibility and control visibility. Follow each arrow and explain what that recipient is expected to do.

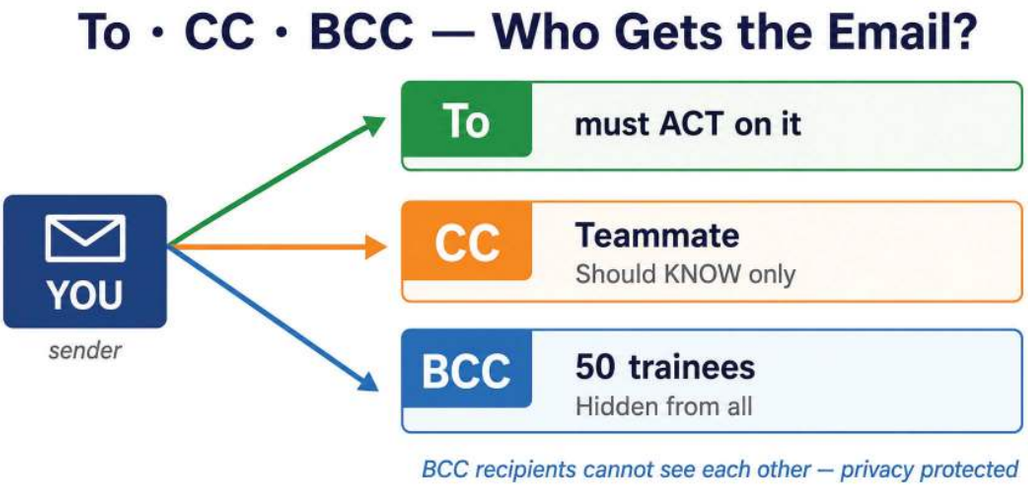


Figure 5.2: To, CC and BCC — the same email reaches three kinds of recipient in three different ways.

Pop-Up Quiz 1

1. **Your supervisor must approve a repair; your teammate should just know about it. The correct setup is:**
   a. both in To
   b. supervisor in To, teammate in CC
   c. both in BCC
   d. teammate in To, supervisor in BCC
2. **WRITING IN ALL CAPITALS in an email is usually understood as:**
   a. politeness
   b. shouting
   c. good formatting
   d. urgency everyone likes

Write your answers here: 1 : ..... **2 :** .....

Topic 3: Phishing — The Email That Bites

Not every email that reaches you is honest. Some are traps, carefully designed to look like a message from your bank, your boss or your IT department. These traps are called phishing, and learning to spot them is one of the most valuable safety skills a modern technician can have. In this topic you will read a real-style story, meet the definition, and learn five warning signs that unmask almost every phishing email.

Recognizing a suspicious message is only the first step. Do not click, reply, download or forward it casually. Report it through the approved channel. If you already clicked, report immediately; hiding the mistake increases the risk.

Phishing is one form of social engineering: manipulating people into revealing information or taking unsafe actions. Attackers often use urgency, authority, curiosity or fear because emotions can bypass careful checking. The safest response is to slow down, inspect the message and use an approved reporting route.

3.1 A True-Style Story

Mechatronics Scenario: The Factory Freeze

A maintenance technician receives an email: 'URGENT: New safety procedure — open the attachment immediately.' The sender looks like the safety manager — but the address is safety-manager@complany-mail.com. Spot the typo: 'complany', not 'company'. In a hurry, he clicks the attachment.

Within minutes, ransomware spreads across the network. Every machine-control PC locks. Production stops for three days, costing the plant an enormous sum. One careful look at the sender's address would have saved the factory. In a modern plant, digital caution IS industrial safety — checking an email is as important as checking a pressure gauge.



Definition - Phishing

Phishing is a fake email that pretends to be from a trusted sender — a bank, a boss, an IT department — and is designed to steal your password, money or data, or to infect the company network with malicious software. The name comes from 'fishing': the attacker dangles bait and hopes you will bite.

3.2 The Five Warning Signs

Almost every phishing email carries at least one of five tell-tale signs. Learn them as a checklist you run in your head before you click anything. The list below explains each sign; then the annotated figure shows all five at once on a single fake email.

1. **Urgency and threats:** 'Your account will be DELETED in 24 hours!' Fear makes people click before thinking.
2. **Strange sender address:** support@micros0ft-help.xyz — look closely, that is a zero, not an 'o'.
3. **Generic greeting:** 'Dear Customer' instead of your real name. A real employer knows your name.
4. **Suspicious links or attachments:** hover over a link (do not click) — does the real destination match the site it claims to be?
5. **Requests for secrets:** a real IT department NEVER asks for your password by email. Ever.

Now look at the annotated email below. It shows a single phishing message with all five warning signs marked and numbered to match the list you just read. Study each red number and connect it to the sign it represents — this is exactly the kind of scanning you should do with any suspicious message.

The annotated message below should be read like a safety inspection. Scan the sender, greeting, urgency, link or attachment, and request for secrets before deciding whether any action is safe.



Figure 5.3: Anatomy of a phishing email — the five warning signs marked 1–5 on one fake message.



Activity 1 *Watch & Spot*

In groups, watch the following video, then do the tasks below.

Video: Spot Phishing Emails

Tasks:

While watching, note TWO warning signs the video mentions.

Your teacher will show a sample email. As a class, vote: phishing or real?

Justify your vote using the five warning signs above.

Warning signs I noted:

Topic 4: When Email Gets Difficult - Professional Responses

So far we have written clear emails and spotted dangerous ones. But some of the hardest moments in a working life happen when an email makes you feel something — anger, embarrassment, frustration. This topic gives you a calm, repeatable method for those moments, so that emotion never writes your emails for you.

A correction email should identify the original message, state the corrected fact clearly and explain whether the correction changes any action. A follow-up should be polite, specific and proportionate. Escalation should follow the workplace chain of responsibility.

4.3 Documenting Corrections and Follow-Up

A difficult email contains two layers: the factual issue and the emotional tone. Professionals answer the factual issue first. They do not copy the sender's anger, speculate about motives or make accusations without evidence.

4.0 Separate the Message from the Emotion

4.1 Not Every Email Is Easy

Picture this. You open your inbox and find an ANGRY message from a colleague, blaming you for a machine fault that was not your fault at all. Your fingers itch to fire back a sharp reply. Stop. This is exactly the moment that separates a professional from an amateur. Difficult email moments happen to everyone; professionals handle them with a method, not with emotion. It is the same problem-solving discipline you use on the workshop floor: UNDERSTAND the situation before you TRY a response.

The good news is that almost every difficult email fits one of a few familiar patterns, and each pattern has a professional response you can prepare in advance. The playbook below is your ready-made guide.

4.2 The Difficult-Situations Playbook

The table below lists four common difficult situations. For each one it shows the wrong, emotional reaction and the professional response beside it. Read each row as a pair — the contrast is the lesson. Notice how every professional response slows the moment down and puts facts and respect ahead of feelings.

Situation	Wrong reaction ×	Professional response ✓
You receive an angry email	Reply angrily within 30 seconds	Wait. Reread it later. Answer only the FACTS, politely; suggest talking in person if needed.
You sent an email with a mistake	Hope nobody notices	Send a short correction quickly: 'Correction: the pressure was 180 bar, not 810. Apologies.'
No reply for two days, work is blocked	Complain about the person to others	Send one polite follow-up: 'Kind reminder — could you confirm...?' Then escalate to your supervisor if still blocked.

Situation	Wrong reaction ×	Professional response ✓
You receive a suspected phishing email	Click to 'check' the link	Do not click. Report it to IT or your teacher immediately.

Look again and you will see the same three-beat rhythm in every row: pause, then think, then respond with facts and respect. The diagram below captures that pattern in three steps. Keep it in mind, because your calm, factual reply is also permanent — and it is what makes YOU look like the professional in the conversation.

The three-step diagram below turns emotional self-control into a repeatable workplace method. The pause creates time, the thinking stage checks facts and consequences, and the response stage produces a calm record.

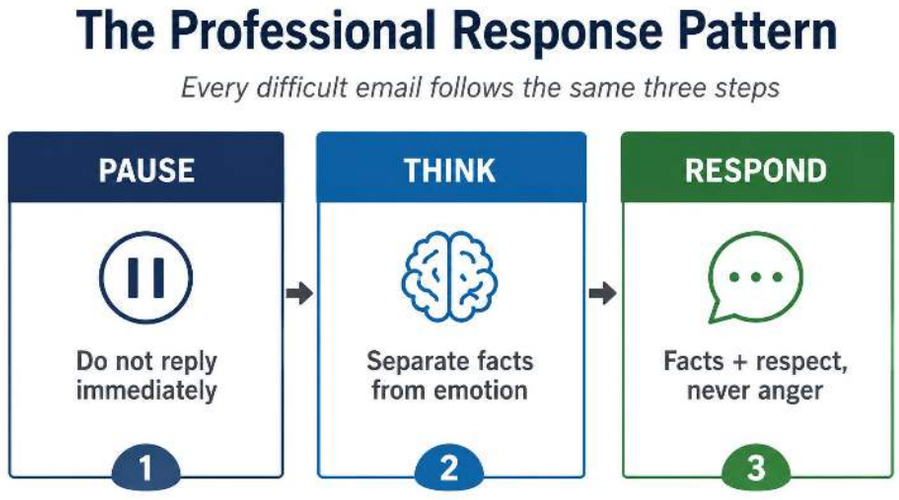


Figure 5.4: The professional response pattern — pause, think, then respond with facts and respect.

Pop-Up Quiz 2

1. An email asks for your password 'for security reasons'. You should:

a. send it quickly

b. never send it — report the email to IT/your teacher

c. send half of it

d. reply asking why

2. You realize you sent a report with a wrong number. The professional action is:

a. stay silent

b. delete your email programme

c. send a short, clear correction with an apology

d. blame the computer

Write your answers:

1 : .....

2 : .....

5 SESSION Writing & Sending a Professional Maintenance-Request Email

Lab Scenario

- This lab applies the four theory topics in one realistic workflow. You will plan who needs the message, compose a maintenance request, attach a report, complete a safety check, send and verify the record, then practice a reply, correction, or follow-up. The goal is not simply to press Send; it is to create a clear, safe and traceable workplace communication.



SAFETY - Email and information boundary

Use only the approved training account, recipient and files. Never include passwords, unnecessary personal data, confidential production information, live-machine access details or unsupported safety claims. Stop and correct the message if the recipient or attachment is wrong.

The lab produces visible evidence. Read the table first so you can check your own work before the teacher records the result.

Evidence	Minimum success condition	Professional purpose
Planning record	Correct To/CC choice, specific subject and clear requested action.	Makes responsibility visible before writing.
Sent maintenance email	Professional body, signature and correct report attached.	Creates a clear and traceable request.
Pre-send and Sent Items check	All five safety checks completed; message and file visible after sending.	Prevents wrong recipients and missing attachments.
Reply/correction/folder evidence	Appropriate response and message stored in Workshop folder.	Shows professional follow-through and record management.

The diagram below shows the complete workflow. Read it from left to right. Each stage protects the next one.

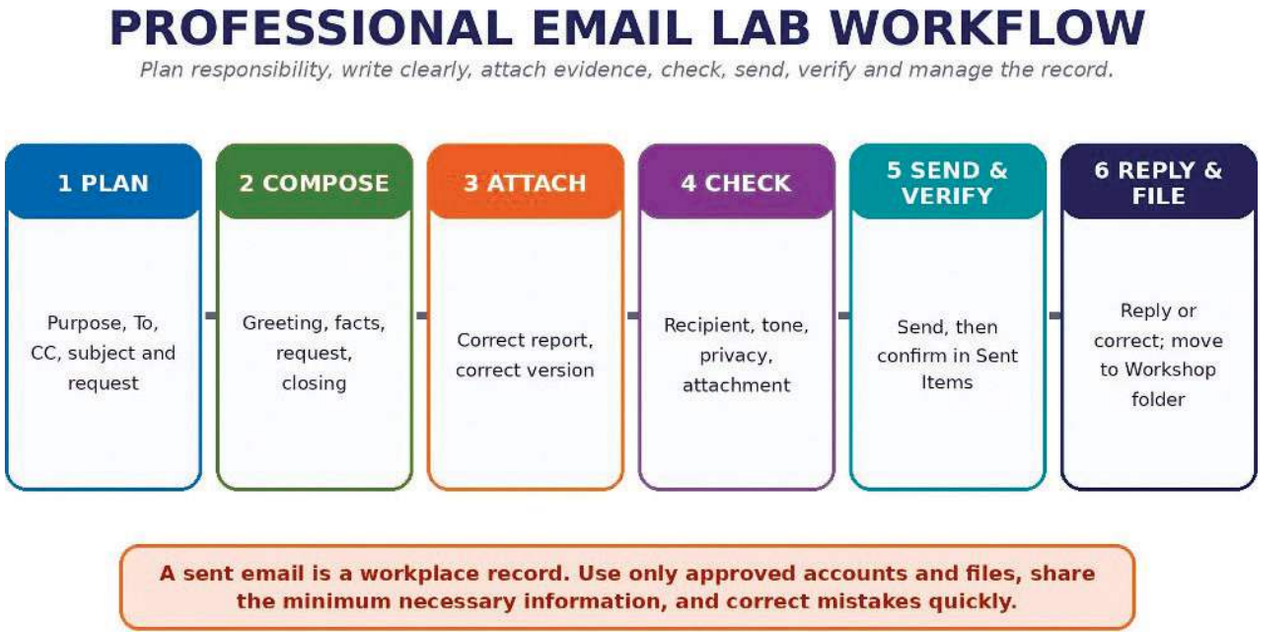


Figure 5.5 — Professional email lab workflow: plan → compose → attach → check → send and verify → reply and file.

Part A: Plan the Message and Choose Recipients

Professional writing begins before the compose window opens. Decide the purpose, the action owner, who only needs awareness, and what evidence must travel with the message.

Planning question	Your decision
Why am I writing?	
Who must ACT? (To)	
Who should KNOW? (CC)	
What is the subject formula?	Request type — Machine — Problem
What verified facts are essential?	
What action am I requesting?	
Which file must be attached?	

Recipient reminder

- To = must act. CC = should know. BCC = hidden and used only when policy requires it, such as a large announcement list.

Part B: Compose, Attach and Send the Maintenance Request

Use your Lesson 2 safety report or the teacher-provided practice report. Keep the email short enough to scan but complete enough to support a decision.

1. Open the approved school email account and select New / Compose.
2. Enter the supervisor in To. Add a teammate in CC only if that person genuinely needs awareness.
3. Write the subject: Maintenance Request — [Machine] — [Problem].
4. Write a professional greeting.
5. Use the three-part body: why you are writing → key verified facts → clear request.
6. Mention the attachment in the body, attach the correct report, and wait for the upload to finish.
7. Close with your full name, programme/group, and role.

Body part	Model sentence
Why I am writing	During this morning's inspection of Hydraulic Press 3, I found a small oil leak near the main cylinder.
Key facts	The emergency stop and pressure gauge worked correctly; one warning label was damaged.
What I request	Please review the attached report and approve an authorised inspection before the machine returns to service.

Part C: Complete the Pre-Send Safety Gate

A professional email is checked twice: before sending and after sending. The diagram below is the final gate. Stop and correct any missing items.

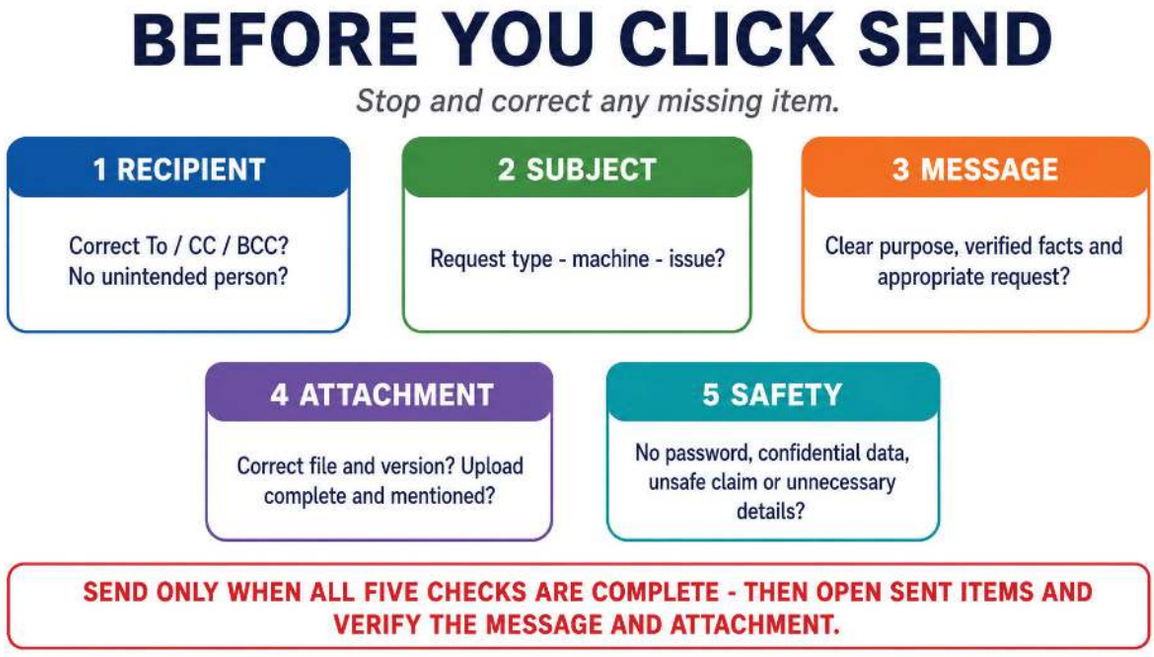


Figure 5.6 — The professional email pre-send gate: recipient, subject, message, attachment and safety.

1. Recipient: confirm every address and remove anyone who does not need the message.
2. Subject: check that the request, machine and problem are visible.
3. Message: confirm professional tone, verified facts, and a clear request.
4. Attachment: open the file from the email preview if policy allows; check the correct version and name.
5. Safety: confirm there is no password, unnecessary personal data, unsafe claim or suspicious link.
6. Send the message, open Sent Items and verify that the correct message and attachment are visible.

Phishing check before opening a received reply

- Trust the sender only after checking the full address, context, link or attachment and any request for secrets. If suspicious: do not click, reply, or download; verify through an official channel and report it.

Part D: Reply, Correct, Follow Up and Organise

Email professionalism continues after the first message. Practice the response that your teacher assigns, then organise the record so it can be found later.

Situation	Professional action	Model opening
Supervisor confirms receipt	Reply to the sender; acknowledge and state the next step.	Thank you for confirming. I will keep the machine isolated until the authorised inspection is completed.
You discover a wrong number	Send a short correction immediately.	Correction to my previous email: the pressure was 180 bar, not 810 bar. I apologize for the error.
No reply and work is blocked	Send one polite, specific follow-up; escalate according to policy if needed.	Kind reminder regarding the Hydraulic Press 3 inspection request sent yesterday. Could you confirm the next authorised action?
Everyone does not need the answer	Use Reply, not Reply All.	Reply only to the person who needs your response.

1. Complete the assigned reply, correction, or follow-up in two or three sentences.
2. Check the recipients before sending; use Reply All only when everyone genuinely needs the answer.
3. Create an email folder called Workshop and move the received or sent practice message into it.
4. Confirm that you can find the message again by subject or folder.

Professional rule

Emails are permanent workplace records. Pause before sending, correct mistakes quickly, protect privacy, and keep the message where authorised colleagues can find it.

Class Discussion — What Have We Learnt?

Which golden rule is most often broken?

When should BCC be used, and why?

Name three phishing warning signs.

How should you respond to an angry email?

What five checks must be completed before clicking Send?

Assessment

Part A: Choose the Correct Answer

1. Netiquette means:

a. network speed **b.** professional behaviour in digital communication

c. a cable **d.** a password.

2. To hide addresses in a large announcement list, use:

a. To **b.** CC **c.** BCC **d.** Subject.

3. "Dear Customer, your account will be deleted, click here" is most likely:

a. maintenance **b.** phishing **c.** a normal reply **d.** a report.

4. The first step after an angry email is:

a. reply in capitals **b.** pause and separate facts from emotion

c. forward it to everyone **d.** delete the account.

5. After sending an attachment, you should:

a. assume it worked **b.** check Sent Items and the attachment

c. share the password **d.** use Reply All.

Part B: True or False

1. A real IT department may ask you to email your password. **True** ☐ **False** ☐

2. A wrong technical number should be corrected quickly. **True** ☐ **False** ☐

3. CC recipients are visible to other recipients. **True** ☐ **False** ☐

Part C: Practical email task

Write a professional subject and a three-part email body requesting authorised inspection of Machine 5 after a conveyor-belt fault. Mention that two photos are attached and include a professional closing.

Lab Performance Checklist

Criterion	2 marks — secure	1 mark — developing	0 marks — not yet
Recipients and subject	Correct To/CC and specific subject.	One minor recipient or subject issue.	Wrong recipient, privacy issue or vague subject.
Email content	Clear three-part body, professional tone and signature.	One omission or unclear sentence.	Incomplete or inappropriate message.
Attachment and safety	Correct file/version; all five checks complete.	One correction needed.	Attachment missing/ wrong or unsafe information included.
Follow-through	Sent Items verified; suitable reply/correction and folder evidence.	One evidence item incomplete.	No verification or record management.

Self-Assessment Checklist

When I finish this lesson, I can:	Yes	Not yet
Apply the 8 golden rules of email etiquette.	☐	☐
Choose correctly between To, CC and BCC.	☐	☐
Identify at least three phishing warning signs.	☐	☐
Respond professionally to anger, mistakes, or no reply.	☐	☐

When I finish this lesson, I can:	Yes	Not yet
Compose and send a professional maintenance-request email with the correct attachment.	☐	☐
Verify Sent Items, reply appropriately, and organise email records.	☐	☐

Reflect & Connect

Write 2–3 sentences. Think of a suspicious or difficult message you have seen. Which warning signs or professional-response steps apply? What will you do differently next time?

Glossary قاموس المصطلحات

English term	Meaning	المصطلح بالعربية
Netiquette	Professional behaviour in digital communication	آداب التعامل الرقمى
Recipient	Person who receives an email	آداب الرسالة
Subject line	Short title explaining the email purpose	عنوان الرسالة
Attachment	File sent with an email	مرفق
Signature	Name and role at the end of an email	التوقيع
CC	Visible copy for people who should know	نسخة كربونية
BCC	Hidden copy used according to policy	نسخة كربونية مخفية
Phishing	Fake message trying to steal data or cause unsafe action	التصيد الاحتيالى
Ransomware	Malicious software that locks files or systems	برمجيات التقنية
Reply All	Reply sent to everyone in To and CC	الرد على الجميع
Follow-up	Polite reminder or request for an update	رسالة متابعة
Correction	Message replacing wrong information with the correct fact	تصحيح
Inbox hygiene	Organizing and managing email records	تنظيم صندوق الوارد
Proofread	Check writing and details before sending	مراجعة/تدقيق

Abbreviations

Abbreviation	Full form
CC	Carbon Copy
BCC	Blind Carbon Copy
IT	Information Technology
SMS	Short Message Service
MFA	Multi-Factor Authentication

Lesson/Session Summary

This summary gathers the key ideas of Lesson 5 and its professional-email practical. Use it as a quick revision map alongside the assessment.

- Professional email is the SHARE stage of the cycle: clear communication that others can act on.
- The 8 Golden Rules of Email keep messages clear, correct and professional.
- A specific subject line – type, machine, problem – lets a reader grasp the message before opening it.
- To, CC and BCC send the same email to different kinds of recipient; BCC hides an address from the others.
- Phishing emails share warning signs – urgency, unusual requests, wrong senders – and are handled by pausing and checking, never rushing.
- A professional response follows a pattern: pause, think, then reply with the right recipients and any attachment.

Reflect and Connect (At Home)

Write two or three sentences: look back at a message you sent recently. Did its subject line tell the reader what it was about? How would the 8 Golden Rules make your everyday messages clearer and more professional?

LESSON 6 Introduction to AI Assistants & Responsible Use



An AI assistant offers ideas while the technician verifies evidence and keeps responsibility.

Core idea

AI can generate useful ideas and drafts, but professional judgement, verification, safety procedures and accountability remain human responsibilities.

Learning Outcomes

By the end of this lesson, you will be able to:

- Explain in simple terms what artificial intelligence, generative AI and a large language model are.
- Describe how an AI assistant produces a response and why fluent language is not proof of accuracy.
- Compare appropriate strengths and important limitations of AI assistants, including hallucinations and bias.
- Apply responsible-use rules covering verification, privacy, honesty, policy, and human accountability.
- Place AI appropriately inside the UNDERSTAND-FIND-TRY-CHECK problem-solving method.
- adopt an ASK ⟶ READ ⟶ VERIFY workflow to critically evaluate AI-generated answers and make the final human decision on what to accept, change, or reject.

Introduction: The Helpful Answer That Was Almost Right

It is 08:10 in the training workshop. A proximity sensor on the conveyor is producing false signals. A trainee asks an AI assistant, "What causes a proximity sensor to trigger incorrectly?" The assistant immediately lists possible causes: incorrect sensing distance, loose wiring, metal interference, vibration, and electrical noise. The list is useful. However, one suggestion refers to a sensor type that is not installed on this conveyor. Another recommendation assumes a supply voltage different from the workshop model. The answer sounds professional, but it cannot be used safely until it is checked against the sensor label, the manufacturer manual, and the real machine.

This short situation shows the correct relationship between a technician and AI. The AI can broaden the search, explain unfamiliar terms, and help organise questions. The technician supplies the local context, protects information, checks the source, works within authority, and tests the result in reality. AI is therefore neither a magic expert nor an enemy. It is a powerful assistant that requires disciplined supervision.

WARM-UP

Human or AI?

Your teacher will read two short maintenance paragraphs using the same neutral voice. One was written by a person, and one was generated by an AI assistant. Vote, discuss the clues you used, and then answer the more important question: can writing style alone prove that information is trustworthy? The expected conclusion is no. Trust must come from evidence, source checking, and applicability to the real task.

1. Vote individually: Text A or Text B?

2. Compare clues with a partner. Consider detail, local context, confidence and wording.

3. Write one method that would verify a technical claim in either paragraph.

Opening principle

- A message can sound human and be generated by AI; it can also sound confident and be wrong. Verification is more reliable than guessing who wrote it.

Topic 1: What Artificial Intelligence Is - and Is Not

Artificial intelligence is a broad field, so beginners can become confused when one word is used for many different technologies. This topic separates the key terms, connects them to familiar examples, and builds a simple mental model of how a language-based AI assistant produces a response. The goal is not to study advanced mathematics. The goal is to understand enough to use the tool intelligently and recognize its limits.

1.1 AI Is Already Part of Everyday Work

AI is not limited to humanoid robots. It appears whenever software recognizes patterns and uses them to classify, recommend, predict, or generate. A phone may recognize a face, an email service may filter spam, a navigation system may estimate traffic, and a factory dashboard may detect an unusual vibration pattern. These systems perform different tasks, but each uses data and computational models to produce an output that would normally require some human judgement.

1.2 Key Definitions

The following table separates terms that are often mixed together. Read each definition beside the workshop example. The important distinction is that "AI" is the broad field, while a generative AI assistant is one particular type of application.

Term	Definition	First-year example
Artificial intelligence (AI)	Software-based systems that perform tasks associated with human intelligence, such as recognizing patterns, understanding language, or making predictions.	A vision system identifies a missing component on an assembly.
Machine learning	A way of building AI in which a model learns patterns from examples or historical data rather than following only fixed hand-written rules.	A model learns normal motor-vibration patterns from past measurements.
Generative AI	AI designed to create new content such as text, images, code, or summaries from a prompt.	An assistant drafts a plain-language explanation of an IP55 rating.
Large language model (LLM)	A generative model trained to process and generate language by modelling patterns in text.	A chatbot answers a question or rewrites a report paragraph.
AI assistant	A user-facing tool that uses one or more AI models to help with tasks.	A chatbot supports brainstorming, explanation, translation, or document review.
Prompt	The instruction, question, and context given to the AI.	"Explain inductive proximity sensors for a first-year technician in 120 words."
Response	The content produced by the AI after the prompt.	A draft explanation that still requires checking.

1.3 How a Language Model Produces a Response

A language model does not write by opening one guaranteed source and copying the correct sentence. During training, it learns statistical relationships between words, concepts, and patterns across a very large collection of examples. When you provide a prompt, it uses those patterns and the prompt context to generate a likely continuation. This explains its strength in language and its vulnerability to plausible mistakes.

Figure 6.1 shows the complete beginner-level process. Read it from left to right. The fifth box is

deliberately included because professional use does not end when the AI finishes writing; it ends when a person has checked and judged the draft.

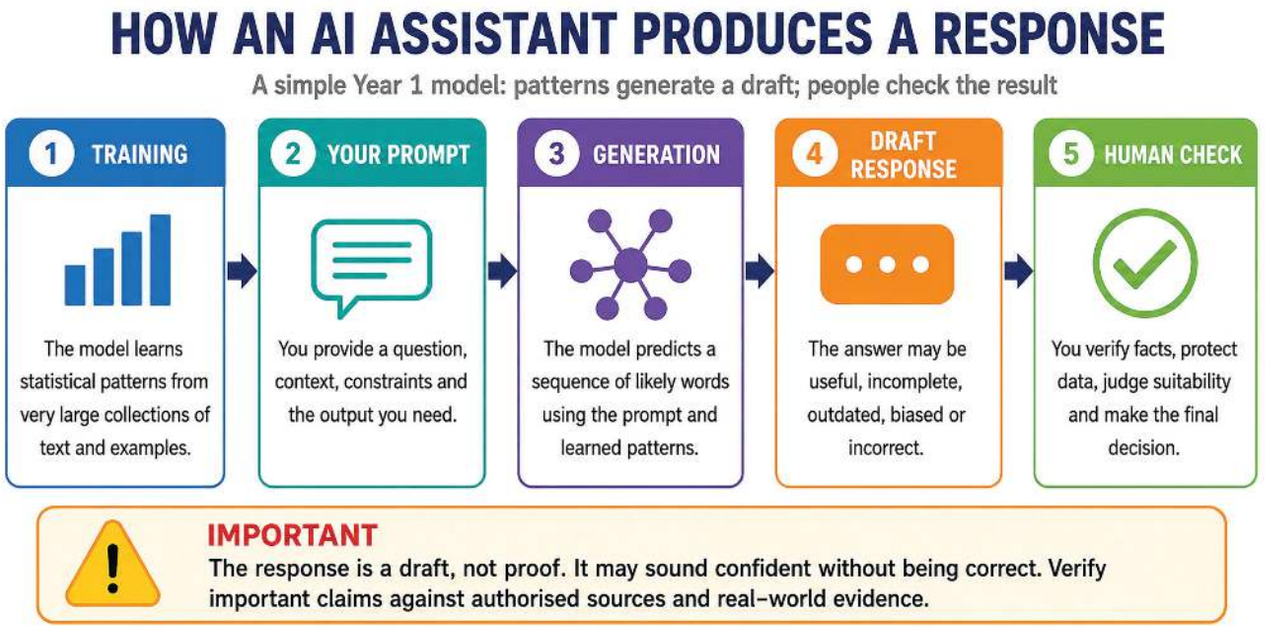


Figure 6.1 - How an AI assistant produces a response and why the response requires human checking.

1.4 What the Model Does Not Automatically Know

An AI assistant may have broad general knowledge, but it does not automatically know your workshop, the exact machine model, the current condition of a component, your school procedures or the authority given to a first-year trainee. Some tools can access current websites or connected files, but that access must never be assumed. Even when a tool can retrieve information, the retrieved source can be incomplete or unsuitable for the task.

1.5 AI, Automation and Human Control

The next table prevents another common confusion. Traditional automation and AI can work together, but they are not the same. The comparison focuses on how each system produces its output and where human oversight remains necessary.

System	How it works	Example	Human responsibility
Fixed-rule automation	Follows explicit programmemed rules or a sequence.	A PLC stops a conveyor when a guard switch opens.	Approve the programmeme, maintain the system, and follow safety procedures.

System	How it works	Example	Human responsibility
Predictive AI	Uses learned patterns to estimate a future condition or identify an anomaly.	A system flags an unusual vibration trend.	Confirm sensor quality, inspect the machine, and decide the authorised action.
Generative AI assistant	Creates a draft response from prompt context and learned language patterns.	Drafts possible inspection questions or a report summary.	Verify content, remove unsafe claims, and take accountability for the final work.

1.6 AI in a Mechatronics Workplace

Industry uses AI for many purposes, but an AI result normally enters a larger system of sensors, software, people, standards and procedures. The table below introduces common uses without suggesting that a trainee should control them. At first-year level, the learning target is to recognise the application and identify the human checking role.

Application	What the system may do	Technician contribution
Predictive maintenance	Identify patterns associated with wear or possible failure.	Check data quality, inspect the asset, and report evidence.
Computer vision	Detect missing parts, surface defects or unsafe conditions in images.	Confirm the camera view and investigate false positives or missed defects.
Anomaly detection	Flag measurements that differ from normal patterns.	Compare the alert with operating context and approved limits.
Energy optimization	Suggest operating patterns that may reduce energy use.	Check production, quality and safety constraints before changes.
Document assistance	Explain terminology, draft text, classify notes or summarize records.	Protect information, verify facts and edit for the intended reader.

Topic 2: Strengths, Limitations and Risk

A responsible user needs a balanced picture. Treating AI as useless prevents learning; treating it as an unquestionable expert creates risk. The best comparison is a fast new assistant who has read widely, communicates confidently and works continuously, but has not inspected your machine and cannot accept responsibility. This topic identifies what to delegate, what to check, and what not to delegate at all.

2.1 What AI Assistants Are Strong At

The strengths table should be read as a list of suitable starting tasks, not automatic final outputs. In every row, the AI creates a draft, option, or explanation that a person can improve.

Strength	Useful example	Required human action
Explaining at different levels	Explain a sensor first in simple language, then with correct technical terms.	Check definitions and remove oversimplification.
Drafting and restructuring text	Turn rough inspection notes into a clear paragraph.	Compare with the notes so no fact is invented or omitted.
Brainstorming possibilities	Generate questions to investigate an intermittent fault.	Prioritize only safe and relevant questions.
Summarizing	Condense a long approved document for revision.	Check that critical conditions and warnings remain.
Language support	Improve grammar or translate a non-sensitive draft.	Verify technical terminology and intended meaning.
Practice and feedback	Create a quiz or role-play questions.	Review answers before using them for study.

2.2 What AI Assistants Are Weak At

The limitations deserve a separate table because they are not simply the opposite of the strengths. Each limitation has a different control: source checking, local context, policy, human approval, or real-world testing.

Limitation	What may go wrong	Control
Factual reliability	The answer may include an invented specification, source, or standard number.	Verify important claims against authoritative sources.
Local context	The model may not know your machine, voltage, procedure, environment or role.	Provide safe context and confirm applicability.
Recency	Information can be outdated, or the tool may not access current sources.	Check dates, versions and live official information.
Bias or incomplete representation	Examples may reflect uneven data or omit groups and alternatives.	Compare sources and ask what may be missing.
Causal judgement	A plausible explanation may not be the true cause of a real fault.	Collect evidence and test safely in the real system.
Accountability	The AI cannot accept legal, academic or safety responsibility.	The authorised person owns the decision and record.

2.3 Hallucination: A Confident False Output

Definition - AI hallucination

An output that presents false, unsupported, or fabricated information as if it were reliable. The term can include invented facts, references, measurements, or connections. It does not mean the system has intentions; it means the generated pattern produced a plausible but incorrect result. Example: An assistant gives a realistic-looking but nonexistent standard number for machine guarding.

Hallucinations are not always dramatic. A small unit error, a wrong model suffix or an omitted condition can be more dangerous because it looks reasonable. A professional verifier checks exact numbers, units, model identifiers, dates, source ownership, exceptions, and safety conditions.

2.4 Confidence Is a Writing Style, Not Evidence

Language models are designed to produce fluent text, so correct and incorrect answers can have the same confident tone. Do not use confidence, length or polished formatting as proof. Evidence comes from authorised documents, reliable measurements, agreement between independent sources and appropriate human approval.

2.5 Match the Checking Level to the Consequence

Figure 6.2 uses a traffic-light model to classify tasks. Green tasks support learning with low consequences. Amber tasks create professional drafts that need verification. Red tasks affect safety, control or authorisation and must remain under approved procedures and qualified human decision-making.

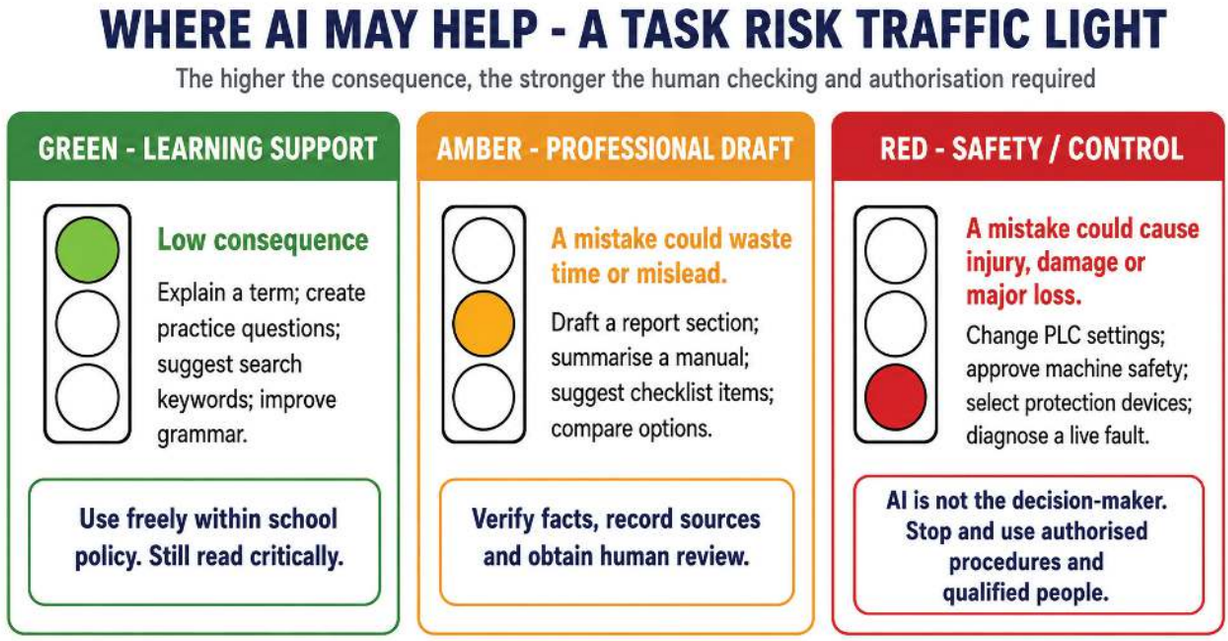


Figure 6.2 - AI task-risk traffic light: learning support, verified professional drafts and prohibited safety/control decisions.

Scenario discussion

An AI assistant suggests a different overload setting for a motor. Which traffic-light category applies? Explain why the answer is red even if the explanation sounds detailed.

Topic 3: Responsible AI Use - A Professional Code

Responsible use is not a list of restrictions added after the technology. It is the method that makes the technology useful. In education, the rules protect learning, privacy, and fairness. In industry, they also protect intellectual property, safety, quality, reputation, and accountability. A professional code converts good intentions into repeatable actions.

3.1 The Five Core Rules

The original five rules from Lesson 5 remain the foundation. Each rule is written as a verb because responsible use is something you do, not something you merely agree with.

1. VERIFY important claims against reliable and applicable sources.

2. PROTECT privacy, personal data, passwords and workplace-confidential information.

3. BE HONEST about meaningful AI assistance and follow assessment requirements.

4. STAY THE DECISION-MAKER: accept accountability and obtain authorised human approval.

5. USE AI TO LEARN: ask for explanations, questions and feedback rather than avoiding your own thinking.

Figure 6.3 places the five rules on one hand so they can be recalled without notes. The centre statement is the controlling principle: the tool proposes, but the user decides and remains accountable.

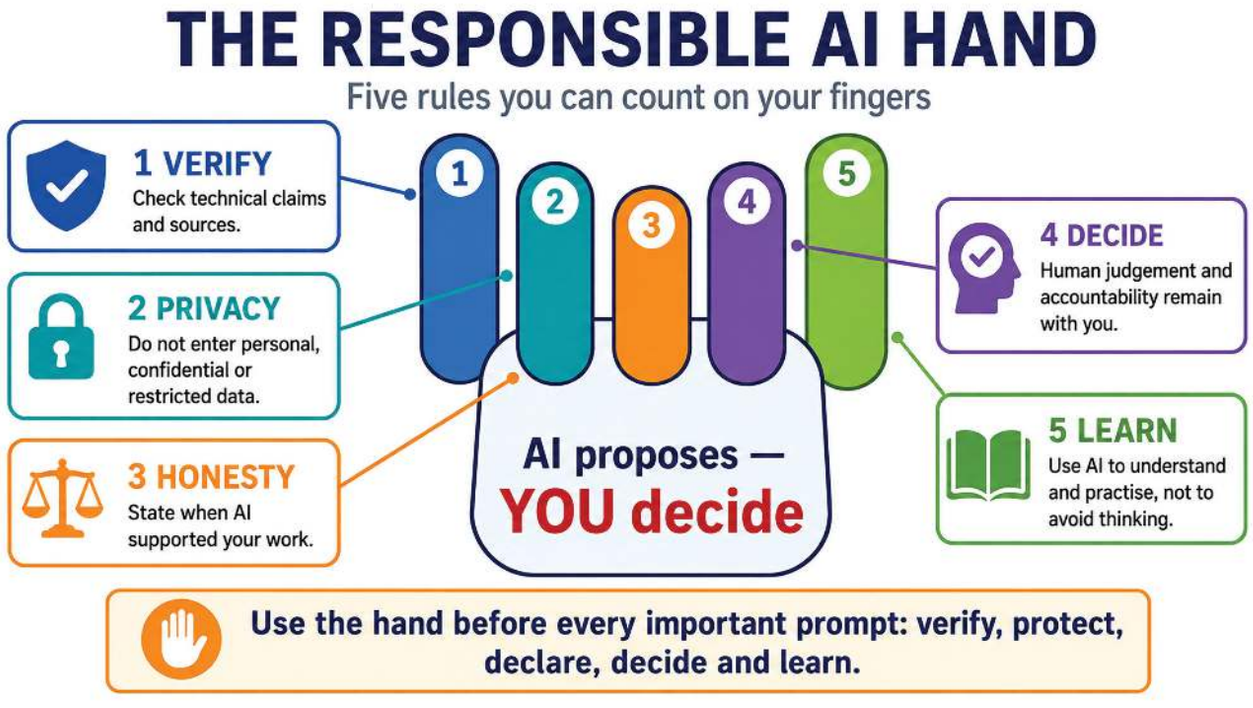


Figure 6.3 - The Responsible AI Hand: verify, protect privacy, be honest, decide and learn.

3.2 Before You Prompt: Data Classification

A safe prompt contains only the information necessary for the task and permitted by policy. Before entering any content, classify it. Public or invented training information may be suitable. Personal data, assessment records, passwords, unreleased designs, customer information, security details, and confidential company records must not be entered into a public tool unless the organisation has explicitly approved the system and the use.

The table below gives practical examples. School and workplace policy always overrides the example, so ask the teacher or supervisor when uncertain.

Data type	Example	Action
Generally suitable	A fictional machine scenario; a public manual paragraph; your own non-sensitive draft.	Use within policy and still verify the output.
Needs caution or permission	A real asset name, internal procedure extract or non-public image.	Ask permission, minimise detail or use an approved internal system.
Do not enter	Passwords, identity numbers, grades, personal contact data, confidential drawings, access codes or incident details.	Stop. Remove the data and use an authorised process.

3.3 Honesty, Authorship and the Decision Record

Honesty means that the final document represents your understanding and judgement. A declaration does not need to be long. For later assessed work such as the capstone, one line records how the tool was used and how the content was verified. Keeping the prompt, response summary, source, and decision also creates traceability: another person can understand how the final checklist was produced.

Example declaration

- "AI assistance was used to generate possible checklist items and improve wording. All final items were selected by the student and verified against the manufacturer manual and teacher-approved safety guidance."

3.4 A Practical Verification Ladder

Verification is stronger when it uses several levels instead of one quick web search. Start by reading the answer critically, then check an authoritative source, confirm that the information applies to the exact task, obtain human approval when required, and finally compare the claim with real-world evidence. Not every classroom explanation needs all five levels, but safety-related content does.

Use the ladder from top to bottom. The later steps do not replace the earlier ones; they add confidence and context.

Level	Question	Evidence
1. Internal check	Is the response clear, consistent and free from obvious unit or logic errors?	Careful reading and calculation.
2. Source check	Does an authoritative and current source support the claim?	Manufacturer, regulator, standards body or approved textbook.
3. Applicability check	Does it match the correct model, environment, version and role?	Nameplate, model number, local procedure and task context.
4. Human approval	Is qualified or supervisory approval required?	Teacher, supervisor, engineer or authorised technician.
5. Reality check	Does observation or safe testing support the conclusion?	Measurement, inspection, controlled test and recorded result.

Topic 4: Using AI Inside a Safe Problem-Solving Method

The UNDERSTAND-FIND-TRY-CHECK method has appeared throughout the unit. AI does not replace this method; it occupies a limited place inside it. The tool is most useful when generating search terms, explaining possibilities or comparing non-sensitive options. Observation and checking remain tied to the real system, and every physical action remains controlled by safety procedures and authority.

4.1 The Role of AI at Each Step

Read the next table across each row. The middle column shows a useful contribution. The final column shows the boundary that prevents the assistant from becoming the decision-maker.

Step	Useful AI contribution	Human boundary
UNDERSTAND	Help organise notes or create questions after the symptoms have been observed.	The person observes the real system, records facts and identifies safety limits.
FIND	Suggest keywords, possible causes, comparison criteria or explanations.	Use AI as one source; verify against authorised information.
TRY	Help compare already authorised options or word a safe test plan.	A qualified person chooses and performs one safe, reversible action.
CHECK	Help structure a result record after evidence is collected.	Reality, measurement, and authorised review determine whether the problem is solved.

4.2 Worked Scenario: False Proximity-Sensor Triggers

A conveyor sensor reports objects when the nearby press starts. The trainee first records that the false trigger occurs only while the press operates. In FIND, the trainee uses the manual, teacher guidance, and an AI assistant to identify possible electromagnetic interference, vibration, incorrect distance, and wiring problems. The manual confirms that cable routing and shielding are relevant. An authorised person selects one safe inspection action. After the change, the system is tested under controlled conditions, and the result is recorded. The AI helped expand the investigation, but the evidence and decision came from the real system and authorised people.

Figure 6.4 places the AI boundary directly on the four-step loop. Notice the central stop condition. It applies before convenience: when a task involves a live machine, electrical risk, restricted settings, confidential data or authority limits, the trainee stops and escalates.

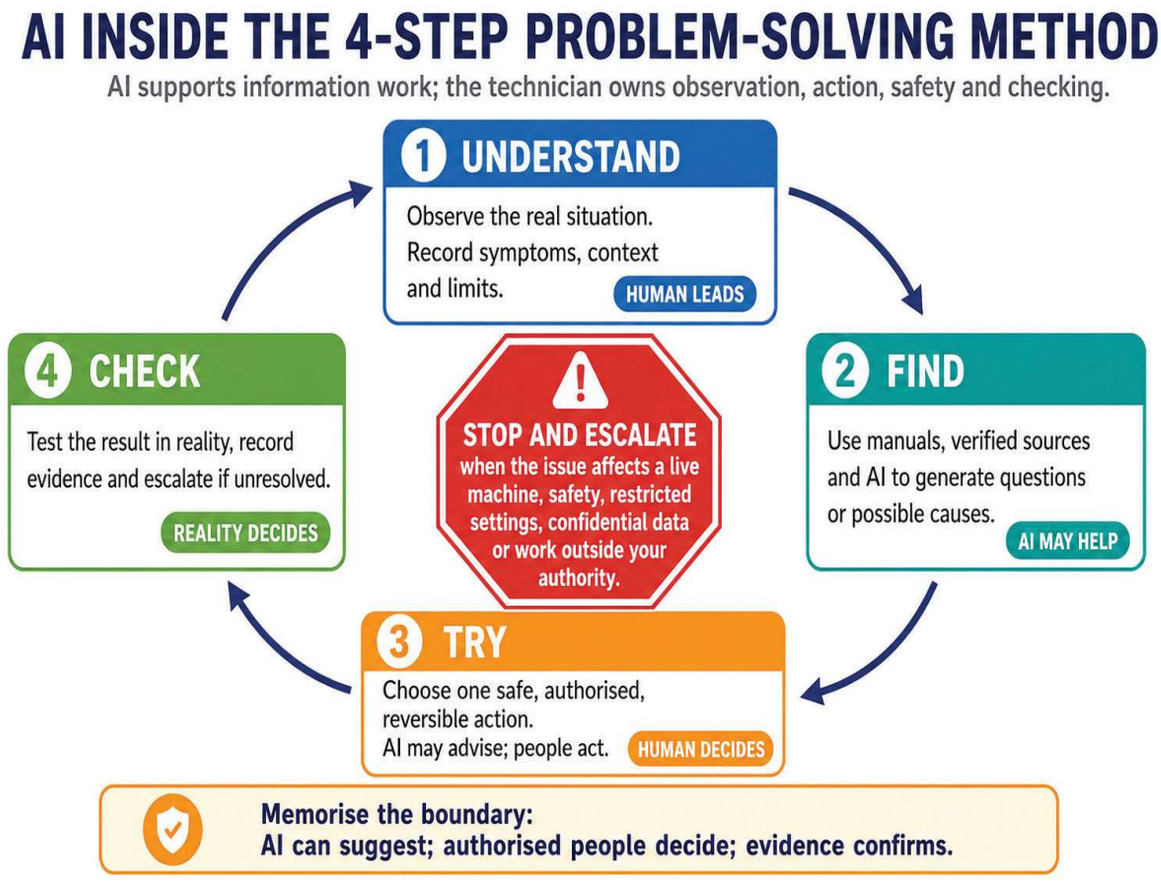


Figure 6.4 - AI inside UNDERSTAND-FIND-TRY-CHECK: support is limited; human responsibility remains central.

4.3 Reporting an AI-Assisted Investigation

A technical record should separate observation, sources, AI suggestions, decisions, and results. This prevents a generated possibility from being mistaken for a confirmed fact. The short evidence log below can be used during the upcoming capstone and later projects.

Each row answers a different question. Keeping the rows separate makes the reasoning auditable and helps another technician continue the investigation.

Record element	What to write	Example
Observed fact	What was seen, heard or measured.	False trigger occurs when the press motor starts.
Authoritative source	Document title, owner, model/ version and relevant section.	Manufacturer manual, wiring section, model PXS-18.
AI suggestion	The idea generated, without presenting it as fact.	Possible electrical interference or cable-routing issue.
Decision	Accept, reject or verify first, with reason.	Verify first - source confirms interference is possible.
Action and result	Authorised action, evidence and outcome.	Cable inspected by instructor; retest completed; result recorded.

Topic 5: Prompting for Technical Learning

A prompt is not a magic phrase; it is a task specification. A strong prompt tells the assistant what to do, what context matters, who the reader is, what limits apply, and what form the answer should take. This reduces guesswork and makes the response easier to evaluate. It does not remove the need for verification.

5.1 A Simple Prompt Formula

The formula below is designed for first-year technical tasks. Not every prompt needs a long sentence, but including the relevant parts improves focus and reveals missing context.

Prompt part	Question to answer	Example
Purpose/role	What support do I need?	Act as a learning assistant, not a machine-control expert.
Context	What safe background can I provide?	This is a Year 1 training task about a pedestal drill.
Task	What exactly should the assistant produce?	Suggest questions for a daily visual inspection checklist.
Constraints	What must it avoid or respect?	Do not give operating instructions; use general language; flag uncertainty.
Format	How should the output be organised?	Return a -3column table: item, reason, verification source needed.

5.2 Weak and Strong Prompt Example

The two prompts below request the same general topic. Compare the amount of safe context and the ease of checking the output.

Weak prompt	Stronger prompt
"Give me drill checks."	"For a first-year mechatronics student, suggest eight candidate items for a daily visual inspection checklist for a pedestal drill in a training workshop. Do not provide operating or repair instructions. For each item, state why it matters and mark whether it must be verified against the manufacturer manual or local procedure."

5.3 Useful Follow-Up Prompts

- "Which parts of your answer are uncertain or depend on the machine model?"
- "Separate observable checks from operating tests that require teacher authorisation."
- "Rewrite the items so a new trainee can understand them without changing the technical meaning."
- "Create a verification checklist for the exact numbers, units, model and source date."
- "Give two alternative search phrases for finding an official manual."

Important caution

- An AI-generated citation or link can itself be wrong. Open the source, identify the publisher, check the title and date, and confirm that the text actually supports the claim.

6 SESSION Using an AI Chatbot to Explain a Technical Concept & Verifying the Answer

Lab Introduction

- You are about to meet a new kind of tool. An AI chatbot — such as ChatGPT, Claude, Gemini or Copilot — is a computer programme you talk to in ordinary language, just as you would ask a knowledgeable colleague. It has learned from an enormous amount of written text and can explain ideas, draft messages, translate and summarise in seconds. Technicians increasingly use chatbots for quick explanations and writing help, and knowing how to use one well is fast becoming a basic workplace skill.
- But here is the crucial warning that shapes this entire lab: an AI can sound 100% confident while being completely wrong. It does not 'know' facts the way a textbook does; it predicts likely words. Most of the time that produces a good answer — but sometimes it produces a confident mistake. The professional skill, therefore, is not simply asking an AI. It is the full cycle: ASK smartly, READ critically, then VERIFY against a reliable source before you trust or act on the answer.
- Before you begin, learn the four key terms for this lab. The table below gives each one a simple meaning. Keep these words in mind — PROMPT and VERIFY are the two most important words of the day.

Term	Meaning
Chatbot	An AI programme you talk to in normal language.
Prompt	The message or instruction you send to the AI.
Response	The AI's answer — fluent, confident... and possibly wrong.
Verify	Check the answer against a reliable source before trusting it.

Explain a Technical Concept with AI — and Verify It

- **Why this lab?** Today you practise the professional AI cycle on a real mechatronics concept — and then use AI to polish the email writing you started in Lesson 4.
- **You will need:** a computer with internet, access to the AI chatbot chosen by your teacher, Microsoft Word, and your folder system from earlier lessons.

The whole lab follows the three-step cycle shown below. Look at it before you start so you always know which stage you are in. Notice that the final stage — VERIFY — is the one most people skip, and it is exactly the stage that will make you better than an AI user who never checks.

The cycle below shows that responsible AI use does not end when an answer appears. The answer must be read critically and compared with evidence before it is copied into technical work or used in a decision.

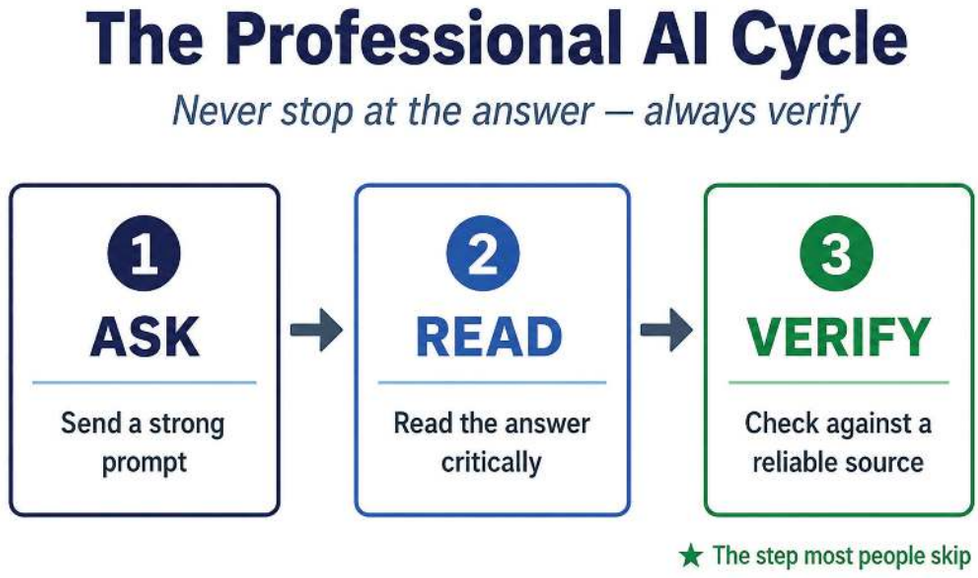


Figure 6.5: The professional AI cycle — ASK, then READ critically, then VERIFY. Never stop at the answer.

Task, audience, topic and context form the foundation. You can improve the result further by adding length, language level, prohibited content and output format.

A Strong Prompt Includes Constraints and Format

An AI assistant can explain, reorganise, suggest and draft. It cannot inspect the real machine, confirm a live safety condition, approve a repair or take professional responsibility for a decision.

What AI Can and Cannot Do

Never paste passwords, personal contact details, student records, confidential company information, unpublished drawings or identifiable incident information into a public AI tool. Replace real names and numbers with safe fictional examples.

AI Safety Boundary: Protect Data Before You Prompt

Part A: Meet the AI and Learn to Prompt

Part A is an orientation practical. Its purpose is to help you notice how an AI conversation works before you depend on it for technical content. Success means that you can explain why one prompt is more useful than another.

The quality of an AI's answer depends almost entirely on the quality of your question. A vague prompt gives a vague, generic answer; a precise prompt gives a precise, useful one. This is the single most important habit to build in Part A.

Definition - Prompt

A prompt is the message you send to an AI. The quality of the answer depends on the quality of your prompt — much like a search engine, but written in full sentences instead of keywords.

Compare the two prompts in the table below. The weak prompt gives the AI almost nothing to work with. The strong prompt tells it four things at once — and those four things are the secret formula you will use all lesson. Read both, then read the explanation of the formula underneath.

Weak prompt ✗	Strong prompt ✓
motor	Explain in simple words, for a first-year mechatronics student, what a proximity sensor is, and give one example of how it is used on a conveyor belt.

The strong prompt tells the AI four things: the **task** (explain), the **audience** (first-year student), the **topic** (proximity sensor) and the **context** (conveyor-belt example). Remember the formula: **Task + Audience + Topic + Context**.

The figure below shows the formula as four building blocks that snap together into one strong sentence. Picture these blocks whenever you write a prompt — build all four and your prompts will be strong every time.

The prompt blocks below show how useful instructions are constructed. Each block adds information that reduces ambiguity and helps the AI produce an answer suited to the learner and workshop context.

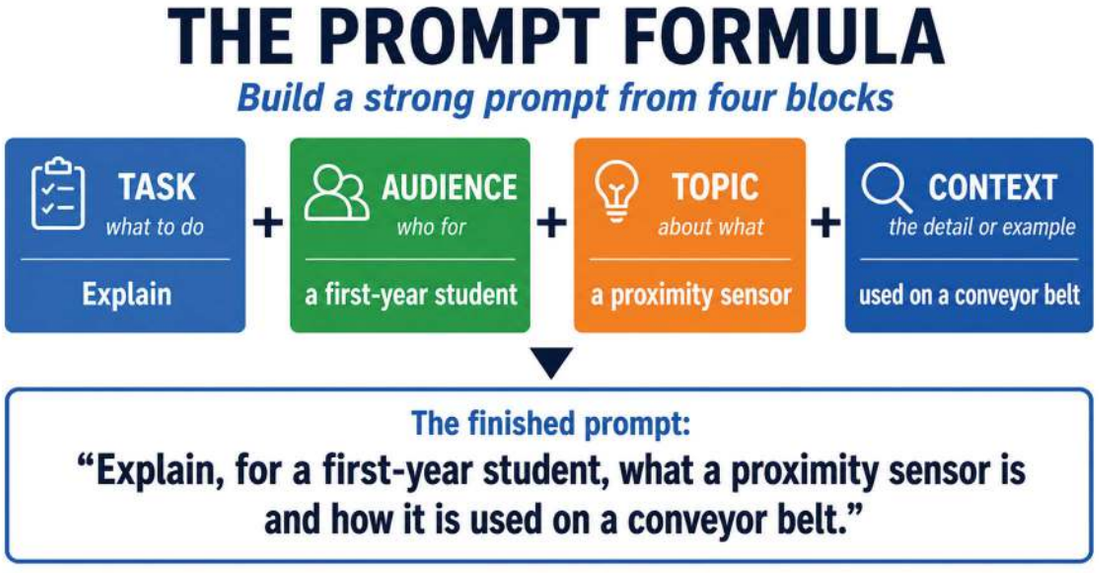


Figure 6.6: The Prompt Formula — Task + Audience + Topic + Context click together into one clear prompt.

- **Step 1:** Open the AI chatbot your teacher assigned. Type a friendly test: 'Hello! Can you help me learn mechatronics?'
- **Step 2:** Read the answer. Notice how polite and confident the AI sounds — keep that word 'confident' in mind for Part C!

Part B: Ask the AI to Explain a Technical Concept

Part B moves from demonstration to guided production. You will build a prompt for a real mechatronics concept, ask a follow-up question and create a saved evidence file that another person can review.

Now you put the prompt formula to work on a real idea from your field. Work with your partner, one screen between you, and talk through each step.

- **Step 1:** Your teacher assigns your pair ONE concept: proximity sensor / hydraulic vs pneumatic systems / what a PLC does / AC vs DC motors / what IP55 means.
- **Step 2:** Write a STRONG prompt using Task + Audience + Topic + Context. Send it.
- **Step 3:** Read the answer together. Ask a follow-up question about anything unclear, for example: 'Can you explain that last part more simply?' A chatbot is a conversation, not a single search.
- **Step 4:** Open Word and save a new file as 2026-XX-XX_AI_ConceptCheck.docx in your 01_Reports folder. Copy the AI's main explanation into it (3–5 sentences) under the heading 'AI Answer'.

Part C: TRUST BUT VERIFY: The Fact-Check

Part C is the assessed reasoning stage. Compare definitions, examples, numbers and operating conditions. A statement can be broadly correct but still too general for a particular machine or safety decision.

This is the heart of the whole lesson — the professional step that most people skip. The question is simple but powerful: is the AI's answer actually correct? You already have every tool you need. From Lesson 4 you have search operators to FIND a reliable source, and the 4-W test (Who, What, When, Why) to JUDGE whether that source can be trusted.

Verifying is not a sign that you distrust technology; it is the mark of a careful professional. You would never install a machine part just because 'someone in a café said so'. Treat an unverified AI answer with exactly the same healthy caution.

- **Step 1:** Use your Lesson 4 search skills to find ONE reliable source on the same concept — a manufacturer's site, GCFGlobal, a standards body, or an .edu page. Apply the 4-W test to it.
- **Step 2:** Compare the AI answer with the reliable source, point by point. In your Word file, under a heading 'Verification', complete the table below.

Fill in the verification table below in your Word file. Each row asks you to place the AI's claim next to what the reliable source says, then decide whether they match. This side-by-side habit is exactly how professionals check facts.

Check	AI answer	Reliable source	Match? ✗ / ✓
Definition			
Example given			
Any numbers / values			

- **Step 3:** Write a ONE-sentence verdict: 'The AI answer was correct / partly correct/incorrect because...' and paste the link of your verification source.
- **Step 4:** Class moment: did ANY pair find a mistake or a too-general answer from the AI? Share it — finding an AI mistake makes you smarter than the user who never checks!

Golden Rule of AI

- AI is a smart assistant, not an authority. For anything technical, safety-related or graded, verify with an official source before you act on it. You would not install a part because 'someone in a café said so' — treat an unverified AI answer exactly the same way.

Part D: AI as Your Writing Coach: Level Up Your Email

Part D uses AI as a writing coach rather than an automatic writer. Every accepted change needs a reason, and rejecting an unsuitable suggestion is evidence of good judgement.
In the final part, you turn the AI into a writing coach. But remember the golden rule you have just learned: the AI proposes, and YOU decide. You are the technician; the AI is only an assistant offering suggestions.



Activity 1 — *The Email Makeover*

Step 1: Copy the disaster email from the warm-up from Lesson 5 (hey, machine 5 is broken...') into the AI with this prompt: 'Rewrite this as a professional maintenance-request email from a technician to a supervisor. Keep it short.'

Step 2: Compare the AI version with the 8 golden rules from Topic 1 — does it follow all of them? Which rule does it apply best?

Step 3: Now improve YOUR OWN Lesson 5 email: paste it in and ask, 'Suggest two improvements to make this email more professional. Explain why.'

Step 4: Decide with your partner: which suggestions do you ACCEPT, and which do you reject? (AI proposes, YOU decide.) Note one accepted improvement below.

Step 5: Save your Word file. Final Ctrl + S!

One improvement I accepted, and why:

Lesson/Session Summary

- AI systems recognize or generate patterns; a language model produces likely text rather than guaranteed facts.
- Fluent, confident writing is not evidence. Hallucinations and missing context require verification.
- Responsible use means verifying, protecting data, being honest, keeping human accountability, and using AI to learn.
- AI is strongest as a helper in FIND and as a limited adviser in TRY. Observation, authorised action, and reality-based checking remain human.
- Why must an AI answer be verified even when it sounds confident?

Assessment

Complete this assessment after the practical or at the beginning of the next lesson. Answer from understanding rather than copying the summary.

Part A: Choose the Correct Answer

1. A language model mainly produces a response by:

a) retrieving guaranteed facts only

b) generating likely text from learned patterns and prompt context

c) asking a hidden human

d) measuring the machine

2. A confident but fabricated technical reference is an example of:

a) version control **b)** hallucination

c) encryption **d)** automation

3. Which prompt content is generally suitable?

a) a password **b)** a classmate's grade

c) a fictional technical scenario **d)** a confidential drawing

4. Which step is the main place for AI-supported searching and explanation?

a) UNDERSTAND **b)** FIND

c) CHECK **d)** none

5. An AI suggests a new safety check. The first professional action is to:

a) add it immediately **b)** verify it against authorised sources and scope

c) hide the use of AI **d)** assume it is correct

Part B: True or False

1. An AI assistant can sound confident while being wrong. **True** ☐ **False** ☐

2. A first-year trainee may use AI to approve a live-machine control change. **True** ☐ **False** ☐

3. Protecting confidential information applies before a prompt is submitted. **True** ☐ **False** ☐

4. declaration of AI assistance supports honesty and traceability. **True** ☐ **False** ☐

Part C: AI and verification:

Write a strong prompt asking for a simple explanation of an emergency stop button for a first-year student, with a workshop example and a four-sentence limit. Then name one suitable source type you would use to verify it and explain why.

Lab Performance Checklist

Criterion	2 marks — secure	1 mark — developing	0 marks — not yet
Prompt quality	All five blocks are clear; no sensitive data.	Prompt is useful, but one block is weak.	Prompt is vague, unsafe, or copied without understanding.
Search and source choice	Precise query and authoritative relevant source.	Source is relevant, but checking is incomplete.	Random or unsuitable source.
Verification reasoning	Claims compared; verdict and correction are evidence-based.	Some comparison is present but lacks detail.	AI answer accepted without checking.
Record and judgement	File is traceable; accept/ modify/reject decisions are justified.	File or reasons need one correction.	Evidence missing or decision unexplained.

Self-Assessment Checklist

When I finish this lesson, I can:	Yes	Not yet
Explain AI, generative AI and an LLM in simple terms.	☐	☐
Describe why an AI response can be fluent but incorrect.	☐	☐
Apply the five responsible-use rules.	☐	☐
Classify AI tasks by consequence and required checking.	☐	☐
Place AI correctly in UNDERSTAND-FIND-TRY-CHECK.	☐	☐
Write a strong AI prompt and verify the response with a reliable source	☐	☐

Reflect and Connect

1. Write 2–3 sentences. Think of a recent online or AI-generated claim. Apply WHO, WHAT, WHEN, and WHY. Does it still pass? What will you do differently before using or sharing similar information?

2. Write two sentences about the full unit: How AI assistant changed your work? What should an AI assistant never decide for you?

Glossary قاموس المصطلحات

The dictionary table supports technical vocabulary development. Use the English term in class and practical work, use the simple meaning to check understanding, and use the Arabic equivalent as a learning support.

English term	Meaning	المصطلح بالعربية
Artificial intelligence	Software systems performing tasks associated with human intelligence.	الذكاء الاصطناعي
Machine learning	Models learning patterns from examples or data.	تعلم الآلة
Generative AI	AI that creates new content from a prompt.	الذكاء الاصطناعي التوليدي
Large language model	A model that processes and generates language patterns.	نموذج لغوي ضخم
Prompt	The instruction and context given to an AI assistant.	موجه / تعليمات للذكاء الاصطناعي
Hallucination	A confident but false or unsupported AI output.	هلوسة الذكاء الاصطناعي
Verification	Checking a claim against reliable evidence.	التحقق
Privacy	Protection of personal or sensitive information.	الخصوصية
Accountability	Responsibility for a decision and its result.	المساءلة / تحمل المسؤولية
Triangulation	Comparing more than one source or evidence route.	التحقق من مصادر متعددة
Authoritative source	A source with recognized responsibility for the information.	مصدر موثوق / رسمي

Abbreviations

These abbreviations appear in AI and industrial digital-work discussions.

Abbreviation	Full form
AI	Artificial Intelligence
LLM	Large Language Model
ML	Machine Learning
IoT	Internet of Things
PdM	Predictive Maintenance

LESSON 7

Unit Assessment

This lesson is the theory checkpoint for the whole unit. It does not teach new content; instead, it revisits the key ideas of Lessons 1 to 6, shows how they connect, and prepares you for the written unit assessment.

Learning Outcomes

By the end of this lesson, you will be able to:

1. Recall and connect the core ideas from all six lessons of the unit.
2. Explain how the FIND ⟶ EVALUATE ⟶ USE ⟶ CREATE ⟶ SHARE cycle runs through every lesson.
3. Answer knowledge questions across digital literacy, file management, Word, internet search, email and AI.
4. Identify your own weak areas so the revision lesson can target them.

Introduction: The Whole Unit at a Glance

You have travelled a long way. You began by learning what digital literacy is and why it matters in a mechatronics workshop; then you mastered the computer and its files, produced professional Word documents, searched and evaluated online information, communicated by professional email, and used AI assistants responsibly. This lesson ties those threads together so you can see the unit as one connected skill – the digitally-literate technician – rather than six separate topics.

Topic 1: The Six Lessons as One Cycle

Every lesson in this unit was really a different part of the same cycle: finding information, evaluating it, using it, creating something from it, and sharing it – always safely and responsibly. Seeing this connection is the key to the assessment, because the questions test whether you can apply the ideas together, not just recite them.

The table below maps each lesson to the cycle stage it strengthens most. Read across each row to see how the unit builds one continuous professional habit.

Lesson	Main skill	Strongest cycle stage
1	Digital literacy & Industry 4.0	The whole cycle, introduced
2	Computers, OS & file management	USE and CREATE (store, organise, protect)
3	Word processing	CREATE (produce clear documents)
4	Internet & safe search	FIND and EVALUATE (locate and trust)
5	Professional email	SHARE (communicate correctly)
6	AI assistants & responsible use	EVALUATE and CREATE (assist, then verify)

Topic 2: Key Ideas to Carry Into the Assessment

Some ideas appeared again and again across the unit because they are the most important. This topic gathers them so you can check you hold each one firmly before the assessment.

The table below lists the recurring big ideas and where you met them. If any row feels uncertain, mark it – that is exactly what Lesson 8's revision will target

Big idea	Where you met it	Confident? (mark)
Accuracy, safety, security, efficiency, communication	Lesson 1 Professional Standard	
A clear folder tree and file names	Lesson 2 file management	
Check before clicking or downloading	Reduces malware and incorrect-source risk.	
Styles build structure (and a table of contents)	Lesson 3 Word	
The 4-W credibility test for a website	Lesson 4 internet	
To / CC / BCC and phishing warning signs	Lesson 5 email	
AI can be confidently wrong – always verify	Lesson 6 AI	

Topic 3: How the Assessment Works

Knowing the shape of the assessment removes anxiety and helps you prepare well. This topic explains what to expect so you can walk in confident.

The written assessment has three parts, described in the table below. Part A checks recall, Part B checks judgement, and Part C checks whether you can apply the ideas to a short workshop scenario – the same blend the whole unit practiced

Part	What it tests	Format
A	Recall of key facts	Multiple choice
B	True/false judgement	Short statements
C	Applying ideas to a scenario	One short written answer

Unit Assessment

Part A – Choose the correct answer (1 mark each):

1. Digital literacy mainly adds, beyond operating a device:

a. faster typing **b.** judgement, safety and responsibility

c. more games **d.** louder speakers

2. The best place to store a finished inspection report is:

a. the Desktop **b.** a clearly named file in the correct project folder

c. Downloads **d.** an email draft

3. A table of contents in Word can be built automatically from:

a. bold text **b.** heading styles

c. coloured text **d.** page numbers typed by hand

4. A trustworthy website is best judged by:

a. how colourful it is **b.** the 4-W test – who, what, when, why

c. how high it ranks **d.** how many adverts it has

5. An AI assistant that invents a confident but false reference is showing:

a. creativity **b.** a hallucination that must be verified

c. a virus **d.** a network error

Part B – True (T) or False (F) (1 mark each):

1. BCC hides a recipient's address from the other recipients. .....

2. It is safe to enter real passwords into a public AI chatbot. .....

3. A clear file name and folder location make evidence easy to find later. .....

Part C – Short scenario answer (2 marks):

Your supervisor asks you to find a motor datasheet online, save it, and email a one-line summary. Describe, in order, which cycle steps you use and one professional standard you apply at each.

.....

.....

.....

Unit Written Assessment Pass Criterion

Duration: 4 minutes - Total: 10 marks

Criterion	Pass Standard	Evidence
Demonstrate understanding of the unit's key concepts	Achieves at least 60% of the total marks in the Unit Assessment, including correct responses to questions covering digital literacy, file management, Word, internet search, email, and AI.	Completed Part A, Part B, and Part C of the written assessment.

7 SESSION Unit Assessment Activity: Building an Equipment Inspection Checklist

The Complete Equipment Inspection Checklist

An inspection checklist is a controlled document that lists the items a person must inspect and provides a consistent place to record status, evidence, and required action. Checklists reduce dependence on memory, improve communication between shifts, and support traceability. However, a checklist is only safe when its content comes from authorised sources and its scope is clear. It must never instruct a first-year student to perform unauthorised testing, bypass a guard, or work on live equipment.

In this capstone, you will create a checklist for a machine assigned by the teacher, such as a pedestal drill, bench grinder, compressor, hydraulic press, or training conveyor. The document should be useful to a new trainee performing a teacher-approved pre-use visual inspection. The exact content must be adapted to the manufacturer's manual, school procedure, and machine condition.

Capstone Deliverable and Success Criteria

The table describes the final evidence. Use it at the beginning to plan and again at the end to check completion.

Deliverable element	Minimum evidence
File management	Professional file name, correct 04_Checklists folder, v01 and final v02, successful reopen.
Planning	Purpose, reader, scope and section order written before drafting.
Research	At least one authoritative source plus an AI-generated candidate list; decisions recorded.
Document structure	Title, machine information, safety boundary, daily-check table, action-if-failed section and footer.
Visual quality	Readable styles, table inside margins, inline image with caption, no overlapping objects.
Verification and honesty	Final items traceable to sources; AI assistance declared accurately.
Usability and delivery	Peer "new trainee" test, corrections applied, professional submission message or teacher-approved handover.

Pro-Tip for Success

To help you excel in your 26-minute Capstone (practical) Assessment, you are allowed to compile all the practical outputs you have completed throughout this unit into a single **Digital Portfolio**.

Your previously developed **Practical Outputs,** such as your problem logs, inspection reports, and verified datasheets, are valuable assets. By organising these artifacts into a professional portfolio, you demonstrate your consistent progress and technical readiness. Think of this portfolio as your **'professional evidence bank'** that you can rely on to effectively complete your final assessment tasks within the provided timeline.

Figure 7.1 shows why this document is the capstone. Each surrounding box represents a skill learned earlier in the unit. The centre checklist is not a separate task; it is where those skills become one professional workflow.

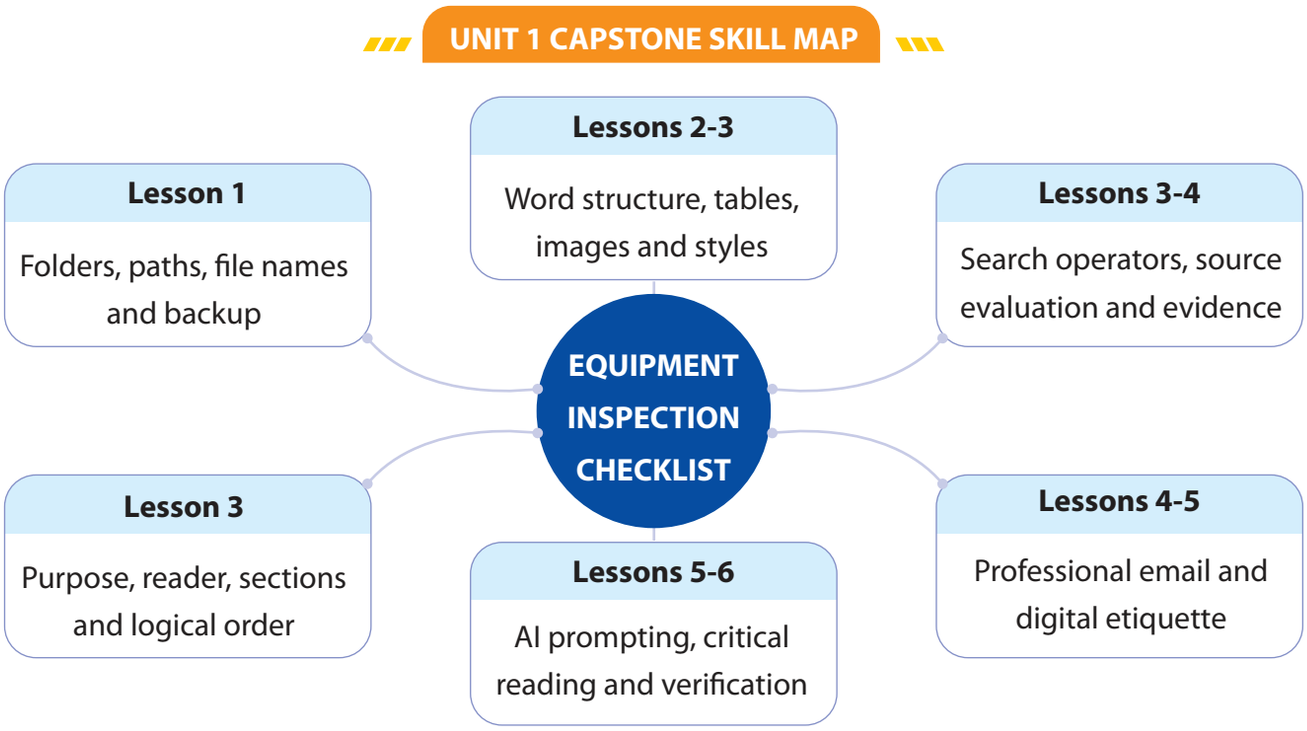


Figure 7.1 - Unit 1 capstone skill map: earlier lesson skills combine into one inspection checklist.

Laboratory Vocabulary

These terms are used repeatedly in the practical instructions. Read them before starting so that the teacher can focus on decisions rather than stopping to define every word.

Term	Meaning in this lab
Scope	The exact type of inspection and actions the document covers.
Candidate item	A possible checklist item that has not yet been accepted.

Term	Meaning in this lab
Triangulation	Comparing more than one source or type of evidence before accepting a claim.
Authoritative source	A source with recognised responsibility, such as the manufacturer, regulator or approved school procedure.
Accept / Reject / Verify First	A decision label used for each AI suggestion.
Usability test	A test of whether the intended reader can follow the document clearly.
Version	A controlled revision of a file, such as v01 before review and v02 after changes.

Figure 7.2 gives the laboratory sequence. Do not jump directly to formatting. Each stage produces evidence needed by the next stage: verified content before document construction, human decisions before final revision and a usability test before submission.

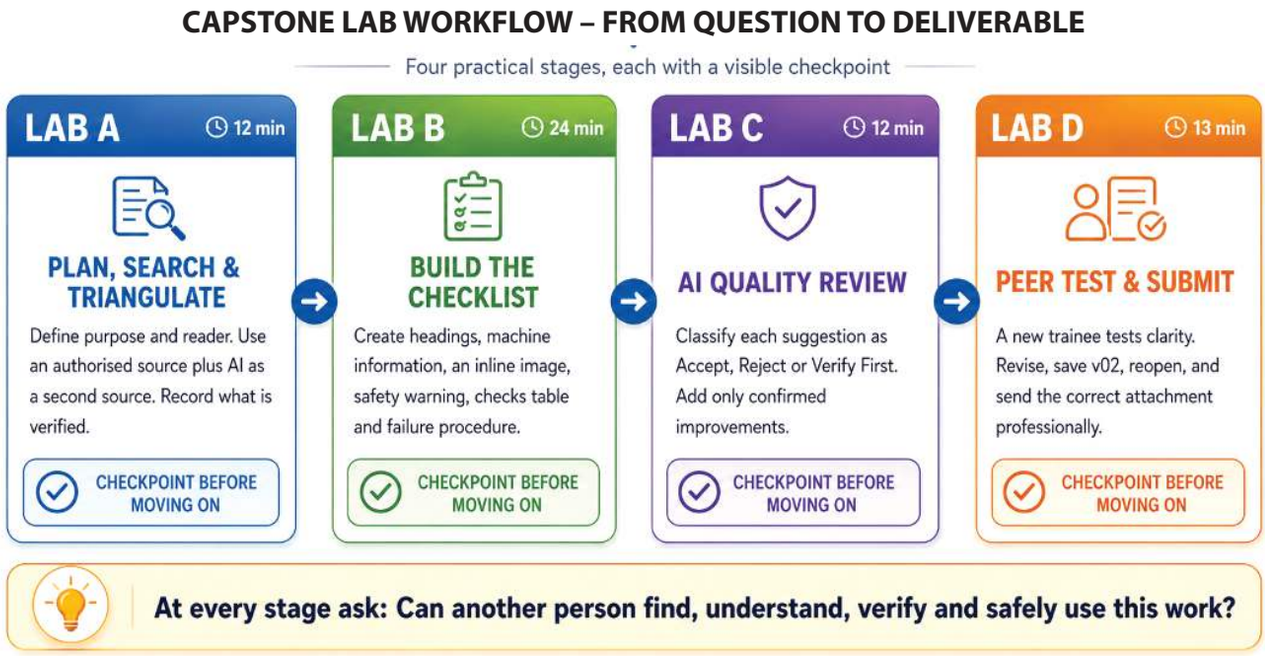


Figure 7.2 - Capstone lab workflow and timing: Plan, Build, Review and Submit.

Lab A (6 minutes): Plan, Search and Triangulate

This practical protects the checklist from becoming a list of guesses. You will define the document before searching, save the file immediately, and collect candidate items from two different routes: an authoritative source and an AI assistant. The AI list is not accepted automatically; it is compared with the source and the machine context.

Before starting

Use only the assigned machine and the teacher-approved inspection scope. Do not search for repair, internal adjustment or live testing procedures. The capstone is a document-design and verification task.

1. **Write the PURPOSE:** support a consistent, teacher-approved pre-use visual inspection.
2. **Write the READER:** a new first-year trainee who needs clear language and clear stop conditions.
3. **Write the SCOPE:** the assigned machine, the inspection frequency, and what the checklist does not authorise.
4. Create and save the document as YYYY-MM-DD_Machine_InspectionChecklist_v01.docx in 04_ Checklists.
5. Find one authoritative source. Record publisher, title, machine/model relevance, date, and section used.
6. Ask the AI for candidate items using the prompt formula. Do not include personal or confidential data.
7. Compare the two lists. Mark each candidate Accept, Reject, or Verify First and record the reason.

Use the research log below. Each row keeps the generated idea separate from the evidence that supports or rejects it.

Candidate check item	AI suggestion	Authoritative-source evidence	Decision and reason
Example: guard or cover condition	Suggested as a visual check.	Source requires guards/ covers to be in place and serviceable.	Accept - within visual-inspection scope.

Lab A checkpoint

The teacher or partner can identify your purpose, reader, scope, saved path, authoritative source and at least six classified candidate items.

Lab B (8 minutes): Build the Checklist Document

This is the main production stage. Structure comes before decoration. Build a document that guides the reader in the order they will use it: identify the machine, read the safety boundary, complete the checks, and know what to do if a check fails. Use styles and tables to reduce reading time, not to make the page colourful.

1. Insert the school/programme header and apply the Title style to "Daily Equipment Inspection Checklist".
2. **Create Section 1:** Machine Information. Include machine name, asset ID if provided, location, inspection type, author, issue date and version.
3. Insert the teacher-provided machine photo using Picture Layout: In Line with Text. Resize with a corner handle and add a factual caption.
4. **Create Section 2:** Safety Boundary. State the stop condition and any teacher-approved PPE or isolation rule. Do not invent a claim that the machine is safe.
5. **Create Section 3:** Daily Checks. Insert a table with columns: No. | Check item | Status | Notes / evidence.

6. Use 6-8 verified items. Write observable items such as condition, presence, damage or housekeeping. Identify any check that requires teacher authorisation.
7. **Create Section 4:** Action if a Check Fails. Use a numbered sequence: stop; prevent use according to procedure; report the exact issue to the authorised person.
8. Add a footer with version and page number. Save with Ctrl + S and check the document in Print Preview.
9. For a pedestal drill, suitable candidate areas may include the condition and presence of guards or belt covers, power cable and plug, chuck key removal, availability of a serviceable work-holding clamp, clear work area, accessible controls and visible damage. The final wording must be confirmed against the assigned machine, manufacturer information and school procedure. Operating tests are included only when the teacher has explicitly authorised and supervised them.

Figure 7.3 shows a model page layout. It is not a universal safety checklist and should not be copied without adaptation. Study the information hierarchy: identification first, safety boundary before the table, clear status columns, a failure procedure and a traceable footer.

10. MODEL: DAILY EQUIPMENT INSPECTION CHECKLIST

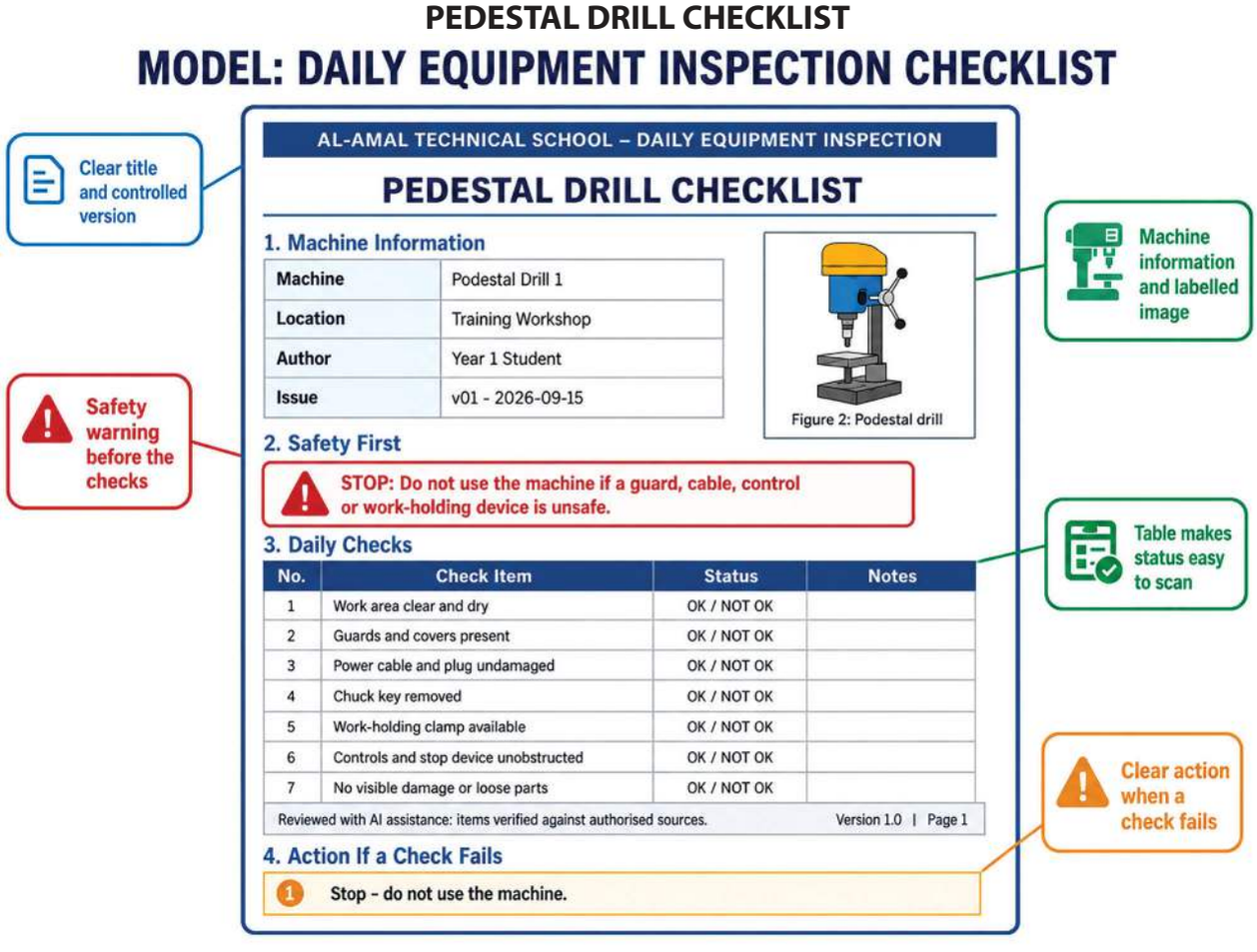


Figure 7.3 - Model layout of a finished inspection checklist. Content must be adapted and verified for the assigned machine.

Lab B checkpoint

Another student can find the machine identity, understand the safety boundary, scan every check and locate the failure action. The image and table remain inside the page with no text overlap.

Lab C (6 minutes): AI Quality Review - Then Your Decision

AI review is useful when it is treated as a second pair of eyes, not an approval authority. You will ask the assistant to identify unclear language, missing categories, and possible contradictions. Every suggestion must then be classified. The important evidence is your reason, especially when you reject an attractive but unsafe or unsupported idea.

1. Remove names, IDs, or sensitive workshop details before sharing any text with the AI.
2. Prompt: "Review these candidate checklist items for clarity and completeness. Do not give repair or operating instructions. Identify any item that requires model-specific verification."
3. For every suggestion, choose Accept, Reject, or Verify First.
4. Verify any new technical content against the authoritative source and local procedure before adding it.
5. Add the honesty statement at the end of the document and save.

The decision table below records human judgement. A completed row should make sense even if the AI conversation is unavailable later.

AI suggestion	Decision	Reason / evidence	Change made
Example: add "check speed setting"	Verify First	Setting depends on task, material, and authorised procedure; not a universal visual check.	Not added until teacher and manual confirm scope.

Lab C checkpoint

Your file contains at least three recorded AI-review decisions, one verified source, and an accurate declaration of how AI was used

Lab D (6 minutes): New-Trainee Test, Revision and Submission

A technically correct document can still fail if the intended reader cannot use it. In the final practical, another pair acts as new trainees. They do not judge your personality or design taste; they test whether the document is findable, readable, unambiguous and safe. Their difficulty is evidence that the document needs revision.

1. Exchange the checklist with another pair. Do not explain it before they read.
2. The reviewers identify the machine, safety boundary, first check, status method, and failure action.
3. Reviewers highlight unclear words and give one specific strength plus one requirement-based improvement.
4. Revise the document and use Save As to create v02. Keep v01 as evidence of development if instructed.
5. Close Word and reopen v02 from File Explorer. Confirm the correct version, image, table, and footer.
6. Submit through the teacher-approved route. When email is used, write a specific subject and verify the correct attachment before sending.

Use the final quality check as a separate table. Do not combine it with the peer-feedback table because the two tables answer different questions.

Final quality check	Yes	Needs correction
Correct folder and professional v02 file name	☐	☐
Purpose, reader and scope are clear	☐	☐
All technical items are verified and applicable	☐	☐
Safety boundary appears before the checks	☐	☐
Table is readable; image is inline and captioned	☐	☐
Failure procedure and reporting route are clear	☐	☐
AI assistance and sources are documented honestly	☐	☐
File closes and reopens correctly	☐	☐

Submission subject example

Capstone Submission - Pedestal Drill Inspection Checklist - Student Name.

Practical Summary

A professional capstone workflow is:
define ⟶ research ⟶ triangulate ⟶ build ⟶ review ⟶ verify ⟶ test ⟶ deliver.

Capstone Practical Rubric (12 marks)

The practical rubric assesses the whole workflow, not only the appearance of the page. Each criterion has equal weight so that verified content, safe scope and human judgement are valued alongside formatting.

Criterion	2 marks	1 mark	0 marks
Planning and scope	Purpose, reader, scope and sections are precise.	One element is vague or incomplete.	No meaningful plan or unsafe scope.
Research and verification	Authoritative source recorded; items verified and applicable.	Some evidence, but gaps remain.	Items are guessed or unsupported.

Criterion	2 marks	1 mark	0 marks
AI decision record	Clear Accept/Reject/ Verify decisions with reasons and honest declaration.	Partial decisions or weak reasons.	AI output copied without judgement.
Document structure	All required sections logically ordered and complete.	One section incomplete or misplaced.	Major sections missing.
Visual usability	Readable styles/table; inline image; no overlap; clear status method.	Minor formatting or layout issue.	Unreadable, overlapping or unstable layout.
Version and delivery	Correct v02, successful reopen and correct professional submission.	One minor file or submission error.	Wrong/missing file or cannot reopen.

Self-Assessment

I can	Yes	Not yet
Create and submit a verified, readable inspection checklist.	☐	☐

Reflect and Connect

Write four sentences: Which capstone decision required the strongest human judgement?

Glossary قاموس المصطلحات

The glossary gives a plain-English meaning and an Arabic equivalent. Use it to understand the concept, then return to the English term used in the lesson.

English term	Meaning	المصطلح بالعربية
Capstone	A final task combining several learned skills.	مشروع ختامي
Inspection checklist	A controlled list of checks and places to record status.	قائمة فحص

LESSON 8

Feedback & Treatment

This final lesson is about consolidation. It gives structured feedback on the whole unit and 'treatment' – targeted support – for any area in the assessment (If needed).

Learning Outcomes

By the end of this lesson, you will be able to:

- revisit and strengthen the ideas you found hardest in the assessment.
- practice the specific skills that need more work, with support.
- confirm that you have met every unit learning outcome.
- set one personal goal for continuing to build digital literacy.

Introduction: Turning Mistakes into Mastery

No one masters a whole unit perfectly on the first pass – and that is exactly what a revision lesson is for. The assessment in Lesson 7 was not only a score; it was a map of what to practice next. This lesson uses that map. Wherever you or the class struggled, we return, re-explain in a new way, and practice until the skill is secure. Mistakes handled this way become mastery.

Topic 1: Diagnose – Where Are the Gaps?

Effective revision starts with honest diagnosis. This topic helps you and your teacher find exactly which parts of the unit need more work, so revision time is spent where it counts rather than re-covering what you already know.

The table below lists the unit's six skill areas. Mark your confidence in each after the assessment; the low marks are your revision priorities.

Skill area	Confident	Needs work
Digital literacy & the professional standard		
File management – folders, names, backups		
Word – styles, structure, a table of contents		
Internet – precise search and the 4-W test		
Email – To/CC/BCC, phishing, professional tone		
AI – responsible use and verifying answers		

Topic 2: Treatment – Targeted Revision by Area

For each skill area, there is a quick way to strengthen it. This topic gives a focused revision action for each, so you can go straight to the ones you marked 'needs work'.

The table below pairs each skill area with a short, concrete revision action. Do the ones matching your gaps first.

If your gap is	Do this focused revision
File management	Rebuild a small folder tree and rename five files professionally
Word	Take a plain document and add a title, headings and a table of contents
Internet search	Run the 4-W test on two websites and score each
Email	Rewrite a rude or vague email into a professional one
AI use	Take one AI answer and verify it against a reliable source
Digital literacy	Re-sort a workshop task list into technical vs digital skills

Topic 3: Confirm Every Outcome Is Met

The final step is confirmation – checking that every unit outcome is now secure. This topic lets you verify your readiness before moving on to Unit 2.

The table below restates the six unit outcomes. Tick each one only when you can genuinely demonstrate it; any unticked row is where to spend your last revision effort.

If your gap is	Met?
Explain why digital literacy is professional competence	
Describe computer, OS and file-management concepts	
Produce clear technical documents in Word	
Search safely and evaluate a source's trustworthiness	
Communicate professionally by email	
Use AI responsibly while checking accuracy	

Reflect & Connect – Your Next Goal

Write two or three sentences: which digital skill from this unit will help you most in your training, and what is one thing you will keep practicing to get better at it?

8 SESSION Repetition of Substandard Activity

Lab Introduction

- This session gives a supported second attempt (if needed) at any practical skill that did not reach the standard in the earlier labs or the capstone. It is not a punishment – it is how professionals build competence: identify the weak step, practice it with guidance, and prove it. Your teacher will direct each student or pair to the specific practical that needs repeating, so different people may work on different tasks in the same session.

How This Session Works

The teacher uses the diagnosis from Lesson 8 and the capstone rubric to assign each student the one practical that most needs a second attempt. This part explains the routine so everyone knows what to expect.

Step 1: Your teacher tells you which practical to repeat, based on your assessment and capstone results.

Step 2: Review the original lab's steps and success criteria for that practical.

Step 3: Attempt the task again, this time with the teacher checking at each checkpoint and giving immediate feedback.

Step 4: Produce the same evidence the original lab required, to the full standard.

The Repeatable Practicals

Any of the unit's practicals can be repeated here. The table below lists them with the single most common reason each one falls below standard, so you know what to focus on in the second attempt.

Practical to repeat	Common weak point to fix this time
Folder system (Lesson 2 lab)	File names not clear, or no verified backup
Word safety report (Lesson 3 lab)	Used bold text instead of heading styles
Datasheet search (Lesson 4 lab)	Did not evaluate the source with the 4-W test
Maintenance email (Lesson 5 lab)	Vague subject line, or missing attachment
AI explain & verify (Lesson 6 lab)	Trusted the AI answer without verifying it
Capstone (Session 7)	One stage of the cycle incomplete

Success Criteria for the Second Attempt

The standard for the repeat is the same as the original – there are no easier marks for a second attempt, only more support to reach the same bar. Check your repeated work against the criteria below.

- The specific weak point from the first attempt is now fully corrected.
- The task meets every success criterion of the original lab.
- You can explain what you did differently this time and why.
- The evidence (file, document, email or folder) is complete and to standard.

A Note on Repetition

- Repeating a task to reach the standard is a normal, respected part of technical training – in a real workshop, work that is not to standard is simply done again, correctly. Treat this session as a professional would: focus on the one weak step, fix it properly, and prove you have mastered it.

Completion Checklist

Confirm the repeated practical is now complete and to standard before the unit closes.

I have	Yes	Not yet
Corrected the specific weak point from my first attempt	☐	☐
Met every success criterion of the original lab	☐	☐
Produced complete, to-standard evidence	☐	☐
Explained what I changed and why	☐	☐

Assessment

The assessment for this session is the successful repeat itself, checked against the original lab's success criteria. Answer the reflection questions below to confirm your understanding.

Short Answer

1. Which practical did you repeat, and what was the single weak point you fixed?
2. What did you do the second time differently, and how do you know it now meets the standard?

Self-Assessment Checklist

Tick honestly. Every row should be 'Yes' before the unit is signed off.

I have	Yes	Not yet
Corrected the specific weak point from my first attempt	☐	☐
Met every success criterion of the original lab	☐	☐
Produced complete, to-standard evidence	☐	☐
Explained what I changed and why	☐	☐

Reflect and Connect (At Home)

Write two or three sentences: how did repeating the task change your confidence? Name one habit from this unit – checking a source, using styles, naming files, staying in your role – that you will keep using in your training.

Glossary قاموس المصطلحات

The key terms for this session, with simple meanings and their Arabic equivalents.

English Term	Meaning	المصطلح بالعربية
Repetition	Doing a task again to reach the standard	تكرار
Standard	The required level of quality	معيار
Success criteria	The points that define a correct result	معايير النجاح
Evidence	Work that proves a skill was demonstrated	دليل
Mastery	Secure, confident command of a skill	إتقان

Abbreviations

The abbreviation table gives the complete wording behind shortened terms used in this unit.

Abbreviation	Full Form
DT	Digital Transformation
ToC	Table of Contents
4-W	Who, What, When, Why (credibility test)

Unit Portfolio Submission Checklist

Use the **'Portfolio Submission Checklist'** below to ensure all your artifacts are organised, verified, and ready for submission. Your portfolio acts as your 'Professional Evidence Bank'; ensure every item is in its correct place.
Please check [√] each item once you have verified its presence in your Digital Portfolio.

Code		Required Artifact / Evidence	Verified	Not Yet
DPC1.1	Create and organise a structured folder system with clear naming conventions for workshop documents	Structured folder tree + professional filenames + correct placement + successful retrieval/search + verified separate backup.		
DPC1.2	Produce a formatted technical report in Word (e.g., a workshop safety inspection report)	Completed Word safety report.		
DPC1.3	Perform an advanced online search to locate and evaluate a component motor datasheet	Evaluated/saved datasheet + source check + extracted specifications.		
DPC1.4	Compose and send a professional maintenance request email with an attachment	A planned and verified maintenance-request email plus a professional follow up record.		
DPC1.5	Using an AI Chatbot to Explain a Technical Concept & Verifying the Answer	prompt/AI response + authoritative verification + judgement record.		
DPC1.6	Build an equipment inspection checklist combining text, tables, and images	Completed inspection checklist including table/ image/source evidence/final version/submission. Then make p.215 summary match this table.		

Unit 1: References and Further Reading

1. What are the Essential Computer Basics for Beginners
2. Video – What is Industry 4.0? (Smart factories introduction)
3. Microsoft Support – Windows help & File Explorer
4. Video – What is Mechatronics? The Very Basics in 7 Minutes
5. Intel Education - Introduction to Computers
6. Intel - What Is Computer RAM and Why Does It Matter?
7. Microsoft Support - System configuration tools in Windows
8. Microsoft Support - Insert tables and pictures in Word
9. Microsoft Support - Add page numbers in Word
10. GCFGlobal - Inside a Computer
11. GCFGlobal - Word tutorials
12. Applying and Modifying Styles in Word.
13. GCF2021/content/word2016/applying-and-modifying-styles/lesson.html
14. Microsoft Support — Insert a Table of Contents.
15. ABB — IEC Low-Voltage Motors (example manufacturer datasheet library)
16. IEC Low voltage motors Google Search Help — Refine Web Searches
17. Google Search Help — search operators (for your verification searches)
18. IEC 60529 — Degrees of Protection Provided by Enclosures (the IP code standard). Summary available at
19. Applying and Modifying Styles in Word.
20. Microsoft Support — Insert a Table of Contents.
21. ABB — IEC Low-Voltage Motors (example manufacturer datasheet library)
22. Google Search Help — Refine Web Searches
23. Introduction to Internet
24. Email Basics (sending email, attachments)
25. Mozilla Developer Network — Transport Layer Security and HTTPS
26. Microsoft Support — Reply, Reply All and Forward
27. Microsoft Support — Add files to email
28. Protect yourself from phishing
29. Video — Spot Phishing Emails | Here's how
30. Email Basics & Etiquette Email etiquette: 15 rules you need to know
31. Statista — Number of emails sent per day worldwide
32. NIST - Artificial Intelligence Risk Management Framework: Generative AI Profile
33. Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile
34. UNESCO - Guidance for Generative AI in Education and Research
35. OSHA - Drill Press Safety Guidance
36. What is an AI assistant? The 6 best examples

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